

MUST-SCORE FACTS

Geography

Consolidated Must-Score Facts and must-score drill cards for every chapter.

UPPCS + UKPCS Study Library · chapter order follows Subject Notes.

Chapters

1. Daily Read — High-Yield Revision Facts
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3. Topic 2 — Climate of India
4. Topic 3 — Drainage System
5. Topic 4 — Lakes, Waterfalls & Water Resources
6. Topic 5 — Soils
7. Topic 6 — Agriculture
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Daily Read — High-Yield Revision Facts

1. Indian Mountain Passes (State-Wise Cheat-Sheet)

State / UT	Mountain Pass	Connects / Geographic Significance
Ladakh	Karakoram Pass	Ancient caravan route between India and Yarkand (Tarim Basin, China)
	Zoji La	Connects Srinagar with Kargil and Leh ; Great Himalayas; Zojila Tunnel
	Khardung La	Gateway to the Shyok and Nubra Valleys
	Chang La	Connects Leh to Pangong Tso lake
	Fotu La	Highest point on the Srinagar-Leh highway (Zaskar Range)
	Umling La	World's highest motorable pass/road (19,024 ft) built by BRO
Jammu & Kashmir	Banihal Pass	Connects Jammu with Srinagar ; Jawahar Tunnel under Pir Panjal Range
	Pir Panjal Pass	Traditional route from Jammu to Srinagar valley (<i>Peer Ki Gali</i>)
	Burzil Pass	Connects Srinagar with Gilgit; ancient gateway to Central Asia
Himachal Pradesh	Rohtang Pass	Connects Kullu Valley with Lahaul and Spiti Valleys; Atal Tunnel (9.02 km) under Rohtang
	Shipki La	Sutlej River enters India from Tibet through this pass; Zaskar Range
	Bara-lacha La	Connects Mandi/Manali to Leh (Zaskar Range); origin of Chandra & Bhaga rivers
	Debsa Pass	Connects Spiti and Parvati Valleys
Uttarakhand (UKPCS Top Priority!)	Lipulekh Pass	Tri-junction of India, Nepal, and Tibet ; primary route for Kailash Mansarovar Yatra (Pithoragarh)
	Mana Pass (Dungri La)	Connects Uttarakhand with Tibet; north of Badrinath (Chamoli)
	Niti Pass	Connects Chamoli (Uttarakhand) with Tibet
	Muling La	High mountain pass in Uttarkashi north of Gangotri connecting with Tibet
	Trail's Pass	Connects Pithoragarh with Bageshwar (Pindari Glacier to Milam Valley)
	Sin La	Connects Darma and Byans valleys in Pithoragarh
Sikkim	Nathu La	Strategic border post on India-China border; Old Silk Route branch into Chumbi Valley; reopened for trade in 2006
	Jelep La	Connects Sikkim with Lhasa (Tibet) passing through the Chumbi Valley
Arunachal Pradesh	Bom Di La	Connects Arunachal Pradesh (Kameng) with Lhasa (Tibet)
	Diphu Pass	Strategic pass at the tri-junction of India, China, and Myanmar
	Bum La	High mountain pass near Tawang on India-China McMahon line
	Dihang Pass	Connects Arunachal Pradesh with Myanmar
	Pangsau Pass	Historic Ledo Road (Stilwell Road) connecting Assam to Myanmar

Western Ghats Gaps (Strict North-to-South Chronology)

- Thalghat (Kasara Ghat):** Connects **Mumbai to Nashik / Nagpur / North India** (National Highway 160 & railway line).
- Bhorghat:** Connects **Mumbai to Pune / Bengaluru / South India** (Mumbai-Pune Expressway).
- Palghat (Palakkad Gap):** Located between **Nilgiri Hills (North) and Anaimalai Hills (South)**; connects **Coimbatore (Tamil Nadu) to Palakkad / Kochi (Kerala)**.

4. **Shenkottai Pass:** Located between **Cardamom Hills and Agasthyamalai Hills**; connects **Kollam (Kerala) to Madurai / Tenkasi (Tamil Nadu)**.

2. Highest Peaks of Major Ranges in India

Mountain Range / Region	Highest Peak	Elevation (m)	Location / Crucial Takeaway
Trans-Himalayas (Karakoram)	K2 (Godwin-Austen)	8,611 m	Second highest peak in the world ; highest point in Indian territory (POK)
Great Himalayas (India)	Kanchenjunga	8,586 m	Highest peak in undisputed Indian territory ; located on Sikkim-Nepal border
Garhwal Himalayas (Uttarakhand)	Nanda Devi	7,816 m	Highest peak located ENTIRELY within India ; Chamoli, UNESCO WHS
Garhwal (Second Highest)	Kamet	7,756 m	Chamoli district, Uttarakhand
Aravalli Range	Guru Shikhar	1,722 m	Mount Abu (Sirohi district, Rajasthan); oldest fold mountains in India
Satpura Range	Dhupgarh	1,350 m	Near Pachmarhi (Hoshangabad/Narmadapuram, Madhya Pradesh)
Vindhya Range	Sadbhavna Shikhar (Kalumar)	752 m	Damoh district, Madhya Pradesh
Western Ghats (Sahyadri)	Anamudi	2,695 m	Anaimalai Hills (Idukki, Kerala) — HIGHEST PEAK OF PENINSULAR / SOUTH INDIA
Nilgiri Hills	Doda Betta	2,637 m	Udhagamandalam / Ooty (Tamil Nadu); meeting point of Western & Eastern Ghats
Eastern Ghats	Jindhagada Peak / Arma Konda	1,690 m	Araku Valley (Visakhapatnam, Andhra Pradesh); Mahendragiri (1,501 m, Odisha) often given in older PCS options
Chhotanagpur Plateau	Parasnath Peak (Sammed Shikhar)	1,366 m	Giridih (Jharkhand); sacred Jain pilgrimage site (20 Tirthankaras attained Moksha)
Andaman & Nicobar Islands	Saddle Peak	732 m	North Andaman Island (Saddle Peak National Park)
Great Nicobar Island	Mount Thullier	642 m	Southernmost peak in India
Naga Hills (Purvanchal)	Saramati Peak	3,826 m	Indo-Myanmar border, Nagaland
Mizo Hills (Lushai Hills)	Phawngpui (Blue Mountain)	2,157 m	Mizoram

3. Panch Prayag & Indian River Drainage System

A. The Panch Prayag of Uttarakhand (Order along Alaknanda River)

Memorize downstream order from source towards plains: 1. **Vishnuprayag: Alaknanda + Dhauliganga** (Chamoli) 2. **Nandaprayag: Alaknanda + Nandakini** (Chamoli) 3. **Karnaprayag: Alaknanda + Pindar** (Chamoli) 4. **Rudraprayag: Alaknanda + Mandakini** (Rudraprayag) 5. **Devprayag: Alaknanda + Bhagirathi** → Official **GANGA** begins here! (Tehri Garhwal)

B. Ganga River Tributaries (Left vs Right Bank)

- Left Bank Tributaries (West to East):** 1. **Ramganga** (originates in Dudhatoli range, Pauri Garhwal; flows through Jim Corbett NP; meets Ganga at Kannauj) 2. **Gomati** (originates in plains at **Gomat Taal / Fulhar Lake, Pilibhit,**

UP — only Himalayan/Ganga tributary originating in the plains!; Lucknow & Jaunpur on its banks; joins Ganga near Saidpur, Ghazipur) 3. **Ghaghra / Karnali** (originates from Mapchachungo glacier near Mansarovar; joined by Sarda/Kali; **largest tributary by water volume**; joins Ganga at Chhapra, Bihar) 4. **Gandak** (originates in Nepal Himalayas; joins Ganga at Sonpur / Hajipur, Bihar) 5. **Kosi** (*Sorrow of Bihar*; 7 streams / Saptakoshi; notorious for course-shifting; joins at Kursela, Katihar) 6. **Mahananda** (originates in Darjeeling hills; **last left-bank tributary of Ganga in India**).

- **Right Bank Tributaries:** 1. **Yamuna** (longest tributary, 1376 km; Yamunotri glacier, Bandarpunch peak; joins at Prayagraj Triveni Sangam; its tributaries: Chambal, Sind, Betwa, Ken, Tons, Hindon) 2. **Son** (originates from **Amarkantak Plateau, MP**; flows north-eastwards through Sonbhadra UP; joins Ganga near Danapur/Patna, Bihar) 3. **Punpun** (originates in Palamu, Jharkhand; joins Ganga at Fatuha near Patna).

C. Peninsular Drainage: West-Flowing vs East-Flowing

- **West-Flowing (into Arabian Sea — form Estuaries, NO deltas):**
- **Narmada** (1312 km): Amarkantak plateau (MP); flows through **Rift Valley** between Vindhyas (North) and Satpuras (South); forms **Dhuandhar Falls** at Marble Rocks (Bhedaghat, Jabalpur); Sardar Sarovar Dam.
- **Tapi / Tapti** (724 km): Multai (Betul district, MP); flows in rift valley south of Satpuras; Ukai Dam & Kakrapar Project (Gujarat).
- **Sabarmati:** Dhebar Lake (Aravalli Range, Udaipur); flows through Ahmedabad and Gandhinagar into Gulf of Khambhat.
- **Mahi:** Vindhyan hills (Dhar, MP); **CUTS THE TROPIC OF CANCER TWICE!**
- **Luni:** Originates near Ajmer (Aravallis) as Sagarmati; **ends in the marshy Rann of Kutch (Inland drainage, never reaches the sea)**; sweet water up to Balotra, then saline.
- **Sharavati:** Shimoga (Karnataka); forms famous **Jog Falls (Gersoppa / Mahatma Gandhi Falls)**.
- **Periyar & Bharathappuzha (Ponnani):** Kerala.
- **East-Flowing (into Bay of Bengal — form large Deltas):**
- **Mahanadi** (851 km): Sihawa hills (Dhamtari, Chhattisgarh); flows through Odisha; Hirakud Dam.
- **Godavari** (1465 km — **Dakshin Ganga / Vridhha Ganga**): Trimbakeshwar (Nashik, Maharashtra); **longest river in Peninsular India**; Tributaries: Pravara, Manjra, Penganga, Wainganga, Wardha, **Pranhita** (largest), Indravati, Sabari.
- **Krishna** (1401 km): Mahabaleshwar (Maharashtra); Tributaries: Koyna, Ghataprabha, Malaprabha, Bhima, **Tungabhadra** (largest), Musi (Hyderabad on its banks), Munneru.
- **Kaveri** (800 km — **Ganga of the South**): Talakaveri (Brahmagiri Range, Coorg, Karnataka); flows year-round (perennial due to SW & NE monsoons); creates river islands: **Srirangapatna, Shivanasamudra, and Srirangam**; Tributaries: Hemavati, Harangi, Shimsha, Arkavathi (left); Lakshmantirtha, Kabini, Bhavani, Noyyal, Amaravati (right).

4. Multipurpose River Valley Projects, Dams & Waterfalls

Project / Dam / Waterfall	River	State	Exam Crucial Fact
Tehri Dam	Bhagirathi + Bhilangana	Uttarakhand	Highest dam in India (260.5 m); rock and earth-fill embankment
Bhakra Nangal Dam	Sutlej	Himachal Pradesh / Punjab	Highest straight gravity dam in India; reservoir is Gobind Sagar
Hirakud Dam	Mahanadi	Odisha (Sambalpur)	Longest earthen dam in India / world (main dam 4.8 km, total 25.8 km)
Sardar Sarovar Dam	Narmada	Gujarat (Navagam)	Beneficiary states: Gujarat, MP, Maharashtra, Rajasthan; Statue of Unity nearby

Project / Dam / Waterfall	River	State	Exam Crucial Fact
Nagarjuna Sagar Dam	Krishna	Andhra Pradesh / Telangana border	World's largest masonry dam; creates Nagarjunasagar reservoir
Rihand Project (Govind Ballabh Pant Sagar)	Rihand (tributary of Son)	Sonbhadra (Uttar Pradesh)	Largest man-made artificial reservoir / lake in India by volume
Mata Tila & Rajghat Dam	Betwa	Lalitpur (Uttar Pradesh / MP)	Joint venture between UP and MP
Kunchikal Falls	Varahi	Karnataka (Shimoga)	Highest tiered waterfall in India (455 m)
Jog Falls (Gersoppa)	Sharavati	Karnataka (Shimoga)	Highest untiered plunge waterfall (253 m)
Dhuandhar Falls	Narmada	Bhedaghat, Jabalpur (MP)	"Smoke cascade" cutting through marble gorge
Chitrakote Falls	Indravati	Bastar (Chhattisgarh)	Known as the "Niagara of India" (widest horseshoe waterfall in India)
Hundru Falls	Subarnarekha	Ranchi (Jharkhand)	98 m drop on Chotanagpur plateau
Nohkalikai Falls	Rain-fed stream	Cherrapunji (Meghalaya)	Tallest plunge waterfall in India (340 m)

5. Major Agricultural Crops & Top Producing States

Crop Category	Specific Crop	Top Producing State (1st Rank)	Other Key States / Prelims Note
Foodgrains	Total Foodgrains	Uttar Pradesh	Followed by Madhya Pradesh, Punjab
	Wheat	Uttar Pradesh	Followed by MP, Punjab (Punjab highest per hectare yield)
	Rice	West Bengal	Followed by Uttar Pradesh, Punjab
	Maize	Karnataka	Followed by Madhya Pradesh, Maharashtra
	Coarse Cereals (Nutri-cereals)	Rajasthan	Followed by Karnataka, Maharashtra
	Total Pulses	Madhya Pradesh	Followed by Maharashtra, Rajasthan
	Gram (Chana)	Maharashtra / MP	Accounts for ~40% of total pulses
Oilseeds	Total Oilseeds	Rajasthan	Followed by Maharashtra, Madhya Pradesh
	Rapeseed & Mustard	Rajasthan	(>45% of national production)
	Soyabean	Maharashtra / MP	MP traditionally called <i>Soyabean State</i>
	Groundnut	Gujarat	Followed by Rajasthan, Tamil Nadu
Cash / Commercial	Sugarcane	Uttar Pradesh	Followed by Maharashtra, Karnataka (Maha highest sugar recovery)
	Cotton	Gujarat	Followed by Maharashtra, Telangana
	Jute & Mesta	West Bengal	(>75% of national production - Golden Fibre)
	Tea	Assam	Followed by West Bengal (Darjeeling - 1st GI tag 2004), Tamil Nadu
	Coffee	Karnataka	(>70% of national production - Arabica & Robusta in Kodagu/Chikkamagaluru)
	Natural Rubber	Kerala	(>75% of national production)
	Tobacco	Gujarat / Andhra Pradesh	CTRI located at Rajahmundry (AP)

6. Major Tribes of India & State-Wise Matrix (Direct Match-Ups)

State / UT	Prominent Indigenous Tribes	Key Ethnographic / Exam Fact
Uttarakhand (UKPCS Top Priority!)	Tharu, Jaunsari, Bhotia, Buksa, Raji (Van Rawat)	Tharu: Largest tribe in UK & UP; historically celebrate Diwali as mourning (<i>Shok Parva</i>); live in Terai (US Nagar). Jaunsari: Second largest; polyandry tradition; Dehradun (Chakrata). Bhotia: Transhumance (<i>Ritu Pravas</i>); seasonal migration to bugyals; Indo-Tibetan trade. Buksa: Claim Rajput descent (Pawar); Terai region. Raji: Smallest tribe (<i>Van Rawat</i> / kings of forest); cave dwellings.
Uttar Pradesh	Tharu, Gond, Kharwar, Sahariya, Parahiya, Baiga, Agariya, Chero	Sonbhadra has highest ST population & diversity in UP; Sahariya in Lalitpur; Agariya (iron smelters) in Sonbhadra.
Madhya Pradesh	Bhil, Gond, Baiga, Sahariya, Kol, Korku, Bharia	Highest ST population in India; Bhils largest tribe of India; <i>Bhagoria</i> festival; <i>Pithora</i> painting.
Rajasthan	Bhil, Meena, Garasia, Sahariya, Damor	Meenas are the most populous tribe in Rajasthan; Sahariya in Baran.
Odisha	Santhal, Bonda, Dongria Kondh, Juang, Saora	Largest number of Particularly Vulnerable Tribal Groups (PVTGs: 13 out of 75 in India) ; Dongria Kondh worship Niyamgiri hills.
Tamil Nadu	Toda, Badaga, Kota, Irula, Kurumba	Toda: Inhabit Nilgiri Hills; pastoral tribe centered on water buffaloes; practice fraternal polyandry; barrel-shaped huts.
Andhra Pradesh / Telangana	Chenchu, Yanadi, Koya, Lambada	Chenchu: Primitive hunter-gatherers in Nallamala forest / Srisailem.
Sikkim	Lepcha, Bhutia, Limbu	Lepchas are indigenous inhabitants; practice sacred groves.
Himachal Pradesh	Gaddi, Gujjar, Kinnaura, Pangwala	Gaddis: Pastoral transhumance shepherds of Dhauladhar range.
Andaman & Nicobar Islands	Great Andamanese, Onge, Jarawa, Sentinelese, Shompen, Nicobarese	Negrito stock (Jarawa, Sentinelese, Onge - North Sentinel isolated); Shompen are Mongoloid stock.

7. Major Sea Ports & National Waterways of India

A. 13 Major Ports of India

- **Western Coast:** 1. **Deendayal Port (Kandla)** (Gulf of Kutch, Gujarat): Tidal port; free trade zone (SEZ); handles crude oil & grains. 2. **Mumbai Port** (Maharashtra): Largest natural port and harbour in India. 3. **Jawaharlal Nehru Port (JNPT / Nhava Sheva)** (Navi Mumbai): **Largest container port in India**; fully mechanized. 4. **Mormugao Port** (Goa - Zuari estuary): Premier port for iron ore exports from Goa. 5. **New Mangalore Port** (Karnataka): Handles iron ore exports from Kudremukh. 6. **Cochin Port** (Kerala - Willingdon Island in Vembanad Lake): Natural harbour; International Container Transshipment Terminal (Vallarpadam).
- **Eastern Coast:** 7. **V.O. Chidambaranar Port (Tuticorin)** (Tamil Nadu - Gulf of Mannar): Artificial deep-sea port; pearl fishery. 8. **Chennai Port** (Tamil Nadu): **Oldest artificial harbour on the east coast**; second busiest. 9. **Kamarajar Port (Ennore)** (Tamil Nadu): **First corporatized port in India** (public company). 10. **Visakhapatnam Port** (Andhra Pradesh): **Deepest landlocked and protected port in India** (Dolphin's Nose hill protects from storms). 11. **Paradip Port** (Odisha - Mahanadi delta): Deep-water port exporting iron ore to Japan. 12. **Syama Prasad Mookerjee Port (Kolkata-Haldia)** (West Bengal): **Only riverine major port in India** (on Hooghly River); twin dock system (Kolkata & Haldia).
- **Island:** 13. **Port Blair Port** (Andaman & Nicobar Islands).

B. Core National Waterways (NW) of India

- **NW-1: Ganga–Bhagirathi–Hooghly River System** from **Prayagraj to Haldia** (1,620 km) — **Longest National Waterway in India**; passes through UP, Bihar, Jharkhand, WB.
- **NW-2: Brahmaputra River** from **Sadiya to Dhubri** (891 km) in Assam.
- **NW-3: West Coast Canal** from **Kottapuram to Kollam** (205 km) in Kerala (includes Champakara & Udyogmandal canals).
- **NW-4: Krishna–Godavari Rivers & Canals** from Kakinada to Puducherry (1,078 km).
- **NW-5: Brahmani River, Deltaic channels of Mahanadi & East Coast Canal** (588 km) in Odisha & WB.
- **NW-16: Barak River** from Lakhimpur to Bhanga (121 km) in Assam.

8. Census 2011 Essential Fact Card (India vs UP vs UK)

Demographic Indicator	All-India	Uttar Pradesh	Uttarakhand
Total Population	121.08 Crore (17.5% of world)	19.98 Crore (16.51% of India - Rank 1)	1.01 Crore (0.83% of India - Rank 20)
Decadal Growth Rate (2001–11)	17.70% (Highest: Meghalaya 27.95%; Lowest/Negative: Nagaland -0.58%)	20.22% (Highest: Gautam Buddha Nagar 49.1%; Lowest: Kanpur Nagar 9.9%)	18.81% (Highest: US Nagar 33.4%; Negative: Almora -1.28%, Pauri Garhwal -1.41%)
Population Density	382 per sq km (Highest: Bihar 1106; Lowest: Arunachal 17)	829 per sq km (Rank 4 among states) • Highest: Ghaziabad (3,971) • Lowest: Lalitpur (242)	189 per sq km (Rank 25) • Highest: Haridwar (801) • Lowest: Uttarkashi (41)
Overall Sex Ratio	943 females / 1000 males (Highest: Kerala 1084; Lowest: Haryana 879)	912 females / 1000 males • Highest: Jaunpur (1,024), Azamgarh (1,019), Deoria (1,017) • Lowest: Gautam Buddha Nagar (851)	963 females / 1000 males • Highest: Almora (1,139), Rudraprayag (1,114) • Lowest: Haridwar (880), Dehradun (902)
Child Sex Ratio (0–6 yrs)	919 (Highest: Mizoram 970; Lowest: Haryana 834)	902 • Highest: Balrampur (950) • Lowest: Baghpat (841)	890 • Highest: Pithoragarh (916) • Lowest: Pithoragarh / Champawat
Literacy Rate (Overall)	73.0% (Male: 80.9%, Female: 64.6%; Gap: 16.3%)	67.7% (Male: 77.3%, Female: 57.2%; Gap: 20.1%) • Highest: Gautam Buddha Nagar (80.1%) • Lowest: Shravasti (46.7%)	78.82% (Male: 87.4%, Female: 70.0%; Gap: 17.4%) • Highest: Dehradun (84.25%) • Lowest: Haridwar (73.43%)
SC Population %	16.6% (Highest %: Punjab 31.9%; Highest No.: UP)	20.7% of state pop (Highest No.: Sitapur; Highest %: Kaushambi 34.7%; Lowest: Baghpat)	18.8% of state pop (Highest %: Bageshwar; Lowest %: Dehradun)
ST Population %	8.6% (Highest %: Lakshadweep/Mizoram; Highest No.: MP; Nil in Punjab, Haryana)	0.6% of state pop (Highest No. & %: Sonbhadra ; Lowest: Baghpat; Tharu is largest tribe)	2.9% of state pop (Highest %: Udham Singh Nagar ; Tharu & Jaunsari are largest)

9. Minerals of Uttar Pradesh & Uttarakhand (state-site map)

National mineral belts stay in the Minerals chapter. Daily sheet needs the **UP / UK places** that otherwise vanish (same gap as Hulas on the Indus table).

Mineral / resource	Uttar Pradesh	Uttarakhand	Trap
Limestone / cement	Sonbhadra (Churk, Dalla, Ghurma); also Banda–Mirzapur belt	Dehradun , Tehri, Pithoragarh, Pauri belts	Do not give Rajasthan's Makrana marble as a UP fact
Coal	Singrauli coalfield (Sonbhadra; shared with MP)	Negligible	Jharia / Raniganj are Jharkhand–WB , not UP
Silica sand / glass	Shankargarh (Prayagraj) , Banda; feeds Firozabad glass	—	Firozabad = glass craft; sand is Prayagraj belt
Bauxite	Banda, Vindhyan pockets (small)	—	Big bauxite = Odisha / Gujarat, not UP
Andalusite / sillimanite	Sonbhadra / Mirzapur (refractory)	—	
Magnesite	—	Jhironi / Bageshwar (Kumaon) — India's important magnesite belt	Magnesite ≠ magnesium metal plant trap
Gypsum	—	Dehradun / Nainital foothill pockets	Rajasthan remains India's gypsum giant
Steatite / soapstone	—	Almora / Pithoragarh (also used in local craft)	
Copper (old)	—	Askot (Pithoragarh) — historic Kumaon copper	Khetri copper is Rajasthan
Rock phosphate	—	Mussoorie / Durmala–Maldeota (Dehradun)	
Building stone / granite	Mirzapur–Sonbhadra; Lalitpur	Granite–quartzite of Lesser Himalaya	
Energy water	Rihand / Govind Ballabh Pant Sagar (Sonbhadra) — largest man-made lake by volume	Tehri (Bhagirathi–Bhilangana) — highest dam	Hydel ≠ mineral ore

Logic: Sonbhadra is UP's mineral hub (coal + limestone + refractories). **Bageshwar magnesite** and **Askot copper** are the UK pair. Do not park Rajasthan or Jharkhand fields in these two states.

10. Major Minerals of India & Famous Mines Master Matrix

Mineral / Resource	Leading Producing State(s)	Famous Mines / Belts (All-India Match Goldmine)	High-Yield Prelims Exam Triggers
Iron Ore (Hematite & Magnetite)	Odisha (1st in production & reserves), Chhattisgarh, Karnataka, Jharkhand	• Odisha: Mayurbhanj (Gorumahisani, Badampahar), Keonjhar (Joda), Sundargarh • Chhattisgarh: Bailadila (Bastar/Dantewada — high-grade hematite exported to Japan via Visakhapatnam Port), Dalli-Rajhara (Durg — feeds Bhilai Steel Plant) • Karnataka: Kudremukh (Chikkamagaluru — horse-face shaped; slurry pipeline to Mangalore), Bababudan Hills, Bellary-Hospet (Sandur) • Jharkhand: Noamundi, Gua, Kiriburu (West Singhbhum) • Goa: Marmagao belt	• India is rich in Hematite (~70%) and Magnetite (~30%). • Dharwar and Cuddapah rock systems are primary sources. • <i>Trap:</i> Bailadila is in Chhattisgarh, NOT MP or Odisha!

Mineral / Resource	Leading Producing State(s)	Famous Mines / Belts (All-India Match Goldmine)	High-Yield Prelims Exam Triggers
Bauxite (Aluminium Ore)	Odisha (>50% of national production), Gujarat, Jharkhand	• Odisha: Panchpatmali (Koraput — India's largest deposit; feeds NALCO refinery at Damanjodi), Kalahandi, Rayagada, Bolangir • Jharkhand: Lohardaga, Gumla, Latehar • Gujarat: Jamnagar, Junagadh, Kutch • Madhya Pradesh: Amarkantak plateau (Shahdol), Balaghat • Maharashtra: Kolhapur, Ratnagiri	• Associated with laterite rocks of tertiary age formed via intense leaching. • NALCO (Damanjodi & Angul) and HINDALCO (Renukoot, UP) are major processors.
Copper	Madhya Pradesh (1st in production), Rajasthan (1st in reserves), Jharkhand	• Madhya Pradesh: Malanjkhand (Balaghat district — largest open-cast copper mine in Asia; accounts for ~50% of Indian production) • Rajasthan: Khetri, Chandmari, Kolihaan (Jhunjhunu), Kho-Dariba (Alwar), Delwara (Udaipur) • Jharkhand: Mosabani, Ghatshila, Rakha (Singhbhum Copper Belt) • Uttarakhand: Askot (Pithoragarh — old workings)	• Hindustan Copper Limited (HCL, est. 1967) holds public monopoly. • India is heavily import-dependent for copper ore (~80% imported).
Manganese	Madhya Pradesh (1st in production), Maharashtra, Odisha	• Madhya Pradesh: Balaghat (Bharveli underground mine), Chhindwara • Maharashtra: Nagpur, Bhandara, Ratnagiri • Odisha: Sundargarh, Keonjhar, Koraput • Karnataka: Shimoga, Bellary	• Vital ferro-alloy: ~10 kg manganese is needed to manufacture 1 tonne of steel. • MOIL (Manganese Ore India Limited) headquartered in Nagpur.
Coal	Jharkhand (1st in reserves), Odisha / Chhattisgarh (1st in production)	• Jharkhand: Jharia (Dhanbad — richest and only storehouse of prime coking coal in India), Bokaro, Giridih, North/South Karanpura, Ramgarh • West Bengal: Raniganj (Asansol/Bardhaman — oldest commercial coalfield in India, started 1774 by Summer & Heatly) • Chhattisgarh: Korba (Hasdeo valley), Tatapani-Ramkola, Chirimiri • Odisha: Talcher (Brahmani valley — feeds NTPC super thermal plant), Ib valley • Madhya Pradesh: Singrauli (Jhingurdah seam — second thickest coal seam in the world; shared with UP Sonbhadra), Pench-Kanhan • Telangana: Singareni (Godavari valley — SCCL) • Tamil Nadu: Neyveli (Cuddalore — premier deposit of Lignite / Brown Coal in India) • Assam: Makum, Nazira, Ledo (Tertiary coal; high sulphur content)	• Gondwana Coal (~98% of India's reserves; 250 million yrs old; low moisture, low sulphur, high ash, non-coking/coking). • Tertiary Coal (~2% reserves; 15–60 million yrs old; NE states & Jammu; high sulphur & volatile matter).
Petroleum & Crude Oil	Offshore (Bombay High) (1st in production), Rajasthan, Gujarat, Assam	• Offshore: Bombay High (1974 discovery by ONGC vessel <i>Sagar Samrat</i>; 160 km off Mumbai), Bassein, Ravva (Krishna-Godavari) • Rajasthan: Barmer Basin (Mangala, Bhagyam, Aishwarya fields — discovered by Cairn/Vedanta; largest onshore producer) • Gujarat: Ankleshwar (<i>Fountain of Prosperity</i> named by Nehru), Kalol, Mehsana, Aliabet Island (Gulf of Khambhat) • Assam: Digboi (oldest producing oilfield & refinery in India and Asia, 1901), Naharkatiya, Moran-Hugrijan, Surma Valley	• Refineries: Jamnagar (Gujarat) is the world's largest grass-root refining complex (private - Reliance). • Indian Oil's Digboi is the oldest operational refinery.

Mineral / Resource	Leading Producing State(s)	Famous Mines / Belts (All-India Match Goldmine)	High-Yield Prelims Exam Triggers
Gold	Karnataka (>90% of national production), Andhra Pradesh	• Karnataka: Kolar Gold Fields (KGF) (shut down in 2001 due to depth & costs), Hutti Gold Mines (Raichur — only active primary gold producing mine in India today) • Andhra Pradesh: Ramagiri (Anantapur) • Jharkhand: Alluvial placer gold in Subarnarekha river sands	• Hutti produces ~99% of India's indigenous mined gold.
Diamond	Madhya Pradesh (holds >90% of diamond resources)	• Madhya Pradesh: Panna (Majhgawan pipe mine) operated by NMDC — the ONLY mechanized active diamond mine in Asia/India • Historic: Kollur Mine (Guntur/Golconda, Andhra Pradesh — origin of the Koh-i-Noor, Hope Diamond, and Daria-i-Noor)	• Kimberly pipe formations in Vindhyan conglomerates.
Uranium	Jharkhand, Andhra Pradesh	• Jharkhand: Jaduguda (East Singhbhum — India's first uranium mine, 1967), Bhatin, Narwapahar, Turamdih, Banduhurang (all run by UCIL) • Andhra Pradesh: Tummalapalle (Kadapa/YSR district — massive global-scale reserves discovered in 2011) • Meghalaya: Domiasiat / Mawthabah (Kasi Hills - sandstone deposit) • Rajasthan: Rohil (Sikar)	• Operated under Department of Atomic Energy (DAE) via Uranium Corporation of India Ltd (UCIL, Jaduguda).
Thorium	Kerala, Odisha, Tamil Nadu	• Kerala: Monazite beach sands of Chavara, Kollam, and Alappuzha • Odisha: Ganjam coast (Chhatrapur beach sands) • Tamil Nadu: Manavalakurichi (Kanyakumari)	• Derived from Monazite sand (contains thorium phosphate, cerium, and rare earths). • Indian Rare Earths Limited (IREL) processes beach sand minerals. India holds the world's largest thorium reserves.
Lead & Zinc / Silver	Rajasthan (near 100% monopoly)	• Rajasthan: Zawar Mines (Udaipur — ancient & modern lead-zinc-silver), Rampura-Agucha (Bhilwara — world-class open-pit zinc deposit), Rajpura-Dariba (Rajsamand), Sindesar Khurd	• Hindustan Zinc Limited (HZL, Vedanta group). • Galena (lead ore) and Sphalerite (zinc ore). Silver is recovered as a by-product.
Mica (Ruby Mica / Muscovite)	Andhra Pradesh (1st in production), Rajasthan, Jharkhand	• Andhra Pradesh: Nellore Mica Belt (Gudur) • Jharkhand: Koderma (historically called the "Mica Capital of the World"), Giridih, Hazaribagh • Rajasthan: Bhilwara, Ajmer, Jaipur	• India was historic world leader in sheet mica (Muscovite / Ruby mica). • Essential electrical insulator resistant to high voltages.

11. Ocean Currents & Planetary Local Winds of the World

A. Major Ocean Currents (Warm vs Cold Master Matrix)

Rule of Thumb: Currents moving from Equator to Poles = Warm; Currents moving from Poles to Equator = Cold.

Ocean Basin	Warm Ocean Currents	Cold Ocean Currents	High-Yield Prelims Exam Takeaways
Atlantic Ocean	• Gulf Stream (powerful warm current) • North Atlantic Drift (keeps NW European ports ice-free year-round) • Florida Current • Antilles Current • Brazil Current (South Atlantic) • Guinea Current (West Africa)	• Labrador Current (flows from Arctic past Greenland) • Canary Current (North Africa coast) • Benguela Current (South-West Africa coast; creates Namib & Kalahari Deserts) • Falkland / Malvinas Current (Patagonia coast) • East Greenland Current	• Grand Banks of Newfoundland: Confluence of warm Gulf Stream and cold Labrador Current produces dense fog and creates world's richest cod-fishing ground. • Sargasso Sea: Clockwise North Atlantic gyre creates a calm, landless sea surrounded by currents, filled with <i>Sargassum</i> seaweed.
Pacific Ocean	• Kuroshio (Japan Current) (warm counterpart of Gulf Stream) • North Pacific Drift • East Australian Current (flows south along eastern Australia) • Alaska Current (keeps Alaskan coast ice-free) • Tsushima Current (Sea of Japan)	• Oyashio (Kurul Current) (cold current from Bering Strait) • California Current (West coast of USA) • Peru / Humboldt Current (flows along Chile & Peru; high upwelling, rich in anchovies; suppressed during El Niño) • Okhotsk Current	• Meeting of warm Kuroshio and cold Oyashio creates dense fog and prime fishing grounds off Japan. • Atacama Desert (driest non-polar desert) created by the desiccating effect of the cold Humboldt/Peru Current.
Indian Ocean	• Mozambique Current (East Africa) • Agulhas Current (South Africa) • South-West Monsoon Drift (summer)	• West Australian Current (flows along West Australia; creates Great Sandy/Victoria Deserts) • Somali Current (reverses direction seasonally; cold upwelling in summer)	• Indian Ocean currents completely reverse their direction of flow twice a year due to the seasonal reversal of the Monsoon winds.
Circumpolar	—	• West Wind Drift (Antarctic Circumpolar Current)	• Largest ocean current by water volume; circles Antarctica unbroken from West to East under roaring westerlies.

B. Famous Local Winds of the World (Hot vs Cold & Nicknames)

Local Wind	Nature (Hot / Cold)	Geographic Region	Distinctive Nickname / Exam Trigger
Chinook	Warm & Dry (Katabatic)	Rocky Mountains (USA & Canada)	"Snow Eater" — descends eastern slopes of Rockies, melting winter snow rapidly and opening pastures for cattle.
Foehn (Föhn)	Warm & Dry	Northern slopes of the Alps (Switzerland)	Accelerates melting of alpine snow, aids grape ripening in Swiss valleys (<i>the climber's wind</i>).
Harmattan	Warm, Dry & Dusty	Sahara Desert blowing south-west to the Gulf of Guinea (West Africa)	"The Doctor" — provides welcome relief from the sultry, humid coastal tropical heat and kills pathogens.
Sirocco	Hot, Dry & Dust-laden	Sahara Desert blowing north across Mediterranean into Southern Europe (Italy/Sicily)	Causes "Blood Rain" (red Sahara dust particles falling with condensation rain in Italy and Greece).
Khamsin	Hot & Dusty	Egypt	Blows from the southern desert for approximately 50 days (<i>Khamsin</i> = fifty in Arabic).
Gibli	Hot & Dry	Libya	Sahara desert wind blowing north towards Libyan coast.
Brickfielder	Hot & Dusty	Southern Australia	Blows from the central Australian desert towards coastal cities (Melbourne/Sydney).

Local Wind	Nature (Hot / Cold)	Geographic Region	Distinctive Nickname / Exam Trigger
Black Roller	Hot & Dust-storm	Great Plains of North America / USA	Intense summer dust squall darkening the sky.
Zonda	Hot & Dry (Foehn-type)	Eastern slopes of the Andes (Argentina)	Blows down Argentine pampas from the Andes.
Santa Ana	Hot, Dry & Gusty	Southern California (USA)	Blows from Mojave desert down coastal mountain canyons; notoriously triggers catastrophic wildfires.
Loo	Scorching Hot & Dry	Northern and North-Western Plains of India (UP, Bihar, Punjab, Rajasthan)	Blows during May–June afternoons; causes severe heatstrokes.
Nor'wester / Kalbaishakhi	Hot-season thunderstorm	West Bengal, Odisha, Assam (Bardoisila)	Violent afternoon squall in Baisakh (April–May); beneficial for early tea, jute, and aus paddy crops.
Mistral	Extremely Cold & Dry	Blows from Alps down the Rhone Valley into Gulf of Lion (France & Spain)	Winter/spring gale-force cold wind; trains often disrupted; houses in Provence face south without north windows.
Bora	Extremely Cold, Gusty & Dry	Blows from Dinaric Alps down towards the Adriatic Sea (Italy, Croatia)	Fierce northeastern catabatic katabatic wind reaching hurricane speeds in winter.
Blizzard	Freezing Cold with Blinding Snow	Canada, Northern USA, Siberia, Polar regions	Gale-force winds with sub-zero temperatures and zero visibility.
Buran / Purga	Bitterly Cold & Fierce Blizzard	Siberia & Central Asian steppes (Russia)	Buran brings icy blizzard conditions; Purga is tundra version.
Pampero	Cold & Violent Wind	Pampas of Argentina and Uruguay	Cold polar squall blowing from southwest across the pampas.

12. Structure of Atmosphere & Key Meteorological Phenomena

Atmospheric Layer	Height Range	Temperature Trend	Landmark Phenomenon / Exam Triggers
1. Troposphere	0 to 18 km (Equator: ~18 km; Poles: ~8 km)	Decreases with altitude (Lapse Rate: 6.5°C per km or 1°C per 165 m)	• Thickest at Equator due to strong vertical convective currents. • Contains ~75% to 80% of total atmospheric mass and almost ALL atmospheric water vapour and dust particles. • ALL weather phenomena (clouds, rainfall, lightning, thunder, cyclones, fog) occur EXCLUSIVELY in this layer. • Upper boundary is the Tropopause (temperature drops to -80°C over Equator and -45°C over Poles).
2. Stratosphere	From Tropopause up to 50 km	Increases with altitude (from -60°C up to 0°C at Stratopause)	• Contains the Ozone Layer (Ozonosphere) between 15 km and 35 km (absorbs harmful solar ultraviolet UV-B and UV-C radiation). • Completely free from clouds and turbulent weather; ideal condition for flying commercial supersonic jet aircraft. • High-altitude Subtropical Jet Streams blow near the tropopause. • Rare iridescent Nacreous Clouds (Mother-of-Pearl clouds) form in polar winter stratosphere.

Atmospheric Layer	Height Range	Temperature Trend	Landmark Phenomenon / Exam Triggers
3. Mesosphere	50 km to 80 km	Decreases rapidly with altitude (reaches -100°C at Mesopause)	• COLDEST LAYER OF THE ATMOSPHERE (-100°C to -130°C). • Meteors burn up in this layer upon entering from space due to friction with thin gases. • Rare Noctilucent Clouds (night-shining polar clouds composed of ice crystals on meteor dust) appear here.
4. Thermosphere / Ionosphere	80 km to 400+ km	Increases rapidly with altitude (exceeds 1,500°C due to intense solar radiation absorption)	• Contains electrically charged ionized gas particles (Ions). • Reflects terrestrial radio waves back to Earth, enabling wireless global radio communication: - D-layer (60–90 km): Reflects low-frequency radio waves, absorbs high-frequency; disappears at night. - E-layer / Kennelly-Heaviside Layer (90–130 km): Reflects medium-frequency radio waves. - F-layer / Appleton-Barnett Layer (140–400 km): Reflects high-frequency short radio waves. • Auroras (polar light displays caused by solar wind hitting charged atmospheric gases): - Aurora Borealis (Northern Lights, Arctic) - Aurora Australis (Southern Lights, Antarctic) • International Space Station (ISS) orbits in this layer (~400 km).
5. Exosphere	>400 km to outer space (~10,000 km)	Extremely high temperature, but imperceptible due to ultra-low density	• Outermost rarefied fringe of Earth's atmosphere merging into interplanetary space. • Dominated by light gases: Hydrogen and Helium. • Artificial communication and weather satellites orbit in higher exosphere.

13. Frontiers, Latitudinal/Longitudinal Framework & Territorial Limits of India

Locational / Border Indicator	Geographic Fact / Value	States / Locations Involved	Crucial Prelims Exam Triggers
Latitudinal Extent	8°4' N to 37°6' N	Mainland southern tip to northern tip	Total span = ~29°2'; Entirely in Northern & Eastern Hemispheres
Longitudinal Extent	68°7' E to 97°25' E	Westernmost point to easternmost point	Total span = ~29°18'; ~2 hours time lag between Arunachal & Gujarat
North–South Distance	3,214 km	Indira Col to Kanyakumari	Greater than East–West distance by 281 km
East–West Distance	2,933 km	Kibithu to Guhar Moti	
Total Land Frontier	15,106.7 km	16 States & 2 UTs share land borders	India shares land borders with 7 sovereign countries
Total Coastline	7,516.6 km	Mainland = 6,100 km; Island territories = 1,416.6 km	9 Coastal States + 4 Coastal UTs
Northernmost Point	Indira Col (37°6' N)	Siachen Glacier (Ladakh)	Near trijunction of India, Pakistan, and China
Southernmost Point (Overall)	Indira Point (Pygmalion / Parsons Point) (6°45' N)	Great Nicobar Island	Submerged partially during 2004 Tsunami; ~150 km from Sumatra (Indonesia)

Locational / Border Indicator	Geographic Fact / Value	States / Locations Involved	Crucial Prelims Exam Triggers
Southernmost Point (Mainland)	Cape Comorin (Kanyakumari) (8°4' N)	Tamil Nadu	Confluence of Arabian Sea, Bay of Bengal, and Indian Ocean
Westernmost Point	Guhar Moti / Sir Creek (68°7' E)	Kutch district, Gujarat	Sir Creek is a 96 km tidal estuary disputed between India & Pak
Easternmost Point	Kibithu (97°25' E)	Anjaw district, Arunachal Pradesh	Lohit River enters India near Kibithu
Tropic of Cancer (23°30' N)	Passes through 8 Indian States	Gujarat → Rajasthan → Madhya Pradesh → Chhattisgarh → Jharkhand → West Bengal → Tripura → Mizoram (West to East)	• Ranchi is the closest state capital to Tropic of Cancer • Mahi River cuts the Tropic of Cancer TWICE! • Does NOT pass through Odisha, Bihar, or UP
Standard Meridian of India (82°30' E)	IST = GMT + 5 hours 30 mins	Passes through 5 States: Uttar Pradesh (Mirzapur / Shankargarh / Naini, Prayagraj) → Madhya Pradesh → Chhattisgarh → Odisha → Andhra Pradesh	• Tropic of Cancer and 82°30' E meridian intersect in Koriya / Surajpur district (Chhattisgarh)

Land Frontiers by Length (Strict Descending Order)

1. **Bangladesh (4,096.7 km) — Longest border**; 5 States: West Bengal (longest border with BD), Assam, Meghalaya, Tripura (surrounded on 3 sides by BD; Zero Line), Mizoram.
2. **China (3,488 km) — 1 UT (Ladakh) + 4 States**: Himachal Pradesh, Uttarakhand, Sikkim, Arunachal Pradesh; McMahon Line (1914).
3. **Pakistan (3,323 km) — 2 UTs (J&K, Ladakh) + 3 States**: Gujarat, Rajasthan (longest border with Pak), Punjab; Radcliffe Line (17 August 1947).
4. **Nepal (1,751 km) — 5 States**: Uttarakhand, Uttar Pradesh (longest border with Nepal), Bihar, West Bengal, Sikkim; Kali River forms western border.
5. **Myanmar (1,643 km) — 4 States**: Arunachal Pradesh (longest), Nagaland, Manipur, Mizoram; Purvanchal hills form watershed.
6. **Bhutan (699 km) — 4 States**: Assam (longest), West Bengal, Sikkim, Arunachal Pradesh.
7. **Afghanistan (106 km) — Shortest border**; Ladakh (POK - Wakhan corridor); Durand Line (1896).

Coastal States by Coastline Length (Descending)

- **Gujarat (1,214.7 km) > Andhra Pradesh (973.7 km) > Tamil Nadu (906.9 km) > Maharashtra (652.6 km) > Kerala (569.7 km) > Odisha (476.4 km) > Karnataka (280 km) > Goa (101 km — shortest among states) > West Bengal (157.5 km).**
- **UTs: Andaman & Nicobar Islands (1,962 km — longest overall territory) > Lakshadweep (132 km) > Puducherry (47.6 km) > Daman & Diu (42.5 km).**

Maritime Channels & Straits Around India

- **8° Channel**: Separates **Minicoy (India)** from **Maldives**.
- **9° Channel**: Separates **Minicoy Island** from the **main Lakshadweep archipelago** (Kalpeni/Suheli).
- **10° Channel**: Separates **Little Andaman** from **Car Nicobar** (Andaman Islands from Nicobar Islands; ~150 km wide).
- **Duncan Passage**: Separates **South Andaman (Rutland Island)** from **Little Andaman**.
- **Sombrero Channel**: Separates **Car Nicobar** from **Central Nicobar (Nancowry)**.
- **Great / Grand Channel**: Separates **Great Nicobar (Indira Point)** from **Sumatra (Aceh, Indonesia)**.

- **Palk Strait & Adam's Bridge (Rama Setu):** Separates **Tamil Nadu (Pamban Island / Dhanushkodi)** from **Sri Lanka (Mannar Island / Talaimannar)**; connects Palk Bay with Gulf of Mannar.
- **Coco Channel:** Separates **North Andaman (Landfall Island)** from **Coco Islands (Myanmar)**.

14. Peninsular & Himalayan Hill Sequences (North to South & West to East)

A. Trans-Himalayan & Himalayan Ranges (North to South Order)

1. **Karakoram Range** (K2 / Godwin-Austen 8,611 m; Siachen, Baltoro, Biafo, Hispar glaciers)
2. **Ladakh Range** (Khardung La pass; Leh lies between Ladakh and Zaskar ranges)
3. **Zaskar Range** (Kamet peak, Nanda Devi; Shipki La, Lipulekh passes)
4. **Great Himalayas (Himadri)** (Everest, Kanchenjunga, Makalu, Dhaulagiri, Nanga Parbat, Namcha Barwa)
5. **Lesser / Middle Himalayas (Himachal)** (Pir Panjal in J&K/HP, Dhauladhar in HP, Mussoorie & Nag Tibba in Uttarakhand, Mahabharat Lekh in Nepal)
6. **Shiwalik Range (Outer Himalayas)** (Youngest foothill range; flat-bottomed structural valleys called **Duns** in west e.g., Dehradun, and **Duars** in east).

B. Western Ghats (Sahyadri) Hill Sequence (Strict North to South)

1. **Northern Sahyadri: Kalsubai** (1,646 m — highest peak of Maharashtra / Northern Sahyadri), Salher, Mahabaleshwar (source of Krishna river), Harishchandra Range.
2. **Central Sahyadri: Mullayanagiri** (1,930 m — highest peak of Karnataka), **Kudremukh** (1,892 m), Baba Budan Hills (first coffee planted in India).
3. **Nilgiri Hills ("Blue Mountains")**: **Doda Betta** (2,637 m); junction meeting point of Western and Eastern Ghats; Nilgiri Tahr and Toda pastoral tribe.
4. *[Palghat / Palakkad Gap separates Nilgiris from Anaimalai Hills]*
5. **Anaimalai Hills ("Elephant Hills")**: **Anamudi** (2,695 m — **HIGHEST PEAK OF PENINSULAR / SOUTH INDIA**; Idukki, Kerala).
6. **Palani Hills**: Eastward spur from Anaimalai into Tamil Nadu; famous hill station **Kodaikanal** located around artificial star-shaped lake.
7. **Cardamom Hills (Yela Mala)**: Southernmost hill range of mainland India (Kerala-Tamil Nadu border); Periyar National Park.
8. *[Shenkottai Pass separates Cardamom Hills from Agasthyamalai]*
9. **Agasthyamalai Hills**: Southernmost biosphere reserve of Western Ghats (Kerala-TN border).

C. Eastern Ghats & Peninsular Discontinuous Hills (Strict North to South)

1. **Garhjat Hills**: Northern Odisha (Mayurbhanj / Keonjhar iron-ore belt).
2. **Nallamala Hills**: Andhra Pradesh (between Krishna and Pennar rivers; Srisailem dam and tiger reserve).
3. **Erramala Hills**: Kurnool district, Andhra Pradesh.
4. **Velikonda Range**: Eastern Andhra Pradesh (Nellore/Kadapa border).
5. **Palkonda Range**: Southern Andhra Pradesh (Tirupati sacred hills / Tirumala located on Seshachalam-Palkonda range).
6. **Nagari Hills**: Chittoor district, Andhra Pradesh.
7. **Javadi Hills**: Northern Tamil Nadu (Vellore / Tiruvannamalai; famous for blue-grey granites and sandalwood).
8. **Shevaroy Hills**: Salem district, Tamil Nadu; famous hill resort **Yercaud**; rich in bauxite deposits.
9. **Panchaimalai Hills**: Central Tamil Nadu (Tiruchirappalli).
10. **Sirumalai Hills**: Dindigul district, Tamil Nadu.
11. **Varushanad Hills**: Southern Tamil Nadu (source of Vaigai River).

D. Central Highlands & Transverse Order (North to South)

- **Vindhya Range** (Sandstone scarplands: Kaimur, Bhandar, Bhander)
↓ [*Narmada River flows west in Rift Valley*]
- **Satpura Range** (Horst block mountain: **Rajpipla Hills** in west → **Mahadeo Hills / Dhupgarh 1,350 m** in middle → **Maikal Range / Amarkantak 1,065 m** in east)
↓ [*Tapi River flows west in Rift Valley south of Satpura*]
- **Ajanta Range & Satmala Hills** (Maharashtra; entirely within single state).

E. North-East Hills & Valleys

- **Arunachal Foothills (West to East):** **Dafila Hills** → **Miri Hills** → **Abor Hills** → **Mishmi Hills** (Daphla-Miri-Abor-Mishmi named after local tribes).
- **Purvanchal Eastern Border Ranges (North to South):** **Patkai Bum** (Arunachal-Myanmar border) → **Naga Hills** (Saramati Peak 3,826 m) → **Manipur Hills** → **Mizo Hills / Lushai Hills** (Phawngpui / Blue Mountain 2,157 m).
- **Meghalaya Plateau (West to East):** **Garo Hills** (Nokrek Biosphere Reserve 1,412 m) → **Khasi Hills** (Shillong Peak 1,961 m; Mawsynram and Cherrapunji) → **Jaintia Hills** → **Mikir Hills** → **Barail Range** (connects Meghalaya plateau to Naga hills).

15. Major Lakes & Inland Wetlands of India Master Matrix

Lake Name	State / UT	Type / Origin	High-Yield Prelims Exam Triggers
Chilika Lake	Odisha (Puri, Khordha, Ganjam)	Brackish coastal lagoon	• Largest brackish water lagoon in India (fed by Daya River) • Designated 1st Indian Ramsar site (1981) along with Keoladeo NP • Habitat of the endangered Irrawaddy Dolphin • Nalabana Island declared Bird Sanctuary
Vembanad Lake	Kerala (Kottayam, Alappuzha, Ernakulam)	Brackish lagoon / backwater	• LONGEST LAKE IN INDIA (96.5 km); largest in Kerala • Famous Nehru Trophy Boat Race (Vallam Kali) held on Punnamada Lake • Willingdon Island (Kochi Port) man-made in it • Kumarakom Bird Sanctuary on its eastern shore
Sasthamkotta Lake	Kerala (Kollam)	Freshwater lake	• Known as the "Queen of Lakes"; largest freshwater lake in Kerala • Contains unique larva <i>Cavaborus</i> which cleanses and purifies lake water
Ashtamudi Lake	Kerala (Kollam)	Brackish estuarine lagoon	• Eight-armed (<i>Ashta-mudi</i>) palm-shaped lake; gateway to Kerala backwaters; Ramsar site
Kolleru Lake	Andhra Pradesh (Eluru/Krishna)	Freshwater lake	• Located between Krishna and Godavari deltas; fed by Budameru and Tammileru streams • Atapaka Bird Sanctuary (pelican paradise); Ramsar site
Pulicat Lake	Andhra Pradesh (96%) & Tamil Nadu (4%)	Brackish water coastal lagoon	• Second largest brackish lagoon in India • Separated from Bay of Bengal by the barrier island of Sriharikota (ISRO Satish Dhawan Space Centre) • Famous annual Flamingo Festival

Lake Name	State / UT	Type / Origin	High-Yield Prelims Exam Triggers
Loktak Lake	Manipur (Moirang, Bishnupur)	Freshwater lake	• Largest freshwater lake in North-East India • Famous for heterogeneous mass of vegetation and soil called Phumdis • Hosts Keibul Lamjao National Park — WORLD'S ONLY FLOATING NATIONAL PARK • Exclusive natural habitat of endangered Sangai (Brow-antlered / Dancing deer)
Lonar Lake	Maharashtra (Buldhana district)	Meteorite Impact Crater Lake	• Created by hyper-velocity meteorite impact during Pleistocene epoch in Deccan basalt • Highly saline and alkaline (pH ~10.5); turns pink due to haloarchaea microbes • Notified National Geo-heritage Monument; Ramsar site (2020)
Wular Lake	Jammu & Kashmir (Bandipora)	Tectonic Freshwater Lake	• LARGEST FRESHWATER LAKE IN INDIA • Formed by tectonic activity; fed and drained by Jhelum River • Tulbul Navigation Project (disputed with Pakistan) located at its mouth
Dal Lake	Jammu & Kashmir (Srinagar)	Freshwater flood-plain lake	• "Jewel in the crown of Kashmir"; famous for Shikaras and floating gardens (<i>Radh</i>)
Sambhar Lake	Rajasthan (Jaipur, Nagaur, Ajmer)	Inland Saline Playa Lake	• LARGEST INLAND SALINE LAKE IN INDIA; represents an endorheic depression • Produces ~9% of India's total salt production; Ramsar site
Didwana, Pachpadra, Lunkaransar	Rajasthan	Inland saline playas	• Pachpadra produces high-grade sodium chloride (98% purity) used locally
Pushkar & Nakki Lake	Rajasthan	Pushkar: Sacred holy lake; Nakki: Mount Abu	• Nakki is highest lake in Rajasthan; legend says dug by gods' nails (<i>nakh</i>)
Pangong Tso	Ladakh (4,225 m elevation)	Endorheic saltwater lake	• World's highest saltwater lake; ~1/3 lies in India and ~2/3 in Tibet (China)
Tso Moriri	Ladakh (Rupshu valley)	High-altitude brackish lake	• Breeding ground for endangered Black-necked Crane and Bar-headed Goose; Ramsar site
Roopkund Lake	Uttarakhand (Chamoli, 5,029 m)	Glacial tarn lake	• Known as the "Mystery Lake" / "Skeleton Lake"; hundreds of human skeletons found
Bhimtal, Nainital, Naukuchiatal, Sattal	Uttarakhand (Kumaon)	Tectonic / fault freshwater lakes	• Bhimtal is the largest lake in Kumaon with a central island; Naukuchiatal has 9 corners
Suraj Tal & Chandra Tal	Himachal Pradesh (Lahaul & Spiti)	High-altitude glacial lakes	• Chandra Tal (source of Chandra river) and Suraj Tal (source of Bhaga river); join to form Chenab
Renuka Lake	Himachal Pradesh (Sirmaur)	Freshwater wetland lake	• Largest natural lake in Himachal Pradesh; shaped like a reclining woman; Ramsar site
Deepor Beel	Assam (Kamrup / Guwahati)	Freshwater oxbow lake	• Former channel of Brahmaputra; vital staging site for migratory waterbirds; Ramsar site
Rudrasagar Lake	Tripura (Melaghar)	Freshwater lowland lake	• Historic Neermahal Palace situated in the center of the lake; Ramsar site

16. Soils of India — ICAR Classification, Characteristics & Soil Health Master Card

A. ICAR 8 Soil Groups (% Share & Leading Characteristics)

- Alluvial Soil (~43.4% of total area — Largest):** * **Genesis:** Deposited by Indus, Ganga, Brahmaputra river systems and coastal waves. * **Nutrient Profile:** **Rich in Potash and Lime;** severely **deficient in Nitrogen, Phosphorus, and Humus.** * **Classification:**
 - **Khadar:** New alluvium deposited by annual floods; fine loamy texture, highly fertile; light color.
 - **Bhangar:** Older alluvium situated above flood levels; clayey texture, less fertile; contains calcareous nodules (**Kankar**).
 - **Bhabar:** Narrow porous pebble-bed at foot of Shiwaliks where rivers disappear underground.
 - **Terai:** Marshy, waterlogged damp tract south of Bhabar with tall elephant grass; re-emerging streams.
 - **Crops:** Wheat, Rice, Sugarcane, Jute, Oilseeds, Maize, Pulses.
 - Red & Yellow Soil (~18.6% — 2nd Largest Group):** * **Genesis:** Weathering of ancient crystalline igneous (granites, gneisses) and metamorphic rocks in low-rainfall areas. * **Color Rationale:** Red color due to wide diffusion of **Iron (Ferric oxides)**; appears yellow when hydrated. * **Nutrient Profile:** **Deficient in Nitrogen, Phosphorus, Lime, and Humus.** * **Regions & Crops:** Tamil Nadu, Southern Karnataka, Telangana, Odisha, Chotanagpur, MP; ideal for millets, pulses, groundnut, tobacco.
 - Black Soil / Regur / Tropical Chernozem (~15.2%):** * **Genesis:** Weathering of basaltic lava rocks of the **Deccan Trap (Cretaceous lava flows).** * **Distinctive Features:** Highly argillaceous (clayey >50%); swells and becomes sticky when wet; shrinks and develops wide deep cracks when dry ("**self-ploughing**" capacity); exceptional moisture-retention capability. * **Color:** Black color attributed to **Titaniferous Magnetite** and iron compounds. * **Nutrient Profile:** **Rich in Lime, Iron, Magnesia, Alumina, and Potash;** severely **deficient in Phosphorus, Nitrogen, and Organic Matter.** * **Regions & Crops:** Maharashtra, Gujarat, Malwa (MP), Northern Karnataka, Telangana; **Ideal for Cotton (Black Cotton Soil), Soyabean, Sugarcane, Citrus fruits.**
 - Laterite Soil (~3.7%):** * **Genesis:** Formed under conditions of **high temperature and alternating heavy wet and dry seasons with intense tropical rainfall.** * **Process:** Intense **leaching (desilication)** washes away silica and lime, leaving behind residual oxides of **Iron and Aluminium (Bauxite).** * **Properties:** Highly acidic (pH 4.5–5.5); hardens like brick upon exposure to atmosphere (Latin *Later* = brick). * **Nutrient Profile:** Poor in Nitrogen, Phosphorus, Potash, Lime, and Humus. * **Regions & Crops:** Summits of Western Ghats (Kerala, Karnataka, Maharashtra), Eastern Ghats, Meghalaya plateau, Rajmahal Hills; **Premier soil for Plantation Crops: Tea, Coffee, Rubber, Cashewnut, Coconut.**
 - Arid / Desert Soil (~4.4%):** * **Properties:** Sandy to gravelly texture, saline in nature; low moisture and very low humus; phosphate normal but nitrogen deficient; with canal irrigation (Indira Gandhi Canal) produces bumper crops of wheat, barley, cotton.
 - Saline & Alkaline Soils (Reh, Kallar, Usar, Rakar, Thur, Chopan) (~1.3%):** * **Genesis:** Excessive canal irrigation in poorly drained arid/semi-arid regions causing high capillary action that pulls soluble sodium, calcium, and magnesium salts to topsoil. * **Reclamation:** Application of **Gypsum ($\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$)** or Iron Pyrites to neutralize alkalinity; growing salt-tolerant green manure crops like **Dhaincha (*Sesbania*).**
 - Peaty and Marshy Soil (Kari) (~2%):** * **Genesis:** Formed in humid regions with heavy rainfall and dense vegetation; high organic matter accumulation (40–50%). * **Properties:** Dark, heavy, highly acidic; found in **Kottayam and Alappuzha districts of Kerala (Kari lands)**, Sundarbans delta.
 - Forest and Mountain Soil (~8.7%):** * Immature loamy soil rich in humus from fallen leaf litter; acidic in higher alpine zones; terrace cultivation of apples, peaches, spices, tea.
-

17. Natural Vegetation & Forests of India (Champion & Seth / ISFR Key Indicators & Species)

A. Core Forest Statistics (India State of Forest Report - ISFR)

- **Total Forest & Tree Cover:** ~24.62% of geographical area (Forest Cover: ~21.71%; Tree Cover: ~2.91%).
- **National Forest Policy (1988) Target:** 33% of total geographical area (60% in mountainous/hilly regions).
- **State with Largest Forest Cover by Area:** **Madhya Pradesh** (1st) > Arunachal Pradesh (2nd) > Chhattisgarh (3rd) > Odisha (4th) > Maharashtra (5th).
- **State with Highest Forest Cover as % of Total Area:** **Mizoram** (84.5%) > Arunachal Pradesh (79.3%) > Meghalaya (76%) > Manipur (74.3%).
- **UT with Highest Forest Cover %:** **Lakshadweep** (90.3%) and **Andaman & Nicobar Islands** (81.7%).

B. Major Forest Types & Indicator Species (Match the Following Goldmine)

Forest Type	Annual Rainfall & Climate	Typical Regions	Dominant Commercial Tree Species (Prelims Match)
Tropical Wet Evergreen	>250 cm; Temp: 25–27°C; High humidity (>77%)	Western slopes of Western Ghats, NE Hills (Assam, Meghalaya), Andaman & Nicobar	• Rosewood, Mahogany, Ebony, Aini, Cinchona (Quinine), Rubber, Wild Mango, Bamboo • Dense multi-layered canopy; trees do not shed leaves simultaneously.
Tropical Semi-Evergreen	200–250 cm	Bordering wet evergreen belts in Western Ghats, Assam, Odisha	• White Cedar, Hollock, Kadam, Kail, Champa
Tropical Moist Deciduous	100–200 cm; distinct dry season	Eastern slopes of Western Ghats, Chotanagpur, foothills of Himalayas (Bhabar-Terai), Odisha, MP, UP	• Teak (Sagan), Sal (Sakhu), Shisham, Sandalwood (Chandan), Mahua, Mulberry, Amla, Kusum, Semul, Bamboo • Economically most valuable timber forests of India; shed leaves for 6–8 weeks in spring.
Tropical Dry Deciduous	70–100 cm; transitional between moist deciduous & thorn	Vast plains of UP, Bihar, MP, eastern Rajasthan, Deccan plateau	• Tendu (Bidi wrapper), Palas / Dhak (<i>Flame of the Forest</i>), Amaltas, Bel, Khair, Axlewood (Dhaora) • MOST WIDESPREAD FOREST COVER IN INDIA (~41.8% of total forest area).
Tropical Thorn & Scrub	<70 cm / <50 cm	Rajasthan, SW Punjab, Haryana, Gujarat, rain-shadow leeward parts of Deccan	• Acacia (Babool), Kikar, Ber, Wild Date Palm, Khair, Cacti, Euphorbia • Deep roots, xerophytic thorns, fleshy stems to conserve water.
Montane Subtropical / Temperate	Altitude: 1,000 to 3,000 m	Himalayas (J&K, HP, Uttarakhand, Sikkim) & Nilgiris/Anamalai (<i>Shola</i> forests)	• 1000–2000 m: Broad-leaved Oak and Chestnut • 1500–3000 m: Coniferous Chir Pine (resin & turpentine), Deodar (<i>Cedrus deodara</i> — durable structural timber, western Himalayas), Blue Pine, Spruce, Silver Fir • <i>Shola</i>: Stunted evergreen temperate grasslands/forests in Nilgiris and Anamalai.
Alpine & Sub-Alpine	Altitude: >3,000 to 4,000 m	High Himalayas above timber line up to snow line	• Silver Fir, Juniper, Birch (Bhojpatra — ancient parchment bark), Rhododendron • Alpine pastures called Bugyals in Uttarakhand and Margs in Kashmir (Gulmarg, Sonamarg).
Littoral / Mangrove / Tidal	Coastal saline estuaries, deltas, tidal creeks	Sundarbans (WB), Mahanadi delta (Bhitarkanika, Odisha), Godavari-Krishna deltas, Pichavaram (TN)	• Sundari (<i>Heritiera fomes</i>), Goran, Kankra, Rhizophora, Sonneratia • Halophytic adaptations: Pneumatophores (breathing roots), stilt/prop roots, vivipary germination.

18. Shifting Cultivation (Jhum) Regional & Global Names Master Match-Sheet

A. Indian Regional Names for Shifting Cultivation

- **Jhum / Jhoom: North-Eastern India** (Assam, Meghalaya, Nagaland, Mizoram, Manipur, Arunachal Pradesh)
- **Podu / Dabi / Koman / Bhringa: Odisha** and Northern **Andhra Pradesh**
- **Bewar / Dahiya: Madhya Pradesh** (practiced by Baiga tribe in Bundelkhand/Baghelkhand)
- **Penda / Dipa: Bastar district (Chhattisgarh)** (practiced by Maria Gonds)
- **Valre / Waltre: South-Eastern Rajasthan** (Aravalli tribal belt)
- **Khil: Himalayan belt** (Himachal Pradesh, Uttarakhand)
- **Kuruwa / Khallu: Jharkhand** (Santhal Parganas)
- **Kumari: Western Ghats** (hilly tracts of Kerala and Karnataka)
- **Zabo: Nagaland** (indigenous forest-clearing + terraced pond water management)

B. International Names for Shifting Cultivation (Direct Exam Matches)

- **Milpa: Mexico and Central America**
- **Ladang: Indonesia and Malaysia**
- **Roca: Brazil** (Amazon basin)
- **Conuco: Venezuela** (Orinoco basin)
- **Taungya: Myanmar (Burma)**
- **Chena: Sri Lanka**
- **Ray: Vietnam and Laos**
- **Tamrai: Thailand**
- **Caingin: Philippines**
- **Masole: Congo Basin (Central Africa)**
- **Chitemene: Zambia, Zimbabwe and Malawi**

19. Agricultural Revolutions, Commodity Boards & Premier National Research Institutes

A. Agricultural Revolutions & Famous Pioneers

- **Green Revolution: Foodgrains (Wheat & Rice)** — Global Father: **Dr. Norman Borlaug** (Nobel Peace Prize 1970; semi-dwarf wheat at CIMMYT Mexico); Indian Father: **Dr. M.S. Swaminathan** (Bharat Ratna); Pantnagar University (1st agri university in India, 1960).
- **White Revolution (Operation Flood): Milk & Dairy** — Father: **Dr. Verghese Kurien**; NDDB Anand, Gujarat (AMUL); launched in 1970.
- **Yellow Revolution: Oilseeds (Mustard & Sunflower)** — Pioneer: **Sam Pitroda** (Technology Mission on Oilseeds 1986).
- **Blue Revolution: Fish & Aquaculture** — Pioneers: **Dr. Hiralal Chaudhari** and **Dr. Arun Krishnan**.
- **Golden Revolution: Horticulture, Honey & Fruits** — Pioneer: **Nirpakh Tutej**.
- **Golden Fibre Revolution: Jute Production.**
- **Silver Revolution: Eggs & Poultry Production** — Pioneer: **Smt. Indira Gandhi**.
- **Silver Fibre Revolution: Cotton Production.**
- **Pink Revolution: Onion, Prawn / Shrimp & Pharmaceuticals** — Pioneer: **Durgesh Patel**.
- **Red Revolution: Meat and Tomato Production** — Pioneer: **Vishal Tewari**.
- **Grey Revolution: Chemical Fertilizers.**
- **Round Revolution: Potato Production.**
- **Brown Revolution: Leather, Cocoa & Non-Conventional Energy.**

- **Evergreen / Rainbow Revolution: Holistic sustainable agriculture** (coined by M.S. Swaminathan).

B. Commodity Boards Headquarters in India (Match Goldmine)

- **Tea Board of India: Kolkata** (West Bengal)
- **Coffee Board of India: Bengaluru** (Karnataka)
- **Rubber Board of India: Kottayam** (Kerala)
- **Spices Board of India: Kochi** (Kerala)
- **Tobacco Board of India: Guntur** (Andhra Pradesh)
- **National Jute Board: Kolkata** (West Bengal)
- **Central Silk Board: Bengaluru** (Karnataka)
- **Coir Board: Kochi** (Kerala)
- **National Fisheries Development Board (NFDB): Hyderabad** (Telangana)

C. Premier National Research Institutes (Geography / Agriculture / Forestry)

- **Uttarakhand (UKPCS Top Triggers!):**
- **Forest Research Institute (FRI): Dehradun** (est. 1906, Greco-Roman architecture)
- **Indian Institute of Remote Sensing (IIRS): Dehradun**
- **Wildlife Institute of India (WII): Dehradun**
- **Wadia Institute of Himalayan Geology (WIHG): Dehradun**
- **Indira Gandhi National Forest Academy (IGNFA): Dehradun**
- **Govind Ballabh Pant Institute of Himalayan Environment & Development: Kosi-Katarmal, Almora**
- **Uttar Pradesh (UPPCS Top Triggers!):**
- **Indian Institute of Sugarcane Research (IISR): Lucknow**
- **National Sugar Institute (NSI): Kanpur**
- **Indian Institute of Vegetable Research (IIVR): Varanasi**
- **Indian Grassland and Fodder Research Institute (IGFRI): Jhansi**
- **Central Institute for Research on Goats (CIRG): Makhdoom, Mathura**
- **Central Institute for Subtropical Horticulture (CISH): Rahmankhera, Lucknow**
- **Indian Institute of Pulses Research (IIPR): Kanpur**
- **National Botanical Research Institute (NBRI): Lucknow**
- **Central Drug Research Institute (CDRI): Lucknow**
- **Central Institute of Medicinal and Aromatic Plants (CIMAP): Lucknow**
- **All-India Premier Geographic & Agricultural Stations:**
- **Central Potato Research Institute (CPRI): Shimla** (Himachal Pradesh)
- **National Rice Research Institute (NRRI / CRRI): Cuttack** (Odisha)
- **Indian Agricultural Research Institute (IARI / Pusa): New Delhi**
- **Central Arid Zone Research Institute (CAZRI): Jodhpur** (Rajasthan)
- **National Dairy Research Institute (NDRI): Karnal** (Haryana)
- **Central Soil Salinity Research Institute (CSSRI): Karnal** (Haryana)
- **National Institute of Oceanography (NIO): Dona Paula, Panaji** (Goa)
- **Indian National Centre for Ocean Information Services (INCOIS): Hyderabad** (Telangana)
- **National Geophysical Research Institute (NGRI): Hyderabad** (Telangana)
- **Central Plantation Crops Research Institute (CPCRI): Kasaragod** (Kerala)
- **Central Tuber Crops Research Institute (CTCRI): Thiruvananthapuram** (Kerala)
- **Central Tobacco Research Institute (CTRI): Rajahmundry** (Andhra Pradesh)
- **Central Sheep and Wool Research Institute (CSWRI): Avikanagar** (Rajasthan)

- **Central Institute for Research on Buffaloes (CIRB): Hisar** (Haryana)

20. World Strategic Straits, Canals & Maritime Chokepoints Master Matrix

Strait / Canal	Water Bodies Connected	Landmasses Separated	High-Yield Prelims Exam Triggers
Strait of Malacca	Andaman Sea (Indian Ocean) & South China Sea (Pacific Ocean)	Malay Peninsula & Sumatra (Indonesia)	• World's busiest shipping chokepoint; shortest sea route between Persian Gulf/India and China/Japan
Strait of Hormuz	Persian Gulf & Gulf of Oman (Arabian Sea)	Iran & Oman / UAE (Musandam Peninsula)	• World's most critical oil transit chokepoint (~20% of global petroleum passes through it)
Bab-el-Mandeb	Red Sea & Gulf of Aden (Indian Ocean)	Yemen (Arabia/Asia) & Djibouti / Eritrea (Horn of Africa)	• Known as the "Gate of Tears"; connects Indian Ocean shipping to the Suez Canal
Suez Canal	Mediterranean Sea & Red Sea	Sinai Peninsula & Mainland Egypt (Africa)	• Opened in 1869; designed by French engineer Ferdinand de Lesseps • Eliminates passage around Cape of Good Hope; Port Said (north) and Port Tewfik/Suez (south)
Strait of Gibraltar	Atlantic Ocean & Mediterranean Sea	Spain (Europe) & Morocco (Africa)	• Known as the "Key to the Mediterranean" / ancient "Pillars of Hercules"; Rock of Gibraltar (UK) on north
Bosporus (Bosphorus) Strait	Black Sea & Sea of Marmara	European Istanbul & Asian Istanbul (Turkey)	• World's narrowest strait used for international navigation; separates Europe from Asia
Dardanelles Strait	Sea of Marmara & Aegean Sea (Mediterranean)	Gallipoli Peninsula (Europe) & Asian Turkey	• Historic naval battle of Gallipoli (WWI); Turkish Straits system regulated by Montreux Convention
Bering Strait	Chukchi Sea (Arctic Ocean) & Bering Sea (Pacific Ocean)	Russia (Siberia/Asia) & Alaska (USA/North America)	• International Date Line (IDL) passes directly through this strait
Palk Strait	Bay of Bengal & Palk Bay / Gulf of Mannar	Tamil Nadu (India) & Jaffna Peninsula (Sri Lanka)	• Named after Robert Palk (Governor of Madras)
Strait of Dover	English Channel & North Sea	England (Great Britain) & France (Mainland Europe)	• Narrowest part of English Channel (34 km); Channel Tunnel (<i>Chunnel</i>) runs beneath it
Panama Canal	Atlantic Ocean (Caribbean Sea) & Pacific Ocean	North America & South America (Isthmus of Panama)	• Opened in 1914; 82 km long; uses 3 sets of freshwater locks (Gatun, Pedro Miguel, Miraflores)
Kiel Canal	North Sea & Baltic Sea	Jutland Peninsula (Germany)	• Bypasses the Danish straits; world's busiest artificial waterway by vessel count
Sunda Strait	Java Sea & Indian Ocean	Java and Sumatra (Indonesia)	• Krakatoa volcanic island sits directly inside this strait
Lombok Strait	Java Sea & Indian Ocean	Bali and Lombok (Indonesia)	• Deeper alternative to Malacca for supertankers; marks the Wallace Line biogeographical boundary
Torres Strait	Arafura Sea & Coral Sea (Pacific Ocean)	Cape York (Australia) & Papua New Guinea	• Shallow tropical strait filled with coral reefs
Bass Strait	Tasman Sea & Southern Ocean / Indian Ocean	Victoria (Mainland Australia) & Tasmania	• King Island and Flinders Island lie in it
Cook Strait	Tasman Sea & South Pacific Ocean	North Island & South Island (New Zealand)	• Connects capital Wellington with South Island

Strait / Canal	Water Bodies Connected	Landmasses Separated	High-Yield Prelims Exam Triggers
Strait of Magellan	Atlantic Ocean & Pacific Ocean	Mainland South America (Chile) & Tierra del Fuego	• Natural passage discovered by Ferdinand Magellan (1520) avoiding turbulent Drake Passage
Drake Passage	South Atlantic Ocean & South Pacific Ocean	Cape Horn (South America) & South Shetland Islands (Antarctica)	• Roughest open water body on Earth; connects Atlantic to Southern Ocean
Florida Strait	Gulf of Mexico & Atlantic Ocean	Florida Keys (USA) & Cuba	• Conveys the powerful warm Gulf Stream northward

21. Major World Rivers, Drainage Features & Riverbank Cities Master Table

A. World Landmark Rivers & Unique Geographic Traits

River Name	Length & Basin	Source & Mouth	Landmark Exam Triggers & Geographic Superlatives
Nile River	6,650 km (Africa)	White Nile (Lake Victoria) + Blue Nile (Lake Tana) → Mediterranean Sea	• LONGEST RIVER IN THE WORLD • Confluence of White & Blue Nile at Khartoum (Sudan) • Forms classic Arcuate Delta in Egypt • Aswan High Dam creates Lake Nasser • Capitals on banks: Cairo (Egypt), Khartoum (Sudan)
Amazon River	6,400 km (South America)	Peruvian Andes (Nevado Mismi) → Atlantic Ocean	• LARGEST RIVER IN THE WORLD BY WATER VOLUME & BASIN AREA (~20% of world river discharge) • Flows across dense tropical rainforests (Selvas — “Lungs of the Earth”) • Has NO bridges across its entire main stem • Spectacular tidal bore known as Pororoca
Yangtze River (Chang Jiang)	6,300 km (Asia / China)	Tibetan Plateau (Tangjula Mtns) → East China Sea (Shanghai)	• LONGEST RIVER IN ASIA (3rd longest in world); flows entirely in China • Hosts Three Gorges Dam (world’s largest hydroelectric plant by capacity) • Cities: Chongqing, Wuhan, Nanjing, Shanghai
Mississippi–Missouri	6,275 km (North America)	Lake Itasca (Minnesota) → Gulf of Mexico	• Longest river system in North America; forms textbook Bird’s Foot Delta • Cities: Minneapolis, St. Louis, Memphis, New Orleans
Danube River	2,850 km (Europe)	Black Forest (Germany) → Black Sea	• Flows through 10 countries (more than any other river in the world) • Passes through 4 national capitals: Vienna (Austria), Bratislava (Slovakia), Budapest (Hungary), Belgrade (Serbia)
Rhine River	1,230 km (Europe)	Swiss Alps (Lake Constance) → North Sea	• Flows through a rift valley between Vosges Mountains (France) and Black Forest (Germany) • Busiest commercial inland waterway in Europe; mouth at Rotterdam Port (Netherlands) • Basel, Strasbourg, Bonn, Cologne on its banks
Congo River (Zaire)	4,700 km (Central Africa)	Katanga Plateau → Atlantic Ocean	• DEEPEST RIVER IN THE WORLD (>220 m); 2nd largest discharge after Amazon • CROSSES THE EQUATOR TWICE! • Boyoma (Stanley) Falls and Livingstone Falls; Kinshasa and Brazzaville face each other across it
Limpopo River	1,750 km (Southern Africa)	Witwatersrand (South Africa) → Indian Ocean (Mozambique)	• CROSSES THE TROPIC OF CAPRICORN TWICE!

River Name	Length & Basin	Source & Mouth	Landmark Exam Triggers & Geographic Superlatives
Mahi River	583 km (India)	Vindhya Range (MP) → Gulf of Khambhat	• CROSSES THE TROPIC OF CANCER TWICE!
Volga River	3,530 km (Russia / Europe)	Valdai Hills → Caspian Sea	• LONGEST RIVER IN EUROPE ; internal drainage into landlocked Caspian Sea; Volgograd (Stalingrad) on its bank
Amur River (Heilong Jiang)	2,824 km (Asia)	Confluence of Shilka & Argun → Sea of Okhotsk	• Forms the natural international border between Russia and China for >1,600 km
Mekong River	4,350 km (Southeast Asia)	Tibetan Plateau → South China Sea	• Known as the “Danube of the East”; flows through 6 countries: China, Myanmar, Laos, Thailand, Cambodia, Vietnam; Vientiane and Phnom Penh on banks
St. Lawrence River	1,197 km (North America)	Lake Ontario → Gulf of St. Lawrence (Atlantic)	• Natural outlet of the Great Lakes; forms part of USA–Canada international border; Montreal and Quebec City on its banks

B. Famous World Cities on Rivers (Direct Match List)

- **London: Thames** (UK)
- **Paris: Seine** (France)
- **Rome: Tiber** (Italy)
- **Berlin: Spree** (Germany)
- **Vienna, Budapest, Belgrade: Danube** (Austria, Hungary, Serbia)
- **Warsaw: Vistula** (Poland)
- **Kyiv (Kiev): Dnieper** (Ukraine)
- **Cairo, Khartoum: Nile** (Egypt, Sudan)
- **Baghdad: Tigris** (Iraq)
- **Washington, D.C.: Potomac** (USA)
- **New York: Hudson** (USA)
- **New Orleans: Mississippi** (USA)
- **Buenos Aires: Río de la Plata** (Argentina)
- **Bangkok: Chao Phraya** (Thailand)
- **Yangon: Yangon River / Irrawaddy delta** (Myanmar)
- **Lahore: Ravi** (Pakistan)
- **Karachi: Indus** (Pakistan)

22. World Lakes, Inland Seas & Salinity Master Matrix

Lake Name	Location / Countries Bordering	Key Geographic Superlative	High-Yield Prelims Exam Triggers
Caspian Sea	Bordered by 5 Countries (mnemonic: TARIK) : Turkmenistan, Azerbaijan, Russia, Iran, Kazakhstan	LARGEST LAKE / INLAND WATER BODY IN THE WORLD	• Surface area ~371,000 km²; slightly saline (~1.2%); largest endorheic basin • Baku (capital of Azerbaijan) situated on its western shore • Fed by Volga, Ural, and Kura rivers; has NO natural outlet
Lake Superior	USA & Canada border (Minnesota, Wisconsin, Michigan, Ontario)	LARGEST FRESHWATER LAKE IN THE WORLD BY SURFACE AREA	• Surface area ~82,100 km²; part of North American Great Lakes • Connected to Lake Huron by St. Mary's River and Soo Locks

Lake Name	Location / Countries Bordering	Key Geographic Superlative	High-Yield Prelims Exam Triggers
Lake Baikal	Russia (Siberia — Irkutsk & Buryatia)	DEEPEST LAKE IN THE WORLD (1,642 m)	• Oldest freshwater lake in the world (25–30 million years) • Largest lake by volume of unfrozen freshwater (~20% of world's surface freshwater) • Drained by only ONE river: the Angara River (tributary of Yenisey); UNESCO WHS
Lake Tanganyika	Shared by Tanzania, DR Congo, Burundi, Zambia	LONGEST FRESHWATER LAKE IN THE WORLD (673 km)	• 2nd deepest lake in the world (1,470 m); situated in East African Great Rift Valley
Lake Victoria	Shared by Tanzania (49%), Uganda (45%), Kenya (6%)	LARGEST TROPICAL LAKE IN THE WORLD	• 2nd largest freshwater lake by surface area (68,800 km²) • Equator passes directly through it • Primary reservoir and source of the White Nile River
Lake Titicaca	Andes Mountains (Border of Peru and Bolivia)	HIGHEST COMMERCIALY NAVIGABLE LAKE IN THE WORLD	• Elevation: 3,812 m above sea level; high alpine lake • Indigenous Uros people live on artificial floating islands made of Totora reeds
Dead Sea	Bordered by Jordan, Israel, and West Bank	LOWEST WATER BODY ON EARTH'S LAND SURFACE	• Surface elevation: ~430 m below sea level • Hypersaline terminal lake (~34% salinity — 10x oceans); high density allows humans to float effortlessly • Fed by Jordan River
Lake Van	Eastern Turkey (Anatolia)	HIGHEST SALINITY AMONG MAJOR WATER BODIES (~330‰ / 33%)	• Saline soda lake with no outlet; water does not freeze even at high elevation
Lake Assal	Djibouti (Afar Depression, Horn of Africa)	Hypersaline lake (~348‰ salinity)	• Lowest point in Africa (-155 m below sea level); 2nd highest salinity on Earth
Lake Maracaibo	Venezuela (South America)	Largest lake in South America	• Connected to Gulf of Venezuela by Tablazo Strait; rich petroleum basin; famous for Catatumbo Lightning
Aral Sea	Border of Kazakhstan and Uzbekistan	Catastrophic environmental shrinkage	• Formerly 4th largest lake; shrunken to <10% original size due to Soviet diversion of Amu Darya and Syr Darya rivers for cotton fields
Lake Balkhash	Kazakhstan (Central Asia)	Unique half-fresh, half-saline lake	• Western part is fresh water (fed by Ili River), while eastern part is saline water
Lake Chad	Shared by Chad, Cameroon, Niger, Nigeria	Endorheic Sahelian lake	• Crucial wetland in the Sahel; shrunken by >90% since 1960s

The Great Lakes of North America (Strict West-to-East Chronology)

- **Superior → Michigan → Huron → Erie → Ontario** (Mnemonic: **SMHEO** or **HOMES**).
- **Lake Michigan** is the **ONLY ONE** of the **5 Great Lakes** situated **entirely within the United States** (the other four are shared between USA and Canada).
- **Niagara Falls** is located on the Niagara River between **Lake Erie and Lake Ontario**!
- **Welland Canal** bypasses Niagara Falls to connect Lake Erie and Lake Ontario for commercial shipping.

23. World Major Grasslands & Deserts Master Cheat-Sheet

A. Grasslands of the World (Tropical Savannas vs Temperate Steppes)

Grassland Name	Type (Temperate / Tropical)	Geographic Region & Countries	High-Yield Prelims Exam Triggers
Prairies	Temperate (Mid-latitude)	North America (USA & Canada)	- Known as the “Granary of the World” (massive commercial wheat belt) - Warm katabatic wind Chinook (“Snow Eater”) melts winter snow on eastern slopes of Rockies.
Steppes	Temperate (Mid-latitude)	Eurasia (Ukraine, Southern Russia to Central Asia & Mongolia)	Extensive temperate grassland plain; dark fertile Chernozem soils; premier wheat & grain farming.
Pampas	Temperate (Mid-latitude)	South America (Argentina & Uruguay)	Fertile alluvial soils; rich in nutritious Alfalfa grass; extensive cattle ranching (Estancias) and cowboys (Gauchos); wheat cultivation.
Veld / Veldt	Temperate (Mid-latitude)	South Africa (High Veld & Bush Veld plateau)	Renowned for rearing Merino sheep for high-grade wool and maize cultivation (Maize Triangle).
Downs	Temperate (Mid-latitude)	Australia (Murray–Darling River Basin)	Rolling grassland plains; premier sheep-farming stations for wool and wheat growing.
Canterbury Plains	Temperate (Mid-latitude)	New Zealand (South Island)	Rich pastoral plains for sheep rearing and dairy farming.
Puszta	Temperate (Mid-latitude)	Hungary (Danube River Basin)	Traditional Hungarian horse-breeding and grain plain.
Savanna / Sudan Type	Tropical	East & Central Africa (Kenya, Tanzania, Sudan)	- Characterized by tall coarse grasses (elephant grass) and scattered umbrella-acacia / baobab trees - Known as “Big Game Country” (abundant herbivore & carnivore diversity); Serengeti; Masai pastoralists.
Llanos	Tropical	Venezuela and Colombia (Orinoco River Basin)	Tropical savanna grassland with distinct wet and dry seasons; extensive cattle ranching.
Campos	Tropical	Brazilian Highlands (Brazil)	Subtropical / tropical savanna plateau grasslands south of the Amazon forest.
Cerrado	Tropical	Central Brazil (Mato Grosso)	Vast tropical savanna ecoregion; converted into world’s largest soyabean and cattle production frontier.
Gran Chaco	Tropical / Semi-arid	Paraguay, Bolivia, Northern Argentina	Scrub woodland and savanna; Quebracho tree (<i>axe-breaker</i>) source of tannin.

B. Major Deserts of the World & Origin Factors

Desert Name	Location & Continents	Type & Origin Factors	Unique Exam Superlative / Highlights
Sahara Desert	North Africa (11 countries)	Subtropical Hot Desert	LARGEST HOT DESERT ON EARTH (~9.2 million km²); covers ~30% of Africa; Ergs (sand seas), Regs (stone plains), Hamadas (rock plateaus).
Atacama Desert	Northern Chile & Southern Peru	Coastal Hyper-Arid Desert	DRIEST NON-POLAR DESERT ON EARTH - Formed by rain-shadow of high Andes + cold desiccating Humboldt (Peru) Current - Vast deposits of Sodium Nitrate (Caliche) and world-class copper mines.
Namib Desert	Namibia (South-West Africa)	Coastal Fog Desert	Formed by the cold Benguela Current offshore; ancient red sand dunes; home to ancient living fossil plant <i>Welwitschia mirabilis</i>.

Desert Name	Location & Continents	Type & Origin Factors	Unique Exam Superlative / Highlights
Kalahari Desert	Botswana, Namibia, South Africa	Semi-Arid Sandy Basin	• Inhabited by the indigenous San (Bushmen) hunter-gatherers; vast red sand dunes (<i>Kgalagadi</i>).
Australian Deserts	Central & Western Australia (Great Victoria, Great Sandy, Gibson, Simpson)	Subtropical Hot / Trade Wind Desert	• Known as the “ Red Centre ” / “ Outback ”; formed by subtropical high-pressure belt + cold West Australian Current ; Great Victoria is largest.
Gobi Desert	Southern Mongolia & Northern China	Cold Continental Rain-Shadow Desert	• Located in the rain-shadow of the Himalayas; extreme winter cold (-40°C); home to indigenous two-humped Bactrian camels .
Taklamakan Desert	Xinjiang, Northwest China (Tarim Basin)	Cold Desert	• Bounded by Tian Shan and Kunlun mountains; known as the “Sea of Death” / “Place of No Return”.
Rub’ al Khali	Southern Arabian Peninsula (Saudi Arabia, Oman, UAE, Yemen)	Continuous Sand Desert (Erg)	• Known as the “ Empty Quarter ”; LARGEST CONTIGUOUS SAND DESERT IN THE WORLD (~650,000 km ²).
Patagonian Desert	Southern Argentina (South America)	Cold Winter Rain-Shadow Desert	• Rain-shadow of the Southern Andes; influenced by the cold Falkland Current ; gravelly plateau.
Dasht-e Kavir & Dasht-e Lut	Iran (Central Plateau)	Salt Deserts (Playa)	• <i>Kavir</i> = great salt swamp; <i>Lut</i> recorded world’s highest satellite land skin temperature (>70.7°C).
Mojave & Sonoran Deserts	Southwestern USA (California, Arizona) & Mexico	Subtropical Hot Desert	• Contains Death Valley (Badwater Basin, -86 m — lowest point in North America; hottest recorded air temp 56.7°C); giant Saguaro cactus .

24. World Volcanoes, Oceanic Trenches & Landmark Islands

A. Famous Volcanoes of the World (Direct Match Sheet)

Volcano Name	Country / Geographic Location	Type / Activity	High-Yield Prelims Exam Triggers
Mount Stromboli	Lipari Islands / Sicily, Italy (Tyrrhenian Sea)	Active Stratovolcano	• Known as the “ LIGHTHOUSE OF THE MEDITERRANEAN ” due to continuous glowing eruptions visible for miles.
Mount Etna	Sicily, Italy	Active Stratovolcano	• Highest active volcano in Europe (~3,357 m); continuously active; UNESCO World Heritage Site.
Mount Vesuvius	Gulf of Naples, Italy	Active / Dormant Stratovolcano	• Catastrophic Plinian eruption in 79 AD completely buried and preserved the Roman cities of Pompeii and Herculaneum .
Mauna Loa & Kilauea	Big Island of Hawaii, USA (Pacific Ocean)	Active Shield Volcanoes	• Mauna Loa is the world’s largest subaerial shield volcano by volume and area • Kilauea is among the world’s most continuously active volcanoes; fluid basaltic lava flows.
Cotopaxi	Andes Mountains, Ecuador (South America)	Active Stratovolcano	• One of the world’s highest active volcanoes (5,897 m); textbook symmetrical snow-capped cone.
Ojos del Salado	Andes Mountains (Border of Chile & Argentina)	Active Stratovolcano	• HIGHEST ACTIVE VOLCANO ON EARTH (6,893 m) ; 2nd highest peak in Western Hemisphere after Mount Aconcagua.
Mount Kilimanjaro	Tanzania (East Africa)	Dormant Stratovolcano	• HIGHEST PEAK IN AFRICA (5,895 m) ; freestanding volcanic mountain with 3 volcanic cones: Kibo, Mawenzi, Shira; permanent snowcap near Equator.

Volcano Name	Country / Geographic Location	Type / Activity	High-Yield Prelims Exam Triggers
Krakatoa (Krakatau)	Sunda Strait (Between Java and Sumatra, Indonesia)	Active Caldera Volcano	• Cataclysmic eruption in 1883 generated massive tsunamis and shockwaves heard 4,800 km away; modern growing cone is Anak Krakatau .
Mount Merapi & Mount Sinabung	Java & Sumatra, Indonesia	Highly Active Stratovolcanoes	• Merapi (<i>Mountain of Fire</i>) is the most active volcano in Indonesia; deadly pyroclastic density currents.
Mount Pinatubo & Mayon	Luzon, Philippines	Active Stratovolcanoes	• Pinatubo's massive 1991 eruption injected huge SO ₂ aerosol clouds causing global temporary cooling of ~0.5°C • Mayon is celebrated for having the world's most perfectly symmetrical volcanic cone .
Mount Fuji (Fujisan)	Honshu Island, Japan	Dormant Stratovolcano	• Highest peak in Japan (3,776 m); sacred mountain; last erupted in 1707 (Hoei eruption).
Mount Erebus	Ross Island, Antarctica	Active Stratovolcano	• SOUTHERNMOST ACTIVE VOLCANO ON EARTH ; features a long-lived persistent phonolitic lava lake.
Barren Island	Andaman Sea, India (~135 km NE of Port Blair)	Active Volcano	• THE ONLY CONFIRMED ACTIVE VOLCANO IN SOUTH ASIA / INDIA ; erupted in 1991, 2005, and continues sporadic activity.
Narcondam Island	Andaman Sea, India	Dormant Volcanic Island	• Extinct/dormant andesite volcano; exclusive habitat of the endemic, endangered Narcondam Hornbill.

B. Deepest Oceanic Trenches of the World

- **Mariana Trench (Challenger Deep): 10,994 m** (Western Pacific Ocean near Guam) — **DEEPEST POINT ON PLANET EARTH**.
- **Tonga Trench (Horizon Deep): 10,882 m** (Southwest Pacific Ocean) — 2nd deepest trench on Earth.
- **Kuril–Kamchatka Trench: 10,542 m** (Northwest Pacific Ocean).
- **Philippine Trench (Galathea Deep): 10,540 m** (Western Pacific Ocean).
- **Puerto Rico Trench (Milwaukee Deep): 8,376 m** (Atlantic Ocean) — **DEEPEST POINT IN THE ATLANTIC OCEAN**.
- **Java / Sunda Trench: 7,450 m** (Eastern Indian Ocean off Sumatra/Java) — **DEEPEST POINT IN THE INDIAN OCEAN**.
- **Factorian Deep (South Sandwich Trench): 7,434 m** (Southern Ocean) — **DEEPEST POINT IN THE SOUTHERN OCEAN**.
- **Molloy Deep: 5,550 m** (Fram Strait, Arctic Ocean) — **DEEPEST POINT IN THE ARCTIC OCEAN**.

25. World Major Industrial Regions & Specialised Manufacturing / Mining Centres

Region / City	Country	Primary Manufacturing / Mineral Specialization	High-Yield Prelims Exam Nickname / Triggers
Ruhr Valley & Saar Basin	Germany	Coal mining, Iron and Steel, Heavy Engineering	• Known as the "Ruhrgebiet" / Industrial Heart of Europe ; Duisburg (world's largest inland port).
Pittsburgh	Pennsylvania, USA	Iron and Steel Industry	• Celebrated as the "STEEL CITY OF THE WORLD" ; confluence of Allegheny and Monongahela rivers.
Detroit	Michigan, USA	Automobile Manufacturing	• Celebrated as the "AUTOMOBILE CAPITAL OF THE WORLD" (Headquarters of Ford, GM, Chrysler).

Region / City	Country	Primary Manufacturing / Mineral Specialization	High-Yield Prelims Exam Nickname / Triggers
Birmingham	West Midlands, England, UK	Iron, Steel, Heavy Metal Hardware	• Historic heart of the Industrial Revolution; central to the English "Black Country" .
Manchester & Lancashire	North-West England, UK	Cotton Textile Manufacturing	• World's premier historic cotton textile spinning cluster due to humid oceanic climate.
Katanga (Shaba) & Copperbelt	DR Congo & Zambia (Central Africa)	Copper and Cobalt Extraction	• World's richest mineral belt for battery-grade Cobalt and high-grade Copper .
Witwatersrand ("The Rand")	Johannesburg, South Africa	Gold Mining	• World's largest gold-bearing reef formation ; produced ~40% of all gold mined in human history.
Kimberley	Northern Cape, South Africa	Diamond Mining	• Historic diamond capital; the "Big Hole" ; origin of the geological term Kimberlite pipe .
Baku	Azerbaijan (Caspian Sea)	Petroleum Extraction & Refining	• Historic birthplace of modern commercial oil extraction; Nobel Brothers and Rothschild oil legacy.
Krivoy Rog & Donbass (Donets Basin)	Ukraine	Iron Ore & Coking Coal	• Krivoy Rog = massive iron-ore basin; Donbass = premier historic coking coalfield.
Magnitogorsk & Kuznetsk (Kuzbass)	Russia (Urals & Siberia)	Iron & Steel Combine	• Magnitogorsk built around Magnetic Mountain (iron ore); Kuznetsk is a giant Siberian coal basin.
Upper Silesia	Poland	Coal Mining, Zinc & Lead, Heavy Industry	• Industrial core of Central Europe around Katowice.
Chuquibambilla	Antofagasta, Atacama Desert, Chile	Copper Mining	• World's largest open-pit copper mine by volume and second deepest open-pit mine on Earth.
Potosí (Cerro Rico)	Bolivia (Andes Mountains)	Silver Mining	• Historic "Rich Mountain" that bankrolled the Spanish Empire with vast silver wealth for centuries.
Kalgoorlie & Coolgardie	Western Australia	Gold Mining	• Historic Australian desert gold rush of 1890s; famous "Golden Mile" open-pit Super Pit gold mine.
Sudbury Basin	Ontario, Canada	Nickel, Copper, Platinum Group Metals	• Ancient 1.85-billion-year-old meteorite impact basin; premier global source of nickel.

Famous Indian Industrial Nicknames & City Specializations

- **Detroit of India: Chennai** (Tamil Nadu — automobile manufacturing hub of South Asia).
- **Manchester of India: Ahmedabad** (Gujarat — premier cotton textile hub).
- **Manchester of North India: Kanpur** (Uttar Pradesh — historic cotton mills and leather industry).
- **Manchester of South India: Coimbatore** (Tamil Nadu — cotton spinning mills).
- **Cottonopolis of India: Mumbai** (Maharashtra — historic concentration of textile mills).
- **Silicon Valley of India / Electronic City: Bengaluru** (Karnataka — premier IT & aerospace cluster).
- **Pittsburgh of India / Steel City: Jamshedpur / Tatanagar** (Jharkhand — TISCO established 1907 at Sakchi, confluence of Subarnarekha and Kharkai rivers).
- **Brass City (Peetal Nagari): Moradabad** (Uttar Pradesh — world-renowned brass handicraft industry).
- **Glass City / Bangle City (Suhag Nagari): Firozabad** (Uttar Pradesh — glass blowing and bangles).
- **Leather City of the World: Kanpur** (Uttar Pradesh).
- **Perfume Capital of India (Itra Nagari): Kannauj** (Uttar Pradesh).
- **Weaver City / City of Handlooms: Panipat** (Haryana).
- **City of Locks: Aligarh** (Uttar Pradesh).

- **Sports Goods Hub: Meerut** (Uttar Pradesh) and **Jalandhar** (Punjab).

Topic 1 — Indian Physical Geography: Mountains & Hills

📋 Must-score facts — belts, peaks, passes

Himalayan belts (N → S)

Trans-Himalaya → Himadri (Greater) → Himachal (Lesser) → Shiwalik

Item	Lock
Youngest belt (usual options)	Himadri (हिमाद्रि)
Oldest fold system (usual options)	Aravalli (अरावली)
Kashmir Valley	Pir Panjal south ; Himadri north

Central India hills (W → E)

Satpura → Mahadeo → Maikal → Chhotanagpur · Vindhya lies **north** of Narmada

State ↔ peak

State	Peak
Tamil Nadu	Doddabetta
Rajasthan	Guru Shikhar (Aravalli)
Nagaland	Saramati

Passes / temple

Item	Lock
Lipulekh / Niti / Mana	All in Uttarakhand (Lipulekh ≠ Ladakh)
Venkateswara / Tirupati	Tirumala / Mallamalla — Eastern Ghats (AP)
Intertrappean beds	Land / freshwater fossils (not marine)
India area rank	7th (~3.28 million km ² ; ~2.4% world land)

Topic 2 — Climate of India

Consolidated — 40 Must-Score Facts

1. India's climate is **tropical monsoon**, not wholly tropical, because the country stretches into the subtropical north. The Himalaya (हिमालय) acts as a winter **wall** against Central Asian cold and forces **orographic** rain on southern slopes in the monsoon.
2. The word monsoon comes from Arabic **mausim** (season). About **75–90%** of India's rain falls with the south-west monsoon from June to September. Onset usually begins first in **Kerala around 1 June**.
3. IMD seasons are Cold Weather (**December–February**, Western Disturbances in the north-west), Hot Weather (**March–May**, Loo, Nor'westers, mango and blossom showers), South-West Monsoon (**June–September**), and Retreating Monsoon (**October–November**, October heat and Bay of Bengal / Andaman (अंडमान) cyclones).
4. South-west monsoon starts when the **ITCZ** shifts north and a thermal low forms over north-west India. South-east trades cross the equator near about **40–60° E**, then the **Mascarene High** and **Somali Jet** feed the two monsoon branches.
5. The **Arabian Sea branch** soaks the Western Ghats (पश्चिमी घाट) windward slope and leaves the leeward Deccan (दक्कन) dry. The Chhattisgarh basin is comparatively weak on this branch. The **Bay of Bengal branch** waters the north-east and the Ganga (गंगा) plain.
6. The Tamil Nadu south-east coast stays **dry in the south-west monsoon** because it lies parallel to the Bay branch and in the rain shadow (छाया) of the Arabian Sea branch. Its main rain comes with the **north-east monsoon** later.
7. Typical onset dates are Kerala about **1 June**, Mumbai and Kolkata about **10 June**, Delhi about **29 June**, and Rajasthan last around **mid-July**.
8. An **active** monsoon keeps the trough on the **Ganga plain** so the plains stay wet. A **break** monsoon shifts the trough onto the **Himalaya** (हिमालय), so central India goes dry.
9. On the northern plains rainfall generally **declines east to west**. The rainy season is **shorter from south to north**. Rajasthan stays dry partly because the Aravalli (अरावली) runs **parallel** to the Arabian Sea branch. A rainy day is counted from about **2.5 mm** of rain.
10. The **north-east monsoon (October–December)** is mainly for Tamil Nadu, south Andhra Pradesh, the south-east Karnataka–Kerala fringe, and Puducherry. It is **not** all-India winter rain.
11. **Western Disturbances** are Mediterranean **extra-tropical** systems that bring winter rain to the north-west. Their rain **decreases west to east**. They are **not** the retreating monsoon. Core belt: Punjab, Haryana, Himachal (हिमाचल), Jammu & Kashmir, Uttarakhand (उत्तराखंड), western Uttar Pradesh (उत्तर प्रदेश), and northern Rajasthan.
12. Mid-latitude jet streams are **westerly**. In winter the **Subtropical Westerly Jet** south of the Himalaya steers Western Disturbances after Tibet splits the jet. The northern branch does **not** drive India's winter rain.
13. The **Tropical Easterly Jet** is a summer **easterly** around about **14° N** and supports the south-west monsoon. The **Somali Jet** is the cross-equatorial feed into that monsoon.
14. **El Niño** warms the eastern Pacific off Peru, usually weakens the Indian monsoon, and **reduces** plankton because warm water suppresses upwelling. **La Niña** cools that ocean and usually strengthens the monsoon.
15. The **Indian Ocean Dipole** compares the western and eastern **Indian Ocean**, not the Pacific. A **positive IOD** helps the Indian monsoon; a **negative IOD** hurts it.
16. Köppen map for India: **Am** = Kerala / Konkan / north-east; **Aw** = Deccan; **As** = Tamil Nadu dry-summer Coromandel; **BWh** = Thar; **Cwg** = Ganga / Great Plains / north Bihar (and most of the Uttar Pradesh plain).
17. The January **18°C** isotherm is the usual tropical–subtropical divide. **Diurnal range** is highest in the Rajasthan desert. **Leh** (लेह) is among the driest class stations; Mawsynram / Cherrapunji are the wettest class.
18. The troposphere (क्षोभमंडल) is heated mainly by **long-wave** radiation from the ground. Condensation releases **latent heat**. Normal lapse rate is about **6.5 °C per 1000 m**. Relative (सापेक्ष) humidity (सापेक्ष आर्द्रता) **falls as temperature rises**. Ozone peaks in the **stratosphere** (समतापमंडल).

19. Circulation cells are **Hadley (0–30°)**, **Ferrel (30–60°)**, and **Polar (60–90°)**. **Doldrums** (डोलड्रम) mark the rainy ITCZ calm belt. **Horse latitudes** (हॉर्स अक्षांश) are the dry subtropical highs near **30°**.
 20. Rain types to fact are **orographic**, **convictional**, and **cyclonic**. An **isohyet** is a line of equal rainfall.
 21. **Loo** is the hot dry wind of the Indo-Gangetic plains in May–June. **Kal Baisakhi / Nor’westers** are violent pre-monsoon thunderstorms of eastern and north-eastern India in April–May.
 22. **Mango showers** are pre-monsoon rains of Kerala and Karnataka. **Blossom showers** help coffee in Tamil Nadu and Kerala. **Sea breeze** blows by day toward land; **land breeze** blows by night toward sea.
 23. **Anabatic** winds climb slopes by day. **Katabatic** winds drain downslope by night. **Foehn / Chinook** (चिनुक) are warm dry lee winds — not the same as cold katabatic drainage.
 24. Tropical cyclones form more often over the **Bay of Bengal** than the Arabian Sea. Sea surface temperature needs about **26–27°C**. The **eye** is calm; the **eyewall** is fiercest. They do **not** form on the equator.
 25. Local cyclone names: **Hurricane** (USA / Atlantic), **Typhoon / Taifu** (NW Pacific / Japan), **Baguio** (Philippines), **Willy-willies** (Australia), and generic **Cyclone** in the North Indian Ocean. Willy-willy (विली-विली) is **not** the Australian hot wind **Brickfielder** (ब्रिकफील्डर).
 26. Thornthwaite’s famous line is that **vegetation is the true index of climate**. That fact is **not** Köppen’s letter-code system.
 27. In Uttar Pradesh, May–June **Loo** heat waves hit the plains. Winter rain and fog track **Western Disturbances**, and **western UP is wetter than eastern UP**.
 28. Uttar Pradesh’s south-west monsoon rain often arrives with **Bay of Bengal depressions**. An active spell needs the trough on the plain. Pre-monsoon **lightning** and winter **inversion smog** (west UP / NCR fringe) are key hazards.
 29. Absolute (निरपेक्ष) humidity (निरपेक्ष) is the **mass of vapour** in air. Relative humidity is the **percentage of saturation**. Relative humidity falls when temperature rises even if vapour mass stays the same.
 30. The north-east monsoon on the Coromandel / Chennai coast is **NE-monsoon dependent**, not a south-west monsoon belt. Do not confuse retreating-monsoon October heat with Western Disturbance (पश्चिमी विक्षोभ) winter rain in the north-west.
-
31. Retreating monsoon months are mainly **October–November**, with October heat and Bay of Bengal / Andaman cyclone risk. Do not call this the Western Disturbance season.
 32. Burst of monsoon is sudden onset with heavy rain; withdrawal is gradual from north-west to south-east. Break monsoon ≠ retreating monsoon.
 33. **Monsoon origin concepts**: The classical **Thermal concept** relies on differential heating rates between the continental landmass and the Indian Ocean. **Flohn’s Dynamic concept** explains monsoon as the seasonal migration of planetary pressure and wind belts with the shifting ITCZ.
 34. **Traditional Indian Calendar (6 Ritus)**: Six bimonthly seasons in chronological order: **Vasanta** (Spring: Mar–Apr) → **Grishma** (Summer: May–Jun) → **Varsha** (Rainy: Jul–Aug) → **Sharad** (Autumn: Sep–Oct) → **Hemant** (Pre-winter: Nov–Dec) → **Shishir** (Winter: Jan–Feb).
 35. **Rainy Day standard**: IMD officially defines a “rainy day” as any 24-hour period recording **≥ 2.5 mm** of precipitation.
 36. **Thirsty Land paradox**: Despite receiving ~1,236 mm average annual precipitation, India faces severe seasonal water stress because ~75–90% is concentrated in 4 monsoon months, with rapid runoff and high evaporation during hot dry months.
 37. **Spatial rainfall hierarchy**: Kochi (~3,932 mm) > Kolkata (~2,451 mm) > Patna (~1,100 mm) > Delhi (~904 mm). Driest station: **Leh** (~4.17 cm / 41.7 mm); Rajasthan driest: **Jaisalmer** (~31.67 cm). Wettest: **Mawsynram / Cherrapunji** (Khasi Hills, Meghalaya > 5,000 mm).
 38. **Pushkar Hills to Balotra flood link**: Heavy rainfall in the Pushkar hills (Ajmer) drains down the Luni River basin, producing severe flash floods downstream in **Balotra** (Barmer) due to the low-lying slope of the Pachpadra depression.

39. **Desert Development Programme (DDP):** Launched in 1977–78 initially in 131 blocks across 21 districts in 5 states (Rajasthan, Gujarat, Haryana, J&K, HP), later expanded to 235 blocks across 40 districts in 7 states.
40. **Winter Anticyclone vs Western Disturbance:** Winter anti-cyclonic high pressure over northern India is a natural result of cold, dense, subsiding continental air (NOT caused by winter rain). Western Disturbances are **extra-tropical** (temperate) depressions from the Mediterranean whose rainfall decreases from **West to East**.

Must-score facts — seasons, monsoon, onset

Frame

Item	Lock
Climate type	Tropical monsoon (not wholly tropical)
Etymology	Arabic mausim (season)
SW monsoon share	~ 75–90% of annual rain

IMD seasons

Season	Months	Tag
Cold	Dec–Feb	Western Disturbances (NW)
Hot	Mar–May	Pre-monsoon / loo
SW monsoon	Jun–Sep	Main rains
Retreating / NE	Oct–Dec	TN coast rain

Onset / branch tags

Place	Typical onset
Kerala	~ 1 June
Mumbai / Kolkata	~ 10 June
Delhi	~ 29 June
Branch	Lock
-----	-----
Arabian Sea	Windward Western Ghats wet; leeward Deccan dry
Tamil Nadu SE coast	Dry in SW monsoon ; rain in NE monsoon (उत्तर-पूर्व मानसून)
Active vs break	Trough on Ganga plain vs shift to Himalaya foothills
Plains rain	Declines east → west

Topic 3 — Drainage System

Consolidated — 48 Must-Score Facts

1. About **77%** of India's drainage **area** faces the **Bay of Bengal**, about **23%** faces the **Arabian Sea**, and inland basins cover about **8%** of area. Over **90% of river water** still goes to the Bay of Bengal.
2. Water-share fact: **Brahmaputra about 40%, Ganga about 25%, Godavari about 6%**. Brahmaputra leads water volume; Ganga leads Indian basin area.
3. The Western Ghats (पश्चिमी घाट) are the main peninsular divide. From **Amarkantak**, drainage is radial — **Narmada** (नर्मदा) west and **Son** toward the Ganga.
4. **Ganga** is the largest basin **inside India**. **Godavari** (गोदावरी) is the largest **peninsular** basin. Among Tapi (ताप्ती) / Narmada / Mahanadi / Cauvery alone, **Mahanadi** (महानदी) is the largest basin.
5. Himalayan rivers are mostly **perennial** (melt plus rain) and include **antecedent** (पूर्ववर्ती) giants. Peninsular rivers are mostly **rain-fed** and graded.
6. Drainage types: **consequent** (अनुवर्ती) follows original slope; **subsequent** (परवर्ती) follows weak belts later; **antecedent** (Indus, Sutlej (सतलुज), Brahmaputra) cuts rising Himalaya (हिमालय); **superimposed** (अध्यारोपित) (Chambal (चंबल)) inherits an older course.
7. **Trans-Himalayan** rivers rise north of the Great Himalaya in Tibet and cut through — fact **Sutlej** (सतलुज), and also Indus and Brahmaputra (Tsangpo). Among Jhelum (झेलम) / Sutlej / Ganga / Ravi (रावी), the trans-Himalayan pick is **Sutlej**.
8. The Indus rises in a Tibet gorge and reaches the Arabian Sea in Pakistan. The five Punjab rivers east to west are **Sutlej–Beas (ब्यास)–Ravi–Chenab (चिनाब)–Jhelum**. Under the Indus Treaty, India gets Ravi, Beas, and Sutlej; Pakistan gets Indus, Jhelum, and Chenab.
9. Dam (दाम –)–river pairs: **Baglihar = Chenab**, **Pandoh = Beas** (not Ravi), **Srisailem = Krishna** (not Tungabhadra (तुंगभद्रा)).
10. The Ganga is named only at **Devprayag** (देवप्रयाग) where Alaknanda (अलकनंदा) meets Bhagirathi (भागीरथी). It is about **2525 km** long. In Bangladesh it becomes the **Padma** (पद्मा), meets the Brahmaputra as **Jamuna** (जमुना), then joins the Meghna system to the Bay of Bengal.
11. Ganga joins: Yamuna (यमुना) at **Prayagraj** (प्रयागराज), Ramganga (रामगंगा) at **Kannauj** (कन्नौज), Ghaghara (घाघरा) at **Chhapra**, Gandak (गंडक) at **Sonpur**, Son at **Patna** (पटना), Kosi (कोसी) at **Kursela**. Left-bank Himalayan tributaries include Ramganga, Ghaghara, Gandak, Kosi, and Mahananda (महानंदा); **Gomti** is a plains-origin left-bank river. Right-bank facts are Yamuna, Son, and Damodar (दामोदर).
12. Yamuna right-bank tributaries west to east are **Chambal–Sind (सिंध)–Betwa–Ken**. Chambal joins at Etawah (इटवा); Betwa at **Hamirpur**; Ken at **Banda**. Betwa does **not** meet at Prayagraj.
13. **Jonk** (जोंक) joins the **Mahanadi**, not the Ganga. **Gandak** (गंडक) is not an Uttar Pradesh river in the usual state list (राज्य सूची). **Punpun** (पुनपुन) meets near Fatuha. Chambal ravines (चंबल) start near **Mhow**.
14. Panch Prayag upstream to downstream: **Vishnu** (Dhuli (धौली)) → **Nanda** (Nandakini) → **Karn** (Pindar (पिंडर)) → **Rudra** (Mandakini) → **Dev** (Bhagirathi) = Ganga. Badrinath (बद्रीनाथ) is on the **Alaknanda** (अलकनंदा); Kedarnath (केदारनाथ) is on the **Mandakini** (मंदाकिनी).
15. The eastern **Dhauliganga** (धौलीगंगा) joins the **Kali** (काली), not the Alaknanda. Do not confuse it with the Dhuli of Vishnuprayag (विष्णुप्रयाग).
16. The Brahmaputra rises as Angsi / Tsangpo, makes a **U-turn at the eastern syntaxial bend near Namcha Barwa**, and enters India as Siang / Dihang (दिहांग). **Majuli** is the classic island. Right-bank north tributaries include Subansiri, Manas, Sankosh, and Teesta; left-bank south tributaries include Burhi Dihing and Dhansiri.
17. **Teesta** was captured from the Ganga system into the Brahmaputra. Barak (बराक) becomes Meghna and reaches the Bay of Bengal — it is **not** inland drainage.
18. Peninsular length fact: **Godavari (~1465 km) > Krishna (~1400) > Narmada (~1312) > Mahanadi (~851) > Tapi (~724)**. Godavari is also called Dakshin / Vriddha Ganga and rises near **Trimbak**.

19. Origins: Yamuna–Yamunotri (यमुनोत्री), Krishna–Mahabaleshwar, Godavari–Nashik, Mahanadi–Sihawa, Narmada–Amarkantak, Tapi–**Multai (Satpura (सतपुड़ा))** not Western Ghats, Kaveri–Brahmagiri.
 20. **Pranhita** (प्रणहिता) joins **Godavari**. **Hemavati** joins Kaveri. **Malaprabha** joins Krishna. Hyderabad (हैदराबाद) stands on the **Musi**, not the Krishna stem. Ludhiana is on the **Sutlej**, not the Ravi.
 21. West-flowing set: Narmada, Tapi, Mahi (माही), Sabarmati (साबरमती), **Periyar** (पेरियार). East-flowing examples include Pennar (पेन्नार) and Palar. Narmada and Tapi form **estuaries** in rift valleys; most east-flowing peninsular rivers build **deltas**.
 22. **Luni** (लूनी) is inland: upper course fresh, lower course **saline** (लवणीय). The classic fault-valley tag is **Damodar** (दामोदर); Narmada and Tapi are also rift valleys.
 23. Delta (डेल्टा) types: **Arcuate** = Sundarbans (सुंदरबन); **Bird's-foot** (पक्षी-पाद) = Mississippi (not Ganga). **Moribund** as a **Bengal subdivision** ≠ **Cauvery** as a **moribund lobe** match — read the stem. **Trellis** (जालनुमा) marks folded structure; **radial** marks Amarkantak; **dendritic** (वृक्षाकार) is the common homogeneous pattern.
 24. National Waterways: **NW-1** Ganga–Hooghly (हुगली), **NW-2** Brahmaputra, **NW-3** Kerala, **NW-4** Krishna–Godavari, **NW-5** Brahmani–Mahanadi.
 25. **Ken–Betwa** is the only National Perspective Plan link under implementation, linking Madhya Pradesh and Uttar Pradesh in Bundelkhand.
 26. Uttar Pradesh points: Sangam (संगम) at **Prayagraj**; **Gomti** is Lucknow (लखनऊ)'s river and the pollution tag; **Ghaghara** (घाघरा) is Ayodhya (अयोध्या)'s river; the Ganga–Yamuna **Doab** (दोआब) is the core plain.
 27. Chambal sanctuary spans **Uttar Pradesh, Madhya Pradesh, and Rajasthan** — not Haryana.
 28. Kaveri island sequence includes **Srirangapatna (श्रीरंगपट्टनम)–Shivasamudram–Srirangam**. Bhubaneswar is **not** a Mahanadi-bank city fact in the usual trap set.
 29. East-flowing peninsular rivers north to south often run **Subarnarekha → Mahanadi → Godavari → Krishna → Pennar → Cauvery → Vaigai**.
 30. Indus is the longest of the four (चातुर्याम) Himalayan systems overall, but most of its course lies in Pakistan. Ganga is the longest river **in India**.
 31. Concordant (संगत) drainage follows structure; discordant (असंगत) drainage cuts across it — Himalayan antecedents are the classic discordant case.
 32. **Kosi** (कोसी) is the Sorrow of Bihar; **Damodar** is the Sorrow of Bengal. Do not swap those titles.
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33. Ganga left-bank Himalayan tributaries include Ramganga, Ghaghara, Gandak, Kosi and Mahananda; **Gomti** is plains-origin. Right-bank facts include Yamuna, Son and Damodar.
 34. Peninsular rivers are older, graded and mostly rain-fed; Himalayan rivers are younger, perennial and often antecedent through rising ranges.
 35. Dendritic rivers include the **Ganga** and **Indus** plains, the **Godavari**, the **Mahanadi**, the **Krishna**, and the **Kaveri**.
 36. The Indian **trellis** ground is the old folded belt of **Singhbhum** (Chotanagpur). **Rectangular** streams follow joints on the **Vindhyan** rocks.
 37. **Centripetal** streams drain into **Loktak** and into **Sambhar** (Mendha, Rupangarh).
 38. The **Sharavati** follows a **parallel** pattern down the Western Ghats.
 39. **Ganga Alluvial Depth**: Depth of Gangetic alluvium below the surface is estimated at **4,000–6,000 metres** according to Oldham, and **~2,000 metres** according to Glennie.
 40. **Son River Origin & Catchment**: Originates from Amarkantak Hill (Sonmuda in Anuppur MP / Sonbachharwar in Gaurela-Pendra-Marwahi CG). Catchment area is **70,055 sq km**, the largest among Ganga's major tributaries (vs Gandak ~45,731, Ramganga ~30,811, Mahananda ~23,700). Flows northward to join Ganga near Patna.
 41. **Kaimur Water Divide**: The **Kaimur Range** acts as the primary watershed dividing the Son system to the south from the Tons/Yamuna system to the north.
 42. **Gandak & Triveni Canal**: In Nepal, the Gandak is known as the **Gandaki or Narayani**, formed by the confluence of the Kali Gandak and Trishulganga. The **Triveni Canal** draws water from the Gandak in West Champaran (Bihar). Gandak joins Ganga at Sonpur (near Patna) and does **not** flow through Uttar Pradesh.

43. **Ganga in Bihar:** Passes through **12 districts**; maximum length is in **Patna (~99 km)**, followed by **Bhagalpur (~97 km)**. Punpun joins the Ganga at **Fatuha**.
44. **Kosi System & Kanwar Lake Trap:** Kosi is formed by 7 Himalayan streams (Arun being principal); it does **not** originate from glaciers. Average elevation of the Kosi plain is **30–40 metres**. **Kanwar Lake** (Begusarai, Bihar) is an oxbow lake formed by a cutoff meander of the **Burhi Gandak**, NOT the Gandak.
45. **Eastern Ghats Origins:** **Nagavali** and **Vamsadhara** rise from the Eastern Ghats. By contrast, **Brahmani** and **Subarnarekha** (origin: Nagri village near Ranchi) originate from the Chotanagpur / Ranchi Plateau.
46. **Punjab Doabs & Chenab Headwaters:** **Chandra** (Chandra Tal) and **Bhaga** (Bara Lacha La) unite at **Tandi** in the Lahaul valley to form the **Chandrabhaga (Chenab)**. Doabs: **Bist** (Beas–Sutlej), **Bari** (Beas–Ravi), **Rechna** (Ravi–Chenab), **Chaj** (Chenab–Jhelum).
47. **Gandikota Canyon & S.P. Chatterjee Division:** The spectacular Gandikota canyon (“Grand Canyon of India”) in Kadapa (AP) was carved by the **Pennar River**. In S.P. Chatterjee’s physiographic division, the Mahanadi Basin is classified under the **Northern Deccan Plateau**.
48. **Key City Riverbank Anchors:** **Kanpur** is the largest city on the Ganga; **Uttarkashi** is on the Bhagirathi (before Ganga forms); **Leh** is on the **right bank** of the Indus; **Pahalgam** is on the **Lidder**; **Hampi** is on the **Tungabhadra** (not Malaprabha); **Pandharpur** is on the **Chandrabhaga (Bhima)**; **Govindghat** is on the **Alaknanda / Lakshman Ganga** (not Mandakini).

Must-score facts — basins, types, dam–river

Area / water share

Item	Lock
Bay of Bengal drainage area	~ 77%
Arabian Sea drainage area	~ 23%
Water volume	Brahmaputra ~ 40% · Ganga ~ 25% · Godavari ~ 6%
Largest basin in India	Ganga
Largest peninsular basin	Godavari
Main peninsular divide	Western Ghats (पश्चिमी घाट)
Amarkantak radial	Narmada west · Son east

Drainage types

Type	Lock
Consequent	Follows original slope
Subsequent	Follows weak belts later
Antecedent	Cuts rising land (Indus, Sutlej, Brahmaputra)
Himalayan vs peninsular	Mostly perennial vs mostly rain-fed / seasonal

Dam ↔ river

Dam	River
Baglihar	Chenab (चिनाब)
Pandoh	Beas (ब्यास) (≠ Ravi)
Srisailem	Krishna (कृष्णा) (≠ Tungabhadra)

Master Table — Panch Prayag of Uttarakhand (पंच प्रयाग)

(Tested heavily in match-the-pair: UPPCS 2023 Q50, UKPCS)

Prayag (प्रयाग)	Headstream 1	Headstream 2 (Tributary)	District / Location	Key Shrine / Geographical Landmark
1. Vishnuprayag (विष्णुप्रयाग)	Alaknanda (अलकनंदा)	Dhauliganga (धौलीगंगा - Western)	Chamoli, Uttarakhand	Nearest prayag to Badrinath & Joshimath; origin from Satopanth glacier
2. Nandaprayag (नंदप्रयाग)	Alaknanda	Nandakini (नंदाकिनी)	Chamoli, Uttarakhand	Confluence named after King Nanda; Nandakini rises from Nanda Ghunti
3. Karnaprayag (कर्णप्रयाग)	Alaknanda	Pindar (पिंडर)	Chamoli, Uttarakhand	Pindar rises from Pindari glacier (Kumaon/Garhwal border)
4. Rudraprayag (रुद्रप्रयाग)	Alaknanda	Mandakini (मंदाकिनी / Kali Ganga)	Rudraprayag, Uttarakhand	Mandakini flows down from Kedarnath (Chorabari glacier)
5. Devprayag (देवप्रयाग)	Alaknanda	Bhagirathi (भागीरथी)	Tehri Garhwal, Uttarakhand	The river is officially named GANGA (गंगा) from this confluence onward ; Bhagirathi rises from Gaumukh (Gangotri glacier)

Master Table — Tributaries of Major Indian River Systems (बाएँ एवं दाएँ तट की सहायक नदियाँ)

(High-yield for list-matching and tributary identification questions)

River System	Origin (उद्गम)	Left-Bank Tributaries (बायाँ तट)	Right-Bank Tributaries (दायाँ तट)	Drainage Mouth / Key Notes
Ganga (2,525 km)	Devprayag (Bhagirathi + Alaknanda)	• Ramganga (meets at Kannauj) • Gomti (origin Gomat Taal/Phulhar Jheel, Pilibhit; meets near Saidpur) • Ghaghara / Saryu (meets at Chhapra) • Gandak (meets at Sonpur) • Burhi Gandak • Kosi ("Sorrow of Bihar"; meets at Kursela) • Mahananda (easternmost left tributary)	• Yamuna (largest tributary; meets at Prayagraj Sangam) • Tamas / Tons (meets near Sirsa) • Son (radial from Amarkantak; meets at Patna/Danapur) • Punpun (meets at Fatuha)	Enters Bangladesh as Padma ; combines with Jamuna (Brahmaputra) and Meghna to form Sundarbans delta
Yamuna (1,376 km)	Yamunotri Glacier (Bandarpunch)	• Tons (largest Himalayan tributary) • Hindon (meets near Noida/Greater Noida) • Rind, Sasur Khaderi	• Chambal (meets near Muradganj/Etawah) • Sindh (meets in Jalaun) • Betwa (meets at Hamirpur) • Ken (meets at Chilla, Banda)	Parallel to Ganga; forms fertile Ganga-Yamuna Doab
Indus (Sindhu, 2,880 km)	Bokhar Chu (near Lake Mansarovar, Tibet)	• Zaskar • Suru, Soan • Panjinad (combined flow of Jhelum, Chenab, Ravi, Beas, Sutlej)	• Shyok (with Nubra) • Gilgit • Hunza • Kabul (with Swat) • Kurram, Gomai	Trans-Himalayan; enters Arabian Sea south of Karachi

River System	Origin (उद्गम)	Left-Bank Tributaries (बायाँ तट)	Right-Bank Tributaries (दायाँ तट)	Drainage Mouth / Key Notes
Brahmaputra (2,900 km)	Angsi / Chemayungdung Glacier (Tibet as <i>Yarlung Tsangpo</i>)	• Lohit • Dibang (Sikang) • Burhi Dihing • Dhansiri (South) • Kopili	• Subansiri (gold-bearing; largest tributary) • Kameng (Jia Bhoreli) • Manas • Sankosh • Teesta (captured from Ganga system in 1787 flood)	Syntaxial U-turn at Namcha Barwa; world's largest river island Majuli (Assam)
Godavari (1,465 km) <i>"Dakshin Ganga"</i> / <i>"Vridhha Ganga"</i>	Trimbakeshwar (Nashik, Maharashtra)	• Dharna • Penganga • Wainganga • Wardha (combined: Pranhita) • Indravati (Chitrakote Falls) • Sabari & Sileru	• Pravara • Mula • Manjra / Manjira (longest right-bank tributary, flows through Nizamabad) • Maner	Largest Peninsular river basin; forms lobate delta in Andhra Pradesh
Krishna (1,400 km)	Near Mahabaleshwar (Western Ghats, Maharashtra)	• Bhima (longest tributary, 861 km) • Dindi • Peddavagu • Musi (Hyderabad sits on its banks) • Paleru, Munneru	• Venna • Koyna (Shivsagar reservoir) • Panchganga • Dudhganga • Ghataprabha (Gokak falls) • Malaprabha • Tungabhadra (Tunga + Bhadra; largest basin tributary)	Second largest east-flowing peninsular river; major dams: Almatti, Srisailem, Nagarjuna Sagar
Cauvery / Kaveri (800 km) <i>"Ganga of the South"</i>	Talakaveri, Brahmagiri Range (Kodagu, Karnataka)	• Harangi • Hemavati • Shimsha • Arkavathi	• Lakshmantirtha • Kabini • Suvarnavathi • Bhavani • Noyyal • Amaravati	Perennial due to dual monsoons (SW monsoon in upper basin + NE monsoon in lower basin); islands: Srirangapatna, Shivanasamudra, Srirangam
Mahanadi (851 km)	Sihawa highlands (Dhamtari, Chhattisgarh)	• Seonath (longest tributary, 383 km) • Hasdeo • Mand • lb	• Jonk (UPPCS repeated trap: Jonk joins Mahanadi, NOT Ganga!) • Ong • Tel	Hirakud Dam (longest earthen dam); drains Chhattisgarh and Odisha into Bay of Bengal
Narmada (1,312 km)	Amarkantak Plateau (Anuppur, MP)	• Sher • Shakkar • Dudhi • Tawa (longest tributary) • Ganjal, Kundi	• Hiran • Barna • Choral • Man, Uri, Orsang	Flows through rift valley between Vindhyas (north) & Satpuras (south); Dhuandhar falls; Sardar Sarovar Dam; forms estuary (no delta)
Tapi / Tapti (724 km)	Multai (Betul district, Satpura Range, MP)	• Purna (main tributary; drains Vidarbha) • Girna • Panjhra • Bori, Waghur	• Suki • Gomai • Arunavati	Parallel rift valley south of Satpuras; Ukai & Kakrapar dams; forms estuary at Surat

Master Table — Major Mountain Passes of India (भारत के प्रमुख पर्वतीय दर्रे)

(High-frequency match-the-pair: Pass ↔ State / Mountain Range)

Pass Name (दर्जा)	State / Union Territory	Mountain Range	Strategic / Historical Connection
Zoji La (ज़ोजिला)	Ladakh (UT)	Great Himalayas	Connects Srinagar to Leh ; Zojila Tunnel under construction
Karakoram Pass	Ladakh (UT)	Karakoram Range	Highest pass in India (~5,540 m); ancient silk route to Tarim Basin / Yarkand
Bara-lacha La (बारालाचा)	Himachal Pradesh	Zaskar Range	Connects Mandi/Manali to Leh ; source of Bhaga river (Chenab)
Shipki La (शिपकी ला)	Himachal Pradesh	Zaskar Range	Sutlej river enters India from Tibet through this gorge; India-China border trade post
Rohtang Pass (रोहतांग)	Himachal Pradesh	Pir Panjal Range	Connects Kullu Valley with Lahaul and Spiti; bypassed by Atal Tunnel (9.02 km)
Mana Pass (माना दर्जा) / <i>Dungri-La</i>	Uttarakhand	Greater Himalayas	Located in Nanda Devi Biosphere; connects Uttarakhand with Tibet; near Badrinath
Niti Pass (नीति दर्जा)	Uttarakhand	Greater Himalayas	Ancient trade route between Uttarakhand (Chamoli) and Tibet
Lipulekh Pass (लिपुलेख)	Uttarakhand (Pithoragarh)	Kumaon Himalayas	Tri-junction of India, Nepal, and Tibet ; pilgrimage route for Kailash-Mansarovar Yatra
Nathu La (नाथुला)	Sikkim	Dongkya Range	Ancient Silk Route to Lhasa (Chumbi Valley, Tibet); reopened for border trade in 2006
Jelep La (जेलेप ला)	Sikkim	Greater Himalayas	Passes through Chumbi Valley connecting Sikkim/Kalimpong with Lhasa (Tibet)
Bomdi La (बोमडिला)	Arunachal Pradesh	Greater Himalayas	Connects Arunachal Pradesh (Tawang) with Lhasa (Tibet)
Diphu Pass (दिफू दर्जा)	Arunachal Pradesh	Mishmi Hills	Tri-junction border of India, China, and Myanmar
Thal Ghat (थाल घाट) / <i>Kasara Ghat</i>	Maharashtra	Western Ghats (Sahyadri)	Links Mumbai to Nashik & Kolkata (NH-3 & Central Railway)
Bhor Ghat (भोर घाट)	Maharashtra	Western Ghats (Sahyadri)	Links Mumbai to Pune & Chennai (Mumbai-Pune Expressway & rail line)
Palghat Gap (पालघाट)	Kerala & Tamil Nadu	Between Nilgiris & Anaimalai	Major gap connecting Palakkad (Kerala) with Coimbatore (Tamil Nadu)
Shencottah Gap (शेनकोट्टा)	Kerala & Tamil Nadu	Between Cardamom & Agasthyamalai	Connects Kollam (Kerala) with Madurai (Tamil Nadu)

Topic 4 — Lakes, Waterfalls & Water Resources

Consolidated — 42 Must-Score Facts

1. First classify a lake as lagoon, inland saline, freshwater, crater, glacial, oxbow, or artificial. Origin pairs: **Wular** tectonic, **Chilika** lagoon, **Lonar** crater, **Sambhar** wind / playa, **Kabartal** oxbow.
2. **Chilika** is the largest **east-coast lagoon** (Odisha / Northern Circars). If a question says "largest saline" and omits "inland," Chilika is often the intended answer.
3. **Sambhar** is the largest **inland** saline lake (Rajasthan). Sister Rajasthan saline names include Didwana, Kuchaman, Sargol, and Khatu.
4. **Wular** is the largest-freshwater volume fact (Jammu & Kashmir, on the Jhelum (झेलम), tectonic). **Dal** is the tourism lake. **Roopkund** is Uttarakhand (उत्तराखंड)'s Mystery / Skeleton Lake.
5. **Loktak** (Manipur) holds **phumdis**, **Keibul Lamjao**, and the Sangai (सांगाई). **Kabartal** is Bihar's oxbow Ramsar lake.
6. **Pulicat** is the second major brackish lagoon (Andhra Pradesh–Tamil Nadu (नाडु), Sriharikota bar). **Kolleru** is mainly **freshwater**, not a lagoon king.
7. **Vembanad** is Kerala's largest lake and India's **longest**; it is a **west-coast** kayal / Ramsar, not India's overall largest lake. **Kayal** means a Kerala coastal lagoon / backwater.
8. **Phulhar** (Pilibhit) is linked to the Gomti source. **Kodaikanal** lake is artificial. Periyar (पेरियार) Lake is **not** a coastal lagoon.
9. World lake traps: **Onega** and **Ladoga** are in **Russia** (not Canada). Michigan is USA-only; Superior is USA and Canada.
10. Rudrasagar is **Tripura**; Sultanpur is **Haryana**; Surinsar–Mansar is **Jammu & Kashmir** — none of these is an Uttar Pradesh (उत्तर प्रदेश) Ramsar.
11. India has more than **17%** of world people, about **4%** of water, and about **2.45%** of land. Precipitation is about **4000 BCM**; available water about **1869 BCM**; utilisable about **1122 BCM** (surface ~690 + groundwater ~433).
12. The Ganga (गंगा)–Brahmaputra (ब्रह्मपुत्र)–Barak (बराक) belt is about **one-third** of area but about **60%** of surface water. Irrigation takes about **89%** of surface water use and about **92%** of groundwater use.
13. Groundwater now covers more than **60%** of irrigated area. Canals dominate the Indo-Gangetic plains; tanks matter on the peninsula. Tamil Nadu leads south-Indian groundwater utilisation.
14. Quality points: **arsenic** in the Ganga belt; **fluoride** in Rajasthan and parts of Maharashtra; **usar** from over-irrigation in the north-west; Yamuna (यमुना)'s Delhi–Etawah (इटवा) stretch is the classic dirty stretch.
15. National Water Policy 2012 puts **drinking first**. PMKSY means har khet plus more crop per drop. **Atal Jal** covers seven states including **Uttar Pradesh** (उत्तर प्रदेश). Ken–Betwa (बेतवा) is the live interlinking project.
16. **Kunchikal** on the **Varahi** is the usual UPPCS **highest** waterfall key. **Nohkalikai** is the tallest **plunge**. **Jog** on the **Sharavati** is famous for **width**, not height.
17. **Shivanasamudra** on the **Cauvery** is the volume king. **Kapildhara** is on the **Narmada** (नर्मदा), not the Godavari (गोदावरी). Sharavati is a **west-flowing** Karnataka river, not a Cauvery tributary.
18. Match pairs: **Dudhsagar–Goa (गोवा) (Mandovi)**; Barkana–Karnataka; Khandadhar–Odisha; Palani–Himachal (हिमाचल) Pradesh.
19. **Hundru** is on the Subarnarekha. **Lodh / Budha Ghagh** is on the **Burha** (Jharkhand's highest), not the Kanchi (कांची). **Dhuandhar** is Narmada at Bhedaghat. **Chitrakote** is on the Indravati.
20. **Chachai–Bihad** is a wrong pair. **Bhalkund** is on the **Bina**, not the Narmada. Other falls: Amritdhara–Hasdeo, Chulia–Chambal (चंबल), Duduma–Machhkund, Gokak–Ghataprabha.
21. **Bhakra–Nangal** is on the **Sutlej** (सतलुज) with **Gobind Sagar** (joint **PB–HR–RJ**). **Tehri** is the tallest rock-fill dam on the **Bhagirathi** (भागीरथी) (+ Bhilangana confluence). **Hirakud** on the **Mahanadi** (महानदी) in Odisha is the longest-earthen narrative (1948–57).
22. **Indira Sagar** (Narmada Sagar) on the Narmada in **Madhya Pradesh** is the volume king (**Harsud** submerged). **Sardar Sarovar** (सरदार सरोवर) is Narmada in **Gujarat** (~163 m, ~1450 MW; beneficiaries GJ–RJ–MH–MP). **Ban**

- Sagar** is on the **Son**, not Narmada. **Idukki** is an arch dam on the **Periyar** (पेरियार). Koyna is linked to reservoir-induced seismicity in **1967**.
23. **Nagarjuna Sagar** (नागार्जुन सागर) and **Srisailem** are both on the **Krishna** (कृष्णा). **Tulbul** is on the Jhelum / Wular. **Baglihar** / **Dulhasti** are on the Chenab (चिनाब). **Pandoh** / **Pong** are on the Beas (ब्यास).
 24. Ramganga project points: length about **825.8 m**, power about **198 MW**, Garhwal (गढ़वाल) location; irrigation is **not** a neat **6 lakh ha**.
 25. **Indira Gandhi (गांधी) Canal** (इंदिरा गांधी नहर) is India's **longest** canal (Harike Barrage; Sutlej–Beas–Ravi (रावी) waters; western Rajasthan). **Gang Canal** (1927, Maharaja (महाराज) Ganga Singh) is among the oldest developed systems. Upper Ganga Canal offtake is at **Bhimgoda / Haridwar (हरिद्वार)**; Lower Ganga Canal at **Narora**.
 26. Himalayan hydropower potential is huge. Eastern Uttar Pradesh and Bihar cannot tap the ranges to their north because those ranges lie in **Nepal**.
 27. India–Bhutan hydro names to fact: **Chukha**, Tala (ताल), Kurichhu, and **Mangdechhu / Mungdechhu**.
 28. **Hokera** is in Jammu & Kashmir, not Punjab. **Sasthamkotta** is Kerala. **Renuka** is Himachal. **Yaya Tso** is a Ladakh (लद्दाख) biodiversity heritage site.
 29. A **dam stores** water; a **barrage diverts** it (**Farakka** 1975 diverts about **40,000 cusec** toward the Hooghly (हुगली) — not a storage dam).
 30. Oxbow lakes are cut-off meanders on floodplains; lagoons are barred coastal water bodies — Chilika is a lagoon, not an oxbow.
 31. Lonar is a **meteorite crater** in Maharashtra basalt, not a volcanic caldera and not a tectonic basin like Wular.
 32. India now has **100 Ramsar** sites; Uttar Pradesh has **13**, with **Surha Tal / JP Narayan Bird Sanctuary, Ballia** as the 100th site.
 33. Irrigation project size by **Culturable Command Area (CCA)**: **minor** up to **2,000 ha** (~**62%** of irrigation potential); **medium 2,000–10,000 ha**; **major** above **10,000 ha** (~**38%** via major + medium together).
 34. **2018–19** source share (provisional): **tubewells ~48.5%**; wells + tubewells together ~**64%**; **canals ~23%**; **tanks ~2.3%**. Wells/tubewells — not canals alone — are the dominant national source.
 35. **Garland Canal** idea = Captain **Dinshaw J. Dastur**. First river-linking thought often tagged to **M. Visvesvaraya**; later **K.L. Rao. Hariyali** (27 Jan **2003**) = watershed / rural water storage. **PMKSY** approved **1 July 2015**.
 36. Peninsula irrigation classic = **tanks / ponds** (hard rock + seasonal rivers). Groundwater resources available for irrigation among major states: **UP** leads (about **40.7 bcm**), then Punjab, Madhya Pradesh, Maharashtra.
 37. **DVC (1948)** is the first multipurpose valley project of independent India; Damodar (दामोदर) = **Sorrow of Bengal**. Tilaiya / Maithon on **Barakar**; Panchet on Damodar.
 38. Oldest hydro = **Sidrapong (Darjeeling, 1897)**; second = **Shivasamudram (Cauvery, 1902)**. Cauvery dispute parties = **TN–KA–KL–Puducherry**.
 39. Chambal chain = **Gandhi Sagar (MP) + Rana Pratap (प्रताप) Sagar + Jawahar Sagar (RJ)**. Telugu Ganga = Krishna water to **Chennai**. Nehru (नेहरू) = **“temples of modern India.”**
 40. **Narmada Bachao (नर्मदा बचाओ) Andolan (नर्मदा बचाओ) / Medha Patkar (मेधा पाटकर)** opposed SSP height. Narmada plan scale ≈ **30 major / 135 medium / 3000 small**.
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41. Multipurpose project logic = flood control + irrigation + power (+ navigation / recreation). **DVC (1948)** is the first multipurpose valley project of independent India.
 42. West-coast kayals / lagoons (Vembanad) are not the same class as inland playa lakes (Sambhar) or tectonic freshwater basins (Wular).
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Must-score facts — lakes, lagoons, Ramsar traps

Lake ↔ type / state

Lake	Lock
Chilika	Largest east-coast lagoon (Odisha)
Sambhar	Largest inland saline (Rajasthan)
Wular	Largest freshwater volume (J&K; Jhelum; tectonic)
Loktak	Phumdis; Keibul Lamjao; Sangai (Manipur)
Pulicat	2nd major brackish lagoon (AP–TN)
Kolleru	Mainly freshwater (AP)
Vembanad	Kerala's largest; India's longest ; west-coast kayal
Kabartal	Bihar oxbow Ramsar
Phulhar	Pilibhit; Gomti source link
Kodaikanal	Artificial

Not in UP (common traps)

Rudrasagar (Tripura) · Sultanpur (Haryana) · Surinsar–Mansar (J&K)

World lake traps

Lake	Lock
Onega / Ladoga	Russia (≠ Canada)
Michigan	Wholly USA
Superior	USA + Canada

Topic 5 — Soils

Consolidated — 42 Must-Score Facts

1. Ancient Indian names: **Urvara** = fertile and **Usara** = sterile. Jenny's five factors are Climate, Living organisms, Relief, Parent rock, and Time (**CLORPT**).
2. **Edaphic** means a **soil** factor — not climate, biotic, or topography. Residual / sedentary soils form on bedrock (regur, red). Transported / azonal soils are the alluvium of the plains.
3. World zonal pairs: **Podzol** = cold temperate woods; **Chernozem** = temperate steppe; **Laterite** = hot humid; **Terra Rossa** = limestone/dolomite weathering; **Sierozem** = mid-latitude desert.
4. Capillary action strongest in **clay** (Clay > silt/loam > sand). Halophytes = saline soils. Contour bunding = **hill slopes**, not desert margins or flood plains.
5. Profile downward is **O → A (topsoil) → E → B → C → R**. **Eluviation** washes material **out** of A/E; **illuviation** deposits it **into B**. Young khadar has weak horizons; mature residual soils show clear horizons.
6. The usual eight-type teaching set is Alluvial, Black / Regur, Red & Yellow, Laterite, Arid, Saline / Usara, Peaty, and Forest / Mountain.
7. Parent material pairs: crystalline rock → red; Cuddapah–Vindhyan → calcareous / clay; Gondwana → immature poor; Deccan Trap → regur; extra-peninsular sediment → alluvium. Climate can override rock — granite becomes laterite if wet, or sandy if arid.
8. **Pedocal** soils accumulate lime in arid tracts such as Rajasthan. Himalayan soils develop better on gentler **north-facing** slopes; many south faces are too steep.
9. **Alluvial** is India's **largest** soil group (about **33.5%–40%**, covering ~11 lakh sq km). It is rich in potash, lime, and phosphoric acid, but **poor in phosphorus, nitrogen, and humus**. Uttar Pradesh is about **90%** alluvial.
10. **Khadar** is new, fine, and fertile. **Bangar** is older, coarser, and has **more kankar**. **Bhur** is sandy ridges of the western Gangetic belt. **Rarh** is older West Bengal / Jharkhand fringe upland.
11. **Bhabar** (about 8–16 km) is pebble fans where streams **disappear**. **Tarai** (about 15–30 km) lies south, where streams **re-emerge**; it is moist and nitrogen-rich but phosphorus-poor.
12. **Black / regur** soils sit mainly on Deccan and Rajmahal Trap (fissure volcanic basalt); Tamil Nadu black may sit on gneiss / schist. Clay is often **≥62%**, montmorillonite is the key clay, and cotton is the crop. Aliases: Tropical Black Earth / Tropical Chernozem. Black soil is **not** Himalayan.
13. Black soil area is about **16.6% / ~5.46 lakh km²**. High moisture retention capacity, sticky when wet, shrinks and develops wide cracks when dry resulting in **"self-ploughing"**. Lacks nitrogen, phosphorus, and organic matter (humus).
14. **Red / yellow** soils form on ancient crystalline and metamorphic rocks (granite, gneiss). Colour comes from **ferric oxides** / iron diffusion; looks yellow in hydrated form. Covers **~8.7 lakh sq km**.
15. **Laterite** was named by **F. Buchanan in 1807** from Latin *later* (brick). Formed under high temperature and heavy rainfall with alternate wet and dry periods (intense leaching). **Rich in iron and aluminium oxides**, poor in nitrogen, lime, phosphorus, and humus. Most widespread in **Kerala (Malabar coast)** followed by Maharashtra.
16. **Desert / arid** soils cover ~29 million ha. Western Rajasthan soils have high **calcium content** (alkaline/saline with calcareous kankar base).
17. **Saline / usara** soils carry sodium, potassium, and magnesium salts. **Gujarat** has the largest saline soil area in India; **Uttar Pradesh** has the largest area of alkaline / sodic wastelands (~2,129 sq km).
18. **Peaty / kari** soils of Kerala (Kottayam–Alappuzha / Kuttanad), plus Sundarbans and north Bihar pockets, grow paddy after rains. Their black colour is **not** Deccan cotton regur.
19. **Forest / mountain** soils cover about **8.67%**. Himalayan soils lack humus due to steep slopes and slow decomposition despite dense forest cover.
20. **Karewa** soils are Kashmir **lacustrine** deposits embedded with moraines, famous for **Zafran (saffron)**, almond, and walnut.

21. Texture: sand drains, clay holds, loam is best for crops. Clay particles are smaller than **0.002 mm**; silt is **0.002 to 0.06 mm**; fine sand is **0.06 to 2.0 mm**.
22. **Micronutrient deficiency**: Among copper, iron, manganese, and zinc, **Zinc (Zn)** is the most deficient micronutrient across Indian soils.
23. Water erosion ladder: **Splash (Raindrop) → Sheet → Rill → Gully → Stream Channel / Ravine**. Gully erosion is responsible for the famous **Chambal Ravines** (badlands), severely affecting **Morena, Bhind, Gwalior, and Shivpuri** in Madhya Pradesh.
24. **Farmland erosion dynamics**: Farmland erodes most rapidly when planted with row crops like **Sorghum (Jowar)**, while cover crops like **Clover** cause minimum soil erosion.
25. **Soil fertility vs crop yield**: Methods like crop rotation, mixed cropping, and multiple cropping replenish soil fertility. **Seed modification** improves crop production/yield but does **not** increase soil fertility.
26. **Agricultural water conservation**: 1. Reduced/zero tillage (prevents moisture loss), 2. Gypsum application (dissolves clay-salt hardpan, enhancing water percolation), 3. Retaining crop residues/mulch (drastically cuts evaporation).
27. **Usar Soil Reclamation**: Usar soils have **pH > 7**, lack bacteria, nitrogen, and zinc. Reclaimed by **Gypsum (CaSO₄·2H₂O)**, pyrites, and green manure (NOT lime). Supports tolerant paddy varieties: **Narendra Shankar Usar Paddy-1 and Usar Paddy-2**.
28. **Soil Acidity & Liming**: Optimal soil pH for crop cultivation is **6.0 to 7.0**. Acidic soils (leached of bases in heavy rainfall areas) are treated with **agricultural lime (CaCO₃ / CaO)**.
29. **Tea soil requirements**: Tea thrives best in **slightly acidic, calcium-free soils** with steep slope drainage (intolerant of waterlogging).
30. **Alluvial sand gradient**: Sand content of alluvial soil **decreases from west to east** across the Indo-Gangetic Plain.
31. **Loamy soil balance**: Standard loam consists of approximately **40% sand, 40% clay, and 20% silt**.
32. **Water retention extremes**: Minimum water retention = **Alluvial sand / sandy soil**; Maximum water retention = **Clay soil**.
33. **Leguminous nitrogen fixation**: Crops like Pea, Soybeans, Alfalfa, Lupins, Peanuts, and **Black Gram (Urd)** fix atmospheric nitrogen via root nodules and restore soil fertility.
34. **Laterite vertical chemical gradient**: In laterite soils, the **upper horizons are more acidic** than the inner/lower horizons.
35. **Excess Iron infertility**: Lateritic soils are rendered relatively infertile by excessive concentrations of iron oxides.
36. **Old Kachhari Clay**: The older alluvium of the Gangetic plain deposited away from flood zones is called **Bhangar** (contains calcareous kankar nodules).
37. **Matasi Soil (Chhattisgarh)**: Red-yellow soil covering **55%–60% of Chhattisgarh**; high in iron oxide; red from ferrous oxide, yellow from ferric oxide; prime soil for paddy.
38. **Soil Leaching Hotspots**: Soil leaching is most severe in **tropical rainforests / high-rainfall monsoon tracts** (Western Ghats, Meghalaya).
39. **Net Sown Area Comparison**: West Bengal has ~60% net sown area, whereas Andhra Pradesh has ~40% (dominated by red and alluvial soils, not laterite).
40. **Mitti Bachao Movement**: Started at **Hoshangabad (Narmadapuram), Madhya Pradesh**. World Soil Day is celebrated on **5 December**.
41. **UP Local Soil Names**: Bundelkhand black = **Mar / Kabar**; Bundelkhand red = **Parua / Rakar**; Vindhyan = **Bhonta / Monta**; Alkaline canal tracts = **Usar / Reh / Kallar / Thur**.
42. **Saline vs Sodic**: Saline soils have high soluble salts; sodic/alkali soils have high exchangeable sodium and pH > 7. Gypsum treats sodic/alkali; lime treats acidic soil.

P

STANDARD SOIL PROFILE & HORIZONS

Key Exam Distinction: Eluviation (E) = Leaching Out | Illuviation (B) = Deposition In



Item	Lock
Profile (down)	O → A → E → B → C → R
Eluviation	Washes out of A/E
Illuviation	Deposits into B
Edaphic	Soil factor
Urvara / Usara	Fertile / sterile (ancient names)
Pedocal	Lime accumulation in arid tracts

So

Soil	Lock
Alluvial	Largest (~ 40%); potash-rich; poor in P & N
Khadar	New, fine, fertile
Bangar	Older; more kankar
Bhur	Sandy doab ridges
Bhabar	Pebble fans; streams disappear (~8–16 km)
Tarai	South of Bhabar; streams reappear (~15–30 km)
Black / regur	Deccan Trap; cotton soil

Soil	Lock
Laterite	High rain leaching; brick-like
Red & yellow	Crystalline parent

Topic 6 — Agriculture

Must-score facts — seasons, crop-state, irrigation

Land-use terms

Term	Lock
NSA	Land sown $\geq$ once a year
GCA	Counts every sowing
Cropping intensity	$(\text{GCA}/\text{NSA}) \times 100$
Irrigation share	Groundwater ~ 64% (tubewell heavy)

Season crops

Season	Lock
Kharif (Jun–Oct)	Rice, maize, millets, cotton, jute, groundnut, soy, tur
Rabi (Oct–Mar)	Wheat, barley, gram, mustard, potato

Crop ↔ leader / need

Crop	Lock
Rice	WB volume often; Punjab yield; hot-wet; clayey alluvium
Wheat	Cool grow / bright ripen; ~50–75 cm; rabi
Jowar / Bajra / Ragi	MH / RJ / KA
Groundnut / Mustard / Soy	GJ / RJ (rabi) / MP (kharif)
Cotton	~21°C+; 50–100 cm; ~210 frost-free days; black soil preferred
Pulses	Rhizobium N-fix; India top producer

Topic 7 — Natural Vegetation & Biodiversity Geography

Consolidated — 36 Must-Score Facts

1. "Vegetation is the true index of climate" is **Thornthwaite (थॉर्नथ्वेट)**'s line, not Köppen (कोपेन)'s letter codes. Controls are climate, soil, relief, biotic factors, and time.
2. India is a **mega-biodiversity** country. The four (चातुर्यम) hotspots touching India are **Himalaya (हिमालय)**, **Western Ghats (पश्चिमी घाट)**–**Sri Lanka**, **Indo-Burma**, and **Sundaland (Nicobar (निकोबार), not Andaman (अंडमान))**. Do not split Himalaya into east and west as if those were the four.
3. Rain ladder: tropical evergreen **>200 cm**; moist deciduous **100–200 cm** (largest area, about **37%**); dry deciduous **70–100 cm** (about **28%**); thorn below about **70–75 cm**. Semi-evergreen is the mix zone. Some notes put wet evergreen above **250 cm** — both figures appear in standard texts.
4. Champion–Seth teaching heads group into five major classes and about **16** sub-types (moist tropical, dry tropical, subtropical montane, temperate montane, alpine).
5. **Dry evergreen** of the Tamil Nadu (ताडु) Coromandel grows on about **100 cm** of **north-east monsoon** with jamun–neem–tamarind. It is not Western Ghat wet rainforest.
6. **Chir pine** belts the western Himalaya at about **1000–2000 m**. Alpine moist belts carry rhododendron–birch; dry alpine juniper sits above about **3500 m**. **Shola** forests are South Indian hill patches.
7. Species diversity (प्रजाति विविधता) **decreases from equator to poles**. Richness is not the same as evenness. Alpha is within one ecosystem (पारिस्थितिकी तंत्र), beta between ecosystems, gamma across a region.
8. India has two realms (Palearctic Himalaya + Indomalayan rest), five biomes, **ten** biogeographic zones, and about **25** provinces.
9. Evergreen forests of the Western Ghats, north-east, and Andamans are multilayered with rosewood, mahogany, and ebony and do **not** shed all leaves in one dry season. Moist deciduous forests (teak, sal, shisham, sandalwood) shed in the dry season and cover the **largest** area.
10. Thorn / scrub of Rajasthan, Gujarat, and rain-shadow tracts holds acacia, babool, ber, and khair (xerophytes). Littoral / mangrove forests of deltas hold **sundari** and **pneumatophores** (वात-मूल) and buffer storms.
11. World vegetation match: **Maquis** = Mediterranean; **Fynbos** = South Africa; **Chaparral** = California; **Matorral** = Chile. Epiphytes peak in equatorial forest; acacia in Sahara; baobab in savanna (सवाना); cedars in Mediterranean.
12. Latest Forest Cover (ISFR 2023 / teaching benchmark for 2025–2026) stands at **21.76%** of India's TGA (Total Forest + Tree Cover = **25.17%**). Top state by forest area is **Madhya Pradesh**; top state by combined green cover gain is **Chhattisgarh**, followed by **Uttar Pradesh (#2 gainer, +559 sq km)**. **Forest cover is not the same as Recorded Forest Area**. National Forest Policy 1988 aims at about **33%**.
13. Uttar Pradesh (उत्तर प्रदेश) latest forest cover is **6.24%** of geographical area (Total Forest + Tree Cover = **9.96%**). In UP, districts above 20% forest cover include **Chandauli (~21.6%)** and **Shravasti (~20.9%)**, not Bahraich (~10.1%). Jaunpur (जौनपुर)–Ghazipur–Ballia have near-nil forest land, yet Ballia holds **Surha Tal** Ramsar. Uttarakhand (उत्तराखंड) forest cover stands at **45.44%** of geographical area (legal RFA is ~**71%**); top forest area is **Pauri Garhwal**, top % is **Nainital**, and lowest is **Udham Singh Nagar**.
14. National Parks are stricter than Wildlife Sanctuaries. Biosphere Reserves have **core + buffer + transition**. India has **18** biosphere reserves; **Nilgiri (1986)** was the first.
15. **In-situ** (स्थान पर) conservation = National Park, Sanctuary, Biosphere Reserve (जैवमंडल आरक्षित क्षेत्र), sacred grove. **Ex-situ** (स्थानांतरित) = zoo, botanical garden (वनस्पति उद्यान), seed bank (बीज बैंक). A botanical garden is **not** in-situ.
16. Chronology fact: Wildlife Protection (वन्यजीव सुरक्षा) Act **1972** → Project Tiger **1973** → Project Elephant (प्रोजेक्ट एलीफेंट) **1992** → Biological Diversity Act **2002**. Project Tiger launched on **1 April 1973**.
17. Park–state pairs: **Simlipal = Odisha** (not Madhya Pradesh); **Nokrek = Meghalaya (मेघालय)** (not Manipur); **Nanda Devi (नंदा देवी) = Uttarakhand (उत्तराखंड)** (not Uttar Pradesh); **Agasthyamalai = Kerala and Tamil Nadu**; Manas = Assam; Keoladeo (केवलादेव) = Rajasthan; Kanha (कान्हा) = Madhya Pradesh.
18. **Dachigam** protects the **Hangul** (Kashmir stag). **Keibul Lamjao** sits on **Loktak** (Manipur) for the Sangai (सांगाई).

19. Uttar Pradesh's **only National Park** is **Dudhwa** (दुधवा) in **Lakhimpur Kheri**. Tiger Reserves are Dudhwa, Pilibhit, Amargarh, and Ranipur — do not invent extra National Parks.
20. Dudhwa Tiger Reserve = Dudhwa National Park (दुधवा राष्ट्रीय उद्यान) + Kishanpur + **Katarniaghat** (Katarniaghat established **1975**). **Chandraprabha** Wildlife Sanctuary is in **Chandauli**.
21. Ramsar traps **not** in Uttar Pradesh: Rudrasagar (Tripura), Sultanpur (Haryana), Surinsar–Mansar (Jammu & Kashmir). Surha Tal / JP Narayan Bird Sanctuary, Ballia is India's **100th Ramsar** and Uttar Pradesh's 13th.
22. **MISHTI** restores mangroves and belongs to Budget **2023–24**, not 2019. Mangrove breathing roots are **pneumatophores**. Sundarbans (सुंदरबन) are the sundari mangrove fact.
23. Coral reefs fact to Andaman & Nicobar, Lakshadweep (लक्षद्वीप), Gulf of Mannar, and Gulf of Kachchh. Coral bleaching (प्रवाल विरंजन) is linked to about **2°C** warming of sea water.
24. **Operation Olivia** is the Coast Guard Olive Ridley programme in Odisha. Nesting season is about **November–May**, not August.
25. Montane vegetation is an altitude elevator: tropical foothills → oak / deodar / chir → alpine → bugyal → snow. Plains rainfall ladders do not replace this altitude logic.
26. Keystone species (कीस्टोन प्रजाति) collapse an ecosystem if lost; flagship species (प्रतीक (फ्लैगशिप) प्रजाति) are public mascots — do not treat them as the same idea.
27. India has about **50,000** rice strains and about **1,000** mango varieties in the usual diversity teaching line.
28. Tea and rubber estates are **plantations**, not climax evergreen forest. Moist and dry deciduous belong to one monsoon-forest family split by rainfall.
29. Tropical evergreen is also called **Selvas**. Typical figures: rain **>200 cm**, mean annual temperature above about **22°C**, humidity near **70%** (many notes write humidity **>77%** and wet evergreen **>250 cm**).
30. **Butea monosperma** is the **flame of the forest**, also called **Dhak** or **Palash**, and is the **state flower of Uttar Pradesh**.
31. **Katha** is obtained from **khair** wood. **Khejri** is the multipurpose “king of the desert” tree used in social forestry (सामाजिक वानिकी).
32. Tree-line is **lower** in the **Western** Himalaya (about **3300–3600 m**) than in the Eastern / Central Himalaya (about **3600–3800 m**). Alpine belts also sit higher in the east because that side is nearer the equator and the sea.
33. **Southern** Himalayan slopes hold **denser** vegetation than northern slopes because they get **more rainfall and sunlight**. Assertion that northern slopes are denser is **false**.
34. **Deodar** is abundant in the **western** Himalayan temperate belt at about **1500–2500 m**. Among common options (deodar, pine/chir, sal, teak), deodar grows at the **highest** elevation.
35. Classic mangrove match: **Achra–Ratnagiri = Maharashtra**; **Coondapur = Karnataka**; **Pichavaram = Tamil Nadu**; **Vembanad = Kerala** (not Karnataka). **Silent Valley (साइलेंट वैली) (Kerala)** = tropical evergreen.
36. **Mahogany** is **not** a Himalayan species (tropical American origin / Indian evergreen belts). Himalayan set = juniper, silver (रूपक) fir, spruce, deodar, birch.

Must-score facts — rain ladder, hotspots, zones

Rain ↔ forest

Forest	Rain / tag
Tropical evergreen	>200 cm
Moist deciduous	100–200 cm (largest area ~37%)
Dry deciduous	Lower rain; teak–sal belts
Dry evergreen (TN Coromandel)	~100 cm NE monsoon (उत्तर-पूर्व मानसून)
Thorn / scrub	Rajasthan, Gujarat, rain-shadow; acacia–babool

Biodiversity tags

Item	Lock
"Vegetation is true index of climate"	Thornthwaite
Hotspots touching India	Himalaya · WG–Sri Lanka · Indo-Burma · Sundaland
Chir pine belt	W. Himalaya ~ 1000–2000 m
Realms	Palearctic (Himalaya) + Indomalayan
Biogeographic zones	About 10 zones

Topic 8 — Minerals, Energy & Industry

Consolidated — 32 Must-Score Facts

1. Most of India's mineral wealth lies **east of the Mangalore–Kanpur (कानपुर) line**. The northern plains are almost empty of major ores. **Chota Nagpur** is the richest mineral belt and is often called India's Ruhr. **Dharwar** rocks are the metallic storehouse; **Gondwana** holds nearly all coal.
2. India is poor in many non-ferrous metals but rich in **bauxite**. Mica's classic dielectric belt is **Koderma–Gaya (गया)–Hazaribagh** in Jharkhand.
3. As per the Ministry of Mines Annual Report **2021–22**, the leading iron-ore states are **Odisha and Chhattisgarh**. Jharkhand and Karnataka do not always lead.
4. Iron-ore mine–state pairs are **Gurumahisani (Odisha)**, **Bailadila (Chhattisgarh)**, **Noamundi (Jharkhand)**, and **Kudremukh (Karnataka)**. Mount Newman in Australia is famous for **iron**, not manganese.
5. The four (चातुर्यम) main iron belts are the Odisha–Jharkhand belt, the Durg–Bastar–Chandrapur belt, the Ballari–Chitradurga belt, and the Maharashtra–Goa (गोवा) belt.
6. Manganese leaders include **Odisha, Maharashtra and Madhya Pradesh**. Postmasburg in South Africa is a manganese centre. **Bhilwara** is mica and zinc (Rampura-Agucha), not manganese.
7. Bauxite feeds aluminium and often sits with **laterite (लेटराइट)**. Odisha leads (Panchpatmali). **MALCO** is at **Mettur, Tamil Nadu (नाडु)**, not Chennai. **HINDALCO** is at **Renukoot, Uttar Pradesh (उत्तर प्रदेश)**. **BALCO** is at Korba; **NALCO** at Damanjodi/Angul.
8. Copper centres are **Khetri (Rajasthan)**, Singhbhum (Jharkhand) and Malanjkhand (Madhya Pradesh). **Panna (Madhya Pradesh)** is diamond. **Sukinda (Odisha)** is chromite. **Zawar** and **Rampura-Agucha** (Bhilwara) are zinc–lead.
9. Uranium's classic mine is **Jaduguda, Jharkhand**. Thorium and monazite sands line the **Kerala–Tamil Nadu** beaches. Lithium finds at **Reasi, Jammu & Kashmir** matter, but India still imports much Li, Ni and Co.
10. About **97%** of Indian coal is **Gondwana** coal in the Damodar (दामोदर)–Son–Mahanadi (महानदी)–Godavari (गोदावरी) valleys. Tertiary coal is mainly in the North-East. Rank rises **peat (<40% C) → lignite (40–55%) → bituminous (60–80%) → anthracite (80–95%)**. Anthracite is scarce (J&K pockets); lignite = **Neyveli**; ~80% of deposits are bituminous **non-coking**.
11. Coalfield centres: **Raniganj (West Bengal)**, **Jharia (Jharkhand, coking)**, Ramgarh (Jharkhand), **Talcher (Odisha)**. **Maolong** coal is in **Meghalaya (मेघालय)**, not Arunachal. **Neyveli** is lignite in Tamil Nadu.
12. Valley–field pairs: Talcher–Mahanadi, Karanpura–Damodar, Singrauli–Son, Singareni–Godavari. **Singrauli** is the oldest large NTPC hub and sits on the **Madhya Pradesh–Uttar Pradesh** border, not northern Chhattisgarh.
13. World coal traps: Ruhr is **Germany**; Donetsk is **Ukraine**; Appalachian is USA; Kuzbass is Russia. **Pegu Yoma (Myanmar)** is mineral oil, not tin. **Kashagan** oil is in **Kazakhstan**, not Kuwait.
14. **Methane** is the main gas in natural gas and **CNG**. **LPG** is propane–butane and is not the same as CNG. The **Black Revolution** means petroleum, not oilseeds.
15. Oil–gas hubs include Digboi/Naharkatiya (Assam), Mumbai High, Ankleshwar (Gujarat), Barmer (Rajasthan) and the Krishna (कृष्णा)–Godavari / Cauvery basins. Digboi is among India's oldest refineries.
16. Refinery–state pairs: Nayara (Gujarat), Manali (Tamil Nadu), Numaligarh (Assam), Tatipaka (Andhra Pradesh), **Mathura (मथुरा) (Uttar Pradesh)**, Barauni (Bihar), Panipat (पानीपत) (Haryana), Jamnagar (Gujarat), Paradip (Odisha), Bina (Madhya Pradesh), Haldia (West Bengal).
17. Conventional energy (पारंपरिक ऊर्जा स्रोत) means coal, oil, gas and nuclear. Non-conventional means solar, wind, tidal, biogas (बायोगैस) and geothermal. Nuclear and geothermal energy (भू-तापीय ऊर्जा) are **not** stored solar energy.
18. Nuclear plant centres (प्रमुख परमाणु ऊर्जा केंद्र): **Tarapur / तारापुर (Maharashtra / महाराष्ट्र)** [first commercial / प्रथम व्यावसायिक, 1969], **Kakrapar / काकरापार (Gujarat / गुजरात)**, **Kudankulam / कुडनकुलम (Tamil Nadu / तमिलनाडु)**, **Kaiga / कैगा (Karnataka / कर्नाटक)**, **Rawatbhata / रावतभाटा (Rajasthan / राजस्थान)**, **Kalpakkam /**

कल्पक्कम (Tamil Nadu / तमिलनाडु), Narora / नरोरा (Uttar Pradesh / उत्तर प्रदेश - बुलंदशहर). Shivanasamudra (शिवणसमुद्र / शिवसमुद्रम) is a **hydel plant** (जलविद्युत परियोजना - कावेरी नदी), not nuclear (परमाणु नहीं).

19. Solar points: electricity from **PV cells**; **Diu** was the first Union Territory to go fully solar-powered; **Cochin** was the first solar airport. The **International Solar Alliance** headquarters is at **Gurugram**, not New Delhi or Paris.
 20. Wind leaders include Tamil Nadu (Muppandal), Gujarat (Kutch) and Rajasthan (Jaisalmer). Tidal potential is higher in the **Gulf of Khambhat** than in Kutch.
 21. The first cotton-mill cluster grew in **Maharashtra–Gujarat** on cotton, humidity and Mumbai port (बंदरगाह). **Mumbai** was traditionally the largest mill centre; Ahmedabad (अहमदाबाद) is in the belt but was not traditionally the largest. Coimbatore is a Tamil Nadu spinning hub; Kanpur is an Uttar Pradesh mill city.
 22. Steel plant centres: **Bhilai (Chhattisgarh)**, **Rourkela (Odisha)**, **Bokaro (Jharkhand)**, **Durgapur (West Bengal)**, Vizag (Andhra Pradesh), Salem (Tamil Nadu), Jamshedpur (Jharkhand). Chotanagpur (छोटानागपुर) is the mineral-heavy industrial heartland.
 23. Agency headquarters: ONGC Delhi, MECL Nagpur, NMDC Hyderabad (हैदराबाद), NALCO Bhubaneswar.
 24. The **Delhi–Mumbai Industrial Corridor (DMIC)** and the **Eastern and Western Dedicated Freight Corridors** are the main corridor schemes. SEZs are export-oriented zones.
 25. In Uttar Pradesh, **Sonbhadra** (सोनभद्र) is the energy capital. **Mathura** (मथुरा) has the major refinery. **Narora** is the nuclear plant. Obra, Anpara and Rihand (रिहंद) are thermal hubs.
 26. Shankargarh silica feeds the **Firozabad** glass industry. The **Chunar (चुनार)–Churk–Dalla** belt is cement. **HINDALCO Renukoot** is aluminium. Sonrai (Lalitpur) has a copper occurrence.
 27. Uttar Pradesh city industries: Kanpur textile and leather, Moradabad brass, Aligarh (अलीगढ़) padlocks and hardware, Firozabad glass, Meerut (मेरठ) sports goods, Noida–Ghaziabad NCR engineering.
 28. Uttar Pradesh coal is mostly **non-coking** Singrauli–Sonbhadra coal. It is not Jharia-type coking coal.
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29. Atomic minerals: **uranium** classic mine = **Jaduguda (Jharkhand)**; **thorium / monazite** sands = **Kerala–Tamil Nadu** beaches. Lithium finds at **Reasi (J&K)** matter, but India still imports much Li, Ni and Co.
 30. Petroleum basins to keep: Assam (Digboi / Naharkatiya), Mumbai High, Ankleshwar (Gujarat), Barmer (Rajasthan), Krishna–Godavari and Cauvery. Digboi is among India's oldest refineries.
 31. Power mix: thermal dominates generation share in the usual teaching line; hydro is strong in Himalayan / Western Ghat pockets; nuclear plants sit where cooling water and security allow (Tarapur, Kakrapar, Kudankulam, Kaiga, Rawatbhata, Kalpakkam, Narora).
 32. Industrial region tags: Chotanagpur = mineral-heavy heartland; Mumbai–Pune = cotton / engineering; Ahmedabad–Vadodara = textile / petrochem; Hugli = jute / engineering; Coimbatore = spinning.
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Must-score facts — ore belts, mine–state, energy

Frame

Item	Lock
Mineral wealth map	Mostly east of Mangalore–Kanpur line
Non-ferrous	Poor many metals; rich in bauxite
Mica belt	Koderma–Gaya–Hazaribagh

Iron / Mn / bauxite

Item	Lock
Iron leaders (2021–22 teaching)	Odisha (ओडिशा), Chhattisgarh
Gurumahisani	Odisha

Item	Lock
Bailadila	Chhattisgarh
Noamundi	Jharkhand
Manganese leaders	Odisha, MH, MP
Bauxite	Feeds aluminium; laterite link

Energy / industry tags

Item	Lock
Coal belts	Damodar, Mahanadi, Godavari, Son
Petroleum	Mumbai High; Assam; Gujarat
Atomic minerals	Beach sands (monazite) South India

Topic 9 — Transport & Communication

Consolidated — 31 Must-Score Facts

1. Transport moves people and goods; communication moves information. About **95%** of India's trade by **volume** moves by sea.
2. India's first passenger train ran **Mumbai–Thane in 1853**. Indian Railways is organised into about **18 zones** (older books still say 16).
3. Zone–HQ pairs: **Northern Railway — New Delhi**; **North Eastern Railway — Gorakhpur (गोरखपुर)**; **South Eastern Railway — Garden Reach, Kolkata** (not Cuttack); **East Central Railway — Hajipur**; **North Central Railway — Prayagraj (प्रयागराज)**; **South Western Railway — Hubballi**; **East Coast Railway — Bhubaneswar**.
4. The **Western Dedicated Freight Corridor** runs **Dadri–JNPT**. The **Eastern Dedicated Freight Corridor** runs **Ludhiana–Dankuni**.
5. **Amrit Bharat** is a **station modernisation** programme, not a new-lines scheme. About **1,275** stations were identified all-India; in **February 2023**, **149 Uttar Pradesh (उत्तर प्रदेश) stations** were listed.
6. The new **Pamban** bridge is an **RVNL** project and India's first **vertical-lift** railway sea bridge. Hubballi holds both the **South Western Railway HQ** and the **August 2020** railway museum.
7. National Highways are only about **2%** of road length but carry about **40%** of traffic. Rural roads are most of the length and are linked to **PMGSY**.
8. The **Golden Quadrilateral** joins Delhi–Mumbai–Chennai–Kolkata (about **5,846 km**). North–South is Srinagar–Kanyakumari (कन्याकुमारी); East–West is Porbandar–**Silchar**. They meet at **Jhansi (झांसी)**.
9. **Bharatmala** is a **highway** programme. **Sagarmala** is **port-led**. Do not swap the two. **PMGSY (2000)** = rural all-weather roads; **PMBJP** = highway BOT corridors.
10. The **Atal Tunnel** runs under **Rohtang (रोहतांग)** in the **Pir (पीर) Panjal (पीर पंजाल)** (Himachal (हिमाचल) / BRO). A bare claim that it is the “world's longest highway tunnel” is not the safe line.
11. The built **Purvanchal (पूर्वांचल) Expressway** alignment is Lucknow (लखनऊ)–Azamgarh–Mau–Ghazipur–Ballia. It **passes Mau**; Basti is off the main spine.
12. India has **12 major ports** plus many non-major ports. **IWAI** dates from **1986**. The 2016 Act notified **111 National Waterways**.
13. National Waterway pairs: **NW-1** = Ganga (गंगा), Haldia–Prayagraj; **NW-2** = Brahmaputra (ब्रह्मपुत्र), Dhubri–Sadiya; **NW-3** = Kerala West Coast Canal; **NW-4** = Godavari (गोदावरी)–Krishna (कृष्णा) / Kakinada–Puducherry; **NW-5** = Brahmani–Mahanadi (महानदी)–East Coast Canal.
14. A July 2020 containership link from Kolkata toward Agartala used the **Chattogram** route.
15. Port (बंदरगाह)–state pairs: **Paradip (Odisha)**, **Tuticorin / VOC (Tamil Nadu (तामिल)), Kakinada (Andhra Pradesh)**, **Alappuzha (Kerala)**. **JNPT** is the container flagship; **Kolkata–Haldia** is riverine.
16. A **harbour** is sheltered water; a **port** is harbour plus cargo facilities. **Igarka** is in **Russia**, not China.
17. **Kushinagar (कुशीनगर)** became an international airport on **24 June 2020**. **Cochin** was India's first solar-powered airport. **Jewar** is Noida International Airport.
18. FLY91's inaugural flight (18 March 2024) was from **Manohar International Airport** (Goa Mopa).
19. The Motihari–**Amlekhganj** (Nepal) petroleum pipeline was inaugurated in **September 2019**.
20. **PM Gati Shakti** was launched on **13 October 2021** with the “seven engines” model. The **National Logistics Policy** (September 2022) complements it. The **Blue Economy** means ocean-based growth.
21. Optical-fibre communication uses **total internal reflection (TIR)**, not simple refraction.
22. Uttar Pradesh has **no major seaport**. Its rail HQ is **NER Gorakhpur**; its waterway fact is **NW-1** on the Ganga.
23. NHA was set up in **1995**. Expressways are access-controlled highways, distinct from ordinary National Highways.
24. West-coast major ports include Kandla (Deendayal), Mumbai, JNPT, Mormugao, New Mangalore and Cochin. East-coast majors include Kolkata–Haldia, Paradip, Vizag, Chennai, Ennore (Kamarajar) and Tuticorin.

25. **Incredible India** is the MoT tourism brand; **ITDC** dates from **1966**. Match hill stations carefully: Chakrata–UK, Haflong–Assam, Kalimpong–WB, Kufri–HP.
26. **City of Lakes = Udaipur (उदयपुर)** (not Jaisalmer). **Pink City = Jaipur**. **Sabarimala = Kerala**. UNESCO tourist set includes Champaner–Pavagarh, CST Mumbai, Mamallapuram and Konark (कोणार्क).
27. **Srisailem = Nallamala**; Omkareshwar (ओंकारेश्वर) and Pushkar are **not** on Satmala / Mahadeo in the IAS 2015 trap. **Biligiriranga** holds the standing Ranganatha note in Karnataka.

28. Railway track gauges in teaching: **broad (1676 mm)**, metre and narrow. Most Indian main lines are broad gauge after unification drives.
29. **Konkan Railway** links Maharashtra–Goa–Karnataka–Kerala along the west coast. **HBJ / HVJ** (Hazira–Bijaipur–Jagdishpur) is a classic gas-pipeline spine in coaching notes.
30. **Deendayal (Kandla)** is a tidal / gulf port on the west; **Ennore / Kamarajar** is a major east-coast port near Chennai. Do not swap west- and east-coast port lists.
31. Communication basics: postal / telecom move information; optical fibre uses **TIR**. Do not treat transport and communication as the same word.

Must-score facts — rail, road, ports, waterways

Rail / freight

Item	Lock
First passenger train	Mumbai–Thane 1853
Zones	~ 18 (older books: 16)
Northern / NER / SER HQ	New Delhi / Gorakhpur (गोरखपुर) / Garden Reach, Kolkata
Western DFC	Dadri–JNPT
Eastern DFC	Ludhiana–Dankuni
Amrit Bharat	Station modernisation (≠ new lines); ~ 149 UP stations
New Pamban	RVNL; first vertical-lift railway sea bridge

Road / schemes

Item	Lock
Golden Quadrilateral	Delhi–Mumbai–Chennai–Kolkata (~5846 km)
NS / EW	Srinagar–Kanyakumari / Porbandar–Silchar
Bharatmala vs Sagarmala	Highways vs port-led
Bharatmala top constructed state (31 Oct 2024)	Rajasthan (~2,241 km)
PMGSY	Rural all-weather (2000)
Atal Tunnel	Under Rohtang (Pir Panjal; BRO)
Gati Shakti	13 Oct 2021 ; NLP Sep 2022

Ports / waterways / air

Item	Lock
Sea trade volume	~ 95% of India's trade volume
Major ports	12 + many non-major
NW-1 / NW-2 / NW-3	Ganga Haldia–Prayagraj / Brahmaputra / Kerala W. Coast Canal

Item	Lock
Paradip / Tuticorin / Kakinada	Odisha / TN / AP
Harbour vs port	Sheltered water vs harbour + cargo facilities
Kushinagar intl	24 Jun 2020 ; Jewar = Noida greenfield
UP	No major seaport ; rail HQ NER Gorakhpur; NW-1

Topic 10 — Tribes & Institutions

Consolidated — 28 Must-Score Facts

1. Scheduled Tribe (अनुसूचित जनजाति) lists are **state-wise** under **Article 342**. Parliament can amend the lists by law. A community listed in one State need not be listed in another.
2. STs are about **8.6%** of India's population (जनसंख्या) (Census 2011). Among the largest groups are **Bhil, Gond (गोंड) and Santhal (संथाल)**. Madhya Pradesh often leads in absolute (निरपेक्ष) ST population.
3. Lokur Committee criteria for ST listing include primitive traits, distinctive culture, geographical isolation, shyness of contact, and economic backwardness.
4. There are **75 Particularly Vulnerable Tribal Groups (PVTGs)** (Dhebar PTG renamed PVTG in 2006). **Odisha (ओडिशा)** has the most PVTG communities. **PM-JANMAN** is the main outreach mission.
5. PVTG criteria stress pre-agricultural technology, stagnant or declining population, extremely low literacy, and a subsistence economy.
6. The **Fifth Schedule** (5वीं अनुसूची) covers Scheduled Areas in many States. The **Sixth Schedule** (6वीं अनुसूची) covers autonomous councils in **Assam, Meghalaya (मेघालय), Tripura and Mizoram** only.
7. State-tribe pairs: **Angami / Rengma = Nagaland**; **Toda / Paliyan = Tamil Nadu (नाडु)**; **Birhor = Jharkhand**; **Khasi (खासी) = Meghalaya** (not Arunachal); **Yanadi = Andhra Pradesh** (not Rajasthan); **Chenchu = Andhra / Telangana**.
8. **Kuki** is a North-East tribe and is **not** a Uttar Pradesh Scheduled Tribe. **Keria / Kharia** fact to Odisha (also Jharkhand for Kharia). **Bodo = Assam**; **Lepcha = Sikkim**; **Meena = Rajasthan**; **Warli (वारली) = Maharashtra**; **Apatani = Arunachal**.
9. **Khasi and Garo (गारो)** of Meghalaya are classic **matrilineal** societies.
10. Uttar Pradesh has **15** notified ST entries. The classic five are **Tharu, Bhotia, Buksa, Jaunsari and Raji**. UP PVTGs are **Buksa and Raji**.
11. In Uttar Pradesh, **Tharu** live in the **Terai (तराई)**; **Buksa** on the Bijnor fringe; **Saharya** in **Lalitpur**; **Agariya** are linked to **iron smelting** in the Sonbhadra (सोनभद्र)–Mirzapur (मिर्जापुर) belt.
12. After the 2003 Act, many UP ST notifications are **district-limited**. Two clusters matter: Terai north versus Vindhyan / Sonbhadra south.
13. Central Indian heartland tribes include **Gond (गोंड)** (with Maria/Muria), **Bhil**, Baiga, Kol, Korku, Sahariya, Halba, Warli and Meena.
14. Jharkhand core tribes include **Santhal, Munda, Oraon**, Ho, Kharia, Birhor, Bhumij, Asur and **Birjia**. Birjia is **not** an Andaman tribe. Santhali uses the **Oi Chiki** script (लिपि); **Birsa Munda** and the **Hul (हूल) of 1855** are pairs.
15. **Janjatiya Gaurav Diwas** is observed on **15 November** (Birsa Munda's birth anniversary).
16. Andaman **Negrito** natives are Great Andamanese, **Onge, Jarawa** and **Sentinelese** (North Sentinel Island). Nicobar (निकोबार) **Mongoloid** groups are Nicobarese and **Shompen** (Great Nicobar PVTG). **Rengma** is not an island tribe.
17. World habitat pairs: **Bushman = Kalahari**; **Eskimo = northern Canada**; **Pygmy = Congo**; **Masai = East Africa** (not West Africa); **Ainu = Japan**; **Punan = Borneo**; Lapps/Sami = Sweden–Finland.
18. Also: Semang = **Malaysia** (not Indonesia); Bedouin = Arabia; Kirghiz = Central Asia (not Sudan); Maori = New Zealand; Zulu = South Africa; Bantu ≠ Sahara. **Transhumance** = seasonal valley–mountain herding.
19. Gypsies' original home = **India**. Africa = cradle of mankind. Cro-Magnon is the most recent among classic fossil-human options.
20. **Khirghiz** are Central Asian, not Sudanese. **Koryak** live in north-east Siberia, not Alaska.
21. **IMD** headquarters is New Delhi (**1875**). **ICAR** headquarters is New Delhi (**1929**). **IARI** is at **Pusa, New Delhi** — it is not the apex ICAR council itself.
22. **FRI** and **ICFRE** are at **Dehradun (देहरादून)**. **WII** is also Dehradun but is not the FRI answer. **NDRI** is at **Karnal** (dairy lab). **IVRI** is at **Izatnagar, Bareilly (UP)**. **NDDB** is at **Anand** and ran **Operation Flood** (ऑपरेशन फ्लड) — it

is not NDRI.

23. **IIHR** is at Bengaluru. **CISH** is at Lucknow (लखनऊ). **CSAUAT** is at Kanpur (कानपुर). **Pantnagar** is remembered as the first State Agricultural University tradition.
24. The **Forest Rights Act** is **2006**. **Adi Karmayogi** (June 2025) is a **Ministry of Tribal Affairs** programme.
25. Main tribal belts are the North-Eastern hills, the central Indian plateau, western Bhil country, and the Andaman–Nicobar (अंडमान) isolates.
26. Krishi Vigyan Kendras (KVKs) are the frontline **extension** arms under the ICAR system.

27. World habitat extras: **Masai** = East Africa (not West Africa); **Ainu** = Japan; **Punan** = Borneo; Lapps / Sami = Sweden–Finland; **Maori** = New Zealand; **Zulu** = South Africa.
28. Gypsies' original home in the usual teaching line = **India**. Transhumance = seasonal valley–mountain herding. Cro-Magnon is the most recent among classic fossil-human options.

Must-score facts — tribe↔state, PVTG, institutes

Frame

Item	Lock
ST lists	State-wise Art. 342
ST share (2011)	~ 8.6% ; largest groups Bhil, Gond, Santhal
PVTGs	75 ; most in Odisha ; PM-JANMAN outreach
5th vs 6th Schedule (6वीं अनुसूची)	Many States' Scheduled Areas vs Assam–Meghalaya–Tripura–Mizoram councils

Tribe ↔ state

Tribe	Lock
Angami / Rengma	Nagaland
Toda / Paliyan	Tamil Nadu
Birhor	Jharkhand
Khasi / Garo	Meghalaya (मेघालय) (matrilineal)
Kuki	NE (not UP ST)
UP classic five	Tharu, Bhotia, Buksa, Jaunsari, Raji (15 notified)
UP PVTGs	Buksa + Raji
Tharu / Buksa / Saharya	Terai / Bijnor fringe / Lalitpur
Jharkhand core	Santhal, Munda, Oraon, Ho...
Andaman Negrito	Great Andamanese, Onge, Jarawa, Sentinelese

World habitats / institutes

Pair	Lock
Bushman / Eskimo / Pygmy / Masai	Kalahari / N. Canada / Congo / East Africa
IMD / ICAR HQ	New Delhi (1875 / 1929)
IARI	Pusa, New Delhi
FRI / ICFRE / WII	Dehradun (WII ≠ FRI answer)
NDRI / IVRI	Karnal / Izatnagar teaching
CISH	Lucknow (लखनऊ); CSAUAT Kanpur (कानपुर)

Pair	Lock
FRA	2006
Janjatiya Gaurav Diwas	15 Nov (Birsa birth)

Topic 11 — Population Geography

Consolidated — 32 Must-Score Facts

1. India's first non-synchronous census was in **1872** (some papers key **1871**). The first **synchronous** all-India census was in **1881**.
2. **1921** is the **Great Divide** year of Indian census history (इतिहास). **2011** was the **15th** census and the **7th** after Independence.
3. Census 2011 India population was about **121.09 crore**. Decadal growth **2001–11** was **17.64%**. India holds roughly one-sixth of world population in that count.
4. Arithmetic density is **population / total area**. India's 2011 arithmetic density is **382** persons per km².
5. Physiological density is **population / net sown area**. Agricultural density is **agricultural population / net sown area**. Do not swap these three density types.
6. Among States, **Bihar** has the highest density (**1106**). West Bengal and Kerala follow high; Uttar Pradesh (उत्तर प्रदेश) is **829**. **Arunachal Pradesh** has the lowest State density (**17**). Delhi UT is very dense but is not a "lowest density State (स्यूनतम जनघनत्व)" answer.
7. Core 2011 figures: sex ratio **943**, child sex ratio **919**, literacy **74.04%** (age **7+**), urban (नगरीय) share **31.16%**. Scheduled Castes about **16.6%**; Scheduled Tribes about **8.6%**.
8. **Uttar Pradesh** (उत्तर प्रदेश) is the most populous State and has the largest rural population. **Sikkim** is the least populous State. **Nagaland** showed negative growth in 2001–11.
9. Highest State sex ratio is **Kerala**. Among States in the usual 2011 set, lowest sex ratio is **Haryana**. Lowest child sex ratio (rural + urban) is also **Haryana**.
10. Literacy is highest in **Kerala** and lowest among States in **Bihar**. In Uttar Pradesh, **Shrawasti** has the lowest female literacy among districts.
11. Absolute (निरपेक्ष) growth is $P_2 - P_1$. Growth rate is the percentage change. **Natural growth = CBR – CDR**. Induced change comes from **migration**.
12. Replacement-level **TFR is 2.1** children per woman, not "per thousand" (that wording belongs to birth/death rates). NFHS-4 (2015–16) reported **2.2**. NFHS-5 reports about **2.0**.
13. The National Population Policy **2000** aimed at population stability by **2045**. World Population Day is **11 July**. The World Population Report is associated with **UNFPA**.
14. The demographic dividend window is the large share of working ages **15–59**, not 60+ or 0–6. Dependency compares young plus aged with workers.
15. Urbanisation acceleration in the classic curve is linked to the **second stage** of demographic transition. Among religions, **Jains** are the most urbanised.
16. Demographic Transition Theory is linked to **Thompson** (with **Notestein**). Optimum population is linked to **Edwin Cannan**. Social mal-adjustment is linked to **Henry George**. Malthus argued population grows **geometrically** while food grows **arithmetically**. The Malthusian Theory is a theory of **population**.
17. The largest internal migration stream is **rural → rural**. Female migration is often for marriage; male migration is often for work. **Immigration** means in-movement; **emigration** means out-movement.
18. Push factors drive people from the origin; pull factors attract them to the destination. Out-migration sources are often Uttar Pradesh–Bihar; destinations are often Maharashtra–Delhi–Gujarat.
19. A **census town** needs population $\geq 5,000$, density $\geq 400/\text{km}^2$, and $\geq 75\%$ of **male main workers** in non-agriculture.
20. Census 2011 listed **53** million-plus urban agglomerations. Kanpur (कानपुर) crossed the million mark in **1971**; Lucknow (लखनऊ) in **1981**. 2011 UP UA order is **Kanpur > Lucknow > Ghaziabad > Agra**.
21. Uttar Pradesh district facts (2011): **Prayagraj** (प्रयागराज) most populous; **Ghaziabad** densest; **Jaunpur** (जौनपुर) among the higher sex-ratio districts; **Shrawasti** lowest female literacy.

22. Uttar Pradesh holds about **16.5%** of India's population. Rural share is about **77.7%** against India's **68.84%**. Urban share is about **22.3%** against India's **31.16%**. State sex ratio is **912** against India's **943**. Literacy is about **67.7%** against India's **74.04%**.
23. Crowded belts are the Ganga (गंगा) plain and coasts. Sparse belts are the Himalaya (हिमालय), North-East hills, Thar (थार) and dry interior pockets. Distribution factors include terrain, climate, soil, water, minerals, industry, transport and history.
24. Keep **Census 2011** numbers for Prelims until **Census 2027** results replace them. UN estimates (प्राकलन समिति) that India became the most populous country around **2023** do not rewrite the 2011 tables.
25. A broad-base population pyramid signals high fertility. Ageing shows earlier in southern States than in the high-fertility northern belt.
26. Occupational structure is taught as primary / secondary / tertiary. Primary still takes a large share of India's workforce, with a slow shift toward secondary and tertiary as development proceeds.
27. Malthus's **positive** checks raise the death rate (famine, disease, war). **Preventive** checks lower the birth rate (delayed marriage, moral restraint). **Karl Marx** treated surplus population as a product of capitalism, not a natural food-arithmetic law. **Ester Boserup** argued that population growth can push agricultural intensification.
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28. **CBR** and **CDR** are expressed **per thousand** of population; **TFR** is children **per woman**. Do not mix the units.
29. Demographic Transition stages: high birth–high death → falling death (population boom) → falling birth → low birth–low death. Urbanisation acceleration is classically linked to the **second** stage.
30. Age structure teaching bands are often **0–14**, **15–59** (workers) and **60+**. A broad-base pyramid = high fertility; a top-heavy pyramid = ageing.
31. Internal migration streams include rural→rural (largest), rural→urban, urban→urban and urban→rural. Inter-state migrants often move from UP–Bihar toward Maharashtra–Delhi–Gujarat.
32. Literacy in Census 2011 is counted for age **7+**. Effective literacy and crude literacy wordings must not be swapped with school-enrolment rates.
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Must-score facts — density, growth, Census tags

Terms

Term	Lock
Density	Population ÷ area
Physiological density	Population ÷ arable land
Agricultural density	Farmers ÷ arable land
Sex ratio	Females / 1000 males
Literacy	7+ who can read & write with understanding

India Census 2011 spine (teaching)

Item	Lock
Highest density state (सर्वाधिक घनत्व) (usual)	Bihar
Lowest density (large state)	Arunachal teaching
Highest literacy (usual)	Kerala
Phase model	High stationary → early expanding → late expanding → low stationary

Topic 12 — Human Geography (Settlements)

Consolidated — 30 Must-Score Facts

- Human geography studies the **man–environment (पर्यावरण) relationship** and the spatial patterns of human life, settlements and economy.
- Environmental determinism** (Ratzel / Semple) says nature controls culture. **Possibilism** (Vidal de la Blache) says humans choose among nature's options. **Neo-determinism** (Griffith Taylor) is stop-and-go determinism.
- Site** is the local ground of a settlement. **Situation** is its wider regional location relative (सापेक्ष) to routes, resources and other places.
- A **census town** needs population $\geq 5,000$, density $\geq 400/\text{km}^2$, and $\geq 75\%$ of **male main workers** in non-agriculture. It need not be a municipality.
- Rural settlements lean on primary activities and lower density. Urban places are statutory towns or census towns.
- Village types are **clustered / nucleated, semi-clustered, hamleted** and **dispersed**. Clustered villages appear on plains, around Rajasthan water points, and for defence in Bundelkhand / Nagaland — do **not** call the Thar (थार) “dispersed by default.”
- Dispersed villages typify **Meghalaya (मेघालय), Uttarakhand (उत्तराखंड), Himachal (हिमाचल) Pradesh, Kerala** and many North-East forest–hill tracts. Hamlet local names include **panna, para, palli, nagla and dhani**.
- Semi-clustered villages often show a dominant-caste centre with lower strata on the flanks (Gujarat / Rajasthan pair). Linear villages follow a road, river, canal or coast.
- Town functions are classed by the **dominant** job: administrative, industrial, transport, commercial, mining, garrison, educational, religious or tourist.
- Class I towns have ≥ 1 **lakh** people (468 towns in 2011 held about **60%** of urban population). Classes II–VI step down from 50–99 thousand to under 5 thousand.
- In the standard Indian size ladder, metropolitan cities are **10 lakh–50 lakh** and mega cities are **above 50 lakh — six** in 2011. A UN megacity is ≥ 1 **crore**. Do not mix these three thresholds.
- 2011 million-plus UA rank starts **Mumbai > Delhi > Kolkata > Chennai > Bengaluru > Hyderabad (हैदराबाद)**. There were **53** million-plus UAs; the smallest in that set is often **Kota**. The **2001** order Mumbai > Kolkata > Delhi is a trap for 2011 stems.
- An urban agglomeration can be a town with outgrowths, two contiguous towns, or a city with adjoining towns and outgrowths. Outgrowths include railway colonies, campuses, ports and cantonments.
- Settlement evolution pairs: ancient **Varanasi (वाराणसी) / Prayag (प्रयाग) / Madurai (मदुरै)**; medieval **Delhi / Agra / Jaipur / Lucknow (लखनऊ)**; modern planned **Chandigarh** (Le Corbusier). Satellite towns include Ghaziabad.
- The **Smart Cities Mission** launched on **25 June 2015** for **100** cities under **MoHUA**, implemented through an SPV. ABD means area-based development (retrofit / redevelopment / greenfield) plus pan-city ICT.
- Uttar Pradesh (उत्तर प्रदेश)'s **Central / Mission** Smart City list is **10**: Lucknow, Kanpur (कानपुर), Prayagraj (प्रयागराज), Agra, Varanasi, Aligarh (अलीगढ़), Bareilly, Jhansi (झांसी), Moradabad and Saharanpur. **Ghaziabad is not** in that Central-10.
- Uttar Pradesh's **State Smart Cities (2019)** are seven Nagar Nigams (including Ghaziabad, Meerut (मेरठ), Gorakhpur (गोरखपुर), Mathura (मथुरा), Ayodhya (अयोध्या), Firozabad, Shahjahanpur (शाहजहाँपुर)). State list (राज्य सूची) does not rewrite Mission-100 keys.
- ISAC-2020 theme pairs: Culture **Indore**, Governance **Vadodara**, Social **Tirupati**, Urban environment **Bhopal**. Best State award went to **Uttar Pradesh** (उत्तर प्रदेश); best cities included Indore and Surat.
- HRIDAY** covers **12** heritage cities. In Uttar Pradesh the pair is **Varanasi and Mathura** — not Prayagraj or Ayodhya.
- SPMRM (Rurban)** was **launched on 21 February 2016** (Cabinet approval 2015). Matching the scheme to **2015** is the classic trap. It is under **MoRD**, not MoHUA.

21. **Rurbanization** is linked to sociologist **G.S. Ghurye**. McLuhan's **Global Village** rests on **transport plus communication**.
22. Scheme chronology: **JNNURM 2005** → Urban Housing Policy **2007** → **AMRUT June 2015** → **Jal Jeevan Mission 2019**. **AMRUT 2.0** is **1 October 2021**. **SAGY** is **2014**.
23. Smart Cities = **MoHUA**; smart village / Rurban = **MoRD**. Keep the ministries separate.
24. **Bhopal** is the classic "not on a major river bank" city trap against Agra, Patna (पटना) or Kolkata.
25. NCR slice of Uttar Pradesh includes Ghaziabad, Noida, Greater Noida and Meerut in the urban-industrial belt.
26. Hamleted villages break the main settlement into secondary units (**panna / para / palli / nagla / dhani**); they are a rural pattern, not a separate census-town category.
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27. Economic activities: **primary** (farming, mining, fishing), **secondary** (manufacturing), **tertiary** (services), with **quaternary** for knowledge / R&D in advanced teaching.
28. Rural settlements lean on primary work and lower density; urban places are **statutory towns** or **census towns**. An **outgrowth** (railway colony, campus, cantonment) can sit inside an urban agglomeration.
29. Functional town tags to keep: administrative, industrial, transport, commercial, mining, garrison, educational, religious and tourist — classed by the **dominant** job.
30. Planned modern city classic = **Chandigarh** (Le Corbusier). Ancient urban tags include **Varanasi / Prayag / Madurai**; medieval tags include Delhi / Agra / Jaipur / Lucknow.
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Must-score facts — approaches, settlements, activities

Nature & scope

Item	Lock
Human geography	Study of human–environment relations / spatial organisation
Approaches	Welfare, behavioural, radical, humanistic (coaching set)
Primary / secondary / tertiary	Extraction · manufacturing · services
Quaternary / quinary	Knowledge / high decision-making

Settlements

Item	Lock
Rural patterns	Compact, dispersed, linear, circular
Urban functions	Administrative, industrial, commercial, transport, garrison
Primate city / rank-size	Coaching definitions

Topic 13 — Disaster Geography

Consolidated — 31 Must-Score Facts

1. Coaching risk fact: **Risk $\approx$ Hazard $\times$ Vulnerability / Capacity**. A disaster happens when a hazard hits exposed people or assets and capacity is too weak.
2. The disaster-management cycle runs **mitigation $\rightarrow$ preparedness $\rightarrow$ response $\rightarrow$ recovery**. Mitigation is before impact; response is during and just after.
3. Earthquake **focus** is the point **inside** the Earth; **epicentre** is the point on the **surface** above it. **Richter** measures magnitude; **Mercalli** measures intensity / damage.
4. Seismic waves split into **body** (through the Earth) and **surface** (along the ground). Body waves arrive **P then S**; surface waves come last and do most damage. **P-wave** shadow (छाया) is about **103–142°**; **S-wave** shadow lies beyond about **103°** (liquid outer core). The Himalaya (हिमालय) is India's highest seismic belt because of **collision and thrust zones**.
5. India uses seismic **Zones II–V** only (Zone I was dropped in 2002). **Zone V is highest**. A draft Zone VI was **withdrawn in March 2026** — maps still follow the 2016 II–V zones.
6. Zone V covers the North-East, Himalayan pockets, **Kutch** and the **Andaman (अंडमान) & Nicobar (निकोबार)** belt. Much of Rajasthan and the Deccan (दक्कन) sits in Zones II–III.
7. About **59%** of India's landmass is earthquake-prone in NDMA note. Famous shocks: Kangra (कांगड़ा) 1905, Bihar–Nepal 1934, **Koyna 1967 (reservoir-induced)**, **Latur 1993 (peninsular / Killari, Zone III)**, Bhuj 2001, and the **26 December 2004** tsunami.
8. A **tsunami** (Japanese "harbour wave") is caused by **seafloor displacement** (quake, landslide or volcano), not by ordinary wind waves or tides. It races fast in deep water and piles up near the coast. India's warning hub is **INCOIS, Hyderabad (हैदराबाद)**. Hall event: **26 Dec 2004** Sumatra megathrust.
9. **Kilimanjaro** sits on the East African Rift and is **not** part of the Pacific Ring of Fire.
10. Landslide belts are the Himalaya, Western Ghats (पश्चिमी घाट) and North-East. pairs include Kedarnath (केदारनाथ) **2013**, Joshimath subsidence **2023**, and Sikkim GLOF **2023**.
11. Flood types include riverine (Ganga (गंगा)–Brahmaputra (ब्रह्मपुत्र)), flash floods in hills, urban (नगरीय) floods and coastal storm surge (तूफानी महोर्मी). Drought is classed as meteorological, hydrological and agricultural. IMD drought criteria often use rainfall deficiency **>25%** of normal.
12. IMD **cloudburst** fact is rainfall **≥ 100 mm in one hour**, typically in Uttarakhand (उत्तराखंड)–Himachal (हिमाचल)–J&K–North-East hill belts around **1000–2500 m**.
13. IMD plains heat-wave gate: when **Tmax $\geq 40^\circ\text{C}$** , a departure of **4.5–6.4°C** is a heat wave and **>6.4°C** is severe; or actual **$\geq 45^\circ\text{C}$ / $\geq 47^\circ\text{C}$** . Criteria also use two stations for two days. Cold-wave plains gate is **Tmin $\leq 10^\circ\text{C}$** with a sharp drop.
14. The **Bay of Bengal** produces **more** cyclones than the Arabian Sea. Peak seasons are roughly **May–June** and **October–December**. Arabian Sea storms can still hit Gujarat–Maharashtra (Tauktae / Kandla note).
15. Cyclone-name pairs: **Baguio = Philippines**; Hurricane = USA; Typhoon / Taifu = China–Japan; **Willy-willy (विली-विली) = Australia**; cyclone = North Indian Ocean; Cordonazo = Mexico.
16. 2019 storm–place pairs: Dorian–Bahamas, Hagibis–Japan, Lekima–China, Mitag–Korea, with Indian Ocean names like Amphan, Fani and Tauktae also in the news map.
17. **NDMA** is chaired by the **Prime Minister**. **SDMA** is chaired by the **Chief Minister**. **DDMA** is chaired by the District Magistrate / Collector.
18. Framework years: **DM Act 2005**, **NPDM 2009**, **NDRF 2006** (MHA), **NDMP 2016**, **Sendai Framework 2015–2030** (after Hyogo 2005–15), **CDRI 2019**. International Day for Disaster Risk Reduction is **13 October**.
19. NEC pair = Union Home Secretary. **NIDM** handles training from Delhi. Its ancestor **NCDM** was set up in **1995** under **IIPA** (not ICSSR / NDMA). India is **not** a disaster-free country.

20. India's multi-hazard note: ~**59%** quake-prone, ~**12%** flood-prone, ~**5,700 km** cyclone/tsunami coast, ~**68%** of cultivable area drought-vulnerable.
21. **Mangroves** cut cyclone and surge impact. **Bhopal 1984** is the classic man-made industrial / chemical disaster.
22. **DPAP** began in **1973–74**. The earlier Community Development Programme pair is **1952**.
23. Uttar Pradesh (उत्तर प्रदेश) sits mainly in seismic **Zones III–IV** (not Zone V as a whole-State label). Eastern districts face river floods; Bundelkhand faces drought. Plains heat waves peak in May–June (**Loo** (लू)).
24. In the 2018 UP paper, the **Gomati** (गोमती) carried the “biological disaster” pollution label among the given rivers.
25. Hazard is potential danger; disaster is when capacity fails. Do not treat the two words as identical.
26. Cold-wave and fog risk in western Uttar Pradesh and the Terai (तराई) peaks in **December–January**; plains heat waves (**Loo**) peak in **May–June**.

27. Surface seismic waves include **Love** and **Rayleigh**; they arrive after body waves and do most damage. **P** waves travel through solids, liquids and gases; **S** waves travel through **solids only**.
28. Hazard classes include geophysical (quake, volcano, tsunami), hydrological (flood), meteorological (cyclone, heat/cold wave), climatological (drought) and technological / industrial (Bhopal-type).
29. Landslide triggers include steep slopes, heavy rain, earthquakes, deforestation and toe-cutting of slopes. Avalanche risk sits mainly in high Himalayan snow belts.
30. Early-warning chain: IMD (weather / cyclone / heat), INCOIS (tsunami / ocean), CWC (floods), GSI / NDMA guidance for landslides — do not dump every warning on one agency.
31. **Sendai Framework** has **seven** global targets and **four** priorities for action (not six targets / three priorities). Priority 4 includes **Build Back Better** in recovery. It was adopted at the **Third** UN World Conference on DRR (Sendai, Japan).

Must-score facts — hazard types, DM cycle, India belts

Terms

Term	Lock
Hazard	Potential threat
Vulnerability	Susceptibility
Risk	Hazard × vulnerability
DM cycle	Mitigation → Preparedness → Response → Recovery

India belts

Hazard	Lock
Earthquake	Himalayan / Indo-Gangetic / peninsular zones
Cyclone	Bay of Bengal > Arabian Sea frequency
Flood	Ganga–Brahmaputra plains; flash floods hills
Drought	Rain-shadow / arid west; monsoon failure
Landslide	Himalaya / WG steep slopes

Topic 14 — Earth & Universe

Consolidated — 50 Must-Score Facts

1. **Big Bang** is linked to **Lemaître / Gamow** (~13.7–13.8 bn yr) with **Hubble** redshift expansion. **Steady State** is **Hoyle–Bondi–Gold**. Solar-system **Nebular** theory is **Kant–Laplace**. **Otto Schmidt** = interstellar dust capture by the Sun.
2. The **Milky Way** is a **barred spiral**. **Andromeda (M31)** is the nearest major spiral and the farthest naked-eye object. **Chandrasekhar limit** $\approx 1.44 M_{\odot}$ is the white-dwarf ceiling; above it the path runs to neutron star / black hole.
3. One **galactic year** for the Sun is ~22.5–25 crore years. IAU lists **88** constellations; **Saptarishi** points to the Pole Star. A **light-year** measures **distance**; star colour tracks **temperature** (blue hottest). **Brittle star** is a sea animal, not a star class.
4. Planet order from the Sun is **MVEMJSUN**. **Venus** is hottest (greenhouse / “veiled planet”) and rotates **retrograde**. **Earth** has the highest mean density. **Jupiter** is largest. **Saturn** has the clearest rings. **Mercury** has the fastest orbit; Mercury and Venus have **no moon**.
5. **Copernicus** is the classic heliocentric name. **Kepler** gave planetary-motion laws; **Newton** explained orbits by universal gravitation. **Uranus** (Herschel, 1781) tilts ~98° (“planet on its side”) and also rotates retrograde. **Neptune** is farthest and coldest among the eight and has the longest year (~165 Earth years).
6. Largest moon is **Ganymede** (Jupiter — not Titan). **Titan** (Saturn) has a thick atmosphere. Mars moons are **Phobos** and **Deimos**. Asteroid belt sits between **Mars and Jupiter**; **Trojans** share Jupiter’s orbit at L4/L5 (~60° ahead/behind).
7. A **meteor** burns in the **mesosphere**; an unburned fragment on ground is a **meteorite**. A **comet** tail always points **away from the Sun**; Halley’s period is ~76 years. **Pluto** became a **dwarf planet** (IAU, Prague, **24 Aug 2006**; MPC number **134340**).
8. A **solar eclipse** occurs at **New Moon**; a **lunar eclipse** at **Full Moon**. Sunlight takes ~8 min 20 s to reach Earth. Sun’s energy is **nuclear fusion**. One **AU** ≈ 149.6 million km. Photosphere $\approx 6000^{\circ}\text{C}$; corona is the outer solar atmosphere.
9. In the 2022 set, Saturn’s atmosphere is linked with **methane**. Mars is the classic life-candidate planet (2018 subsurface-lake news). **Mangalyaan / MOM** = India’s first Mars orbiter. **Aditya-L1** = solar; **Chandrayaan** = lunar — do not swap.
10. Earth **rotates** ~24 h (day–night, west→east) and **revolves** ~365¼ days (year / leap year). Shape is an **oblate spheroid**. **Perihelion** $\approx$ 3–4 January; **aphelion** $\approx$ 4 July.
11. Seasons come from **23½° axial tilt**, not perihelion distance. Equinoxes $\approx$ **21 March / 23 September**; solstices $\approx$ **21 June / 22 December**. On equinox, day and night are about equal everywhere; Arctic / Antarctic Circles mark 24-hour day or night at solstice.
12. One degree of latitude $\approx$ **111 km**. Tropics = **23½°**; polar circles = **66½°**. Heat zones: torrid between tropics, temperate to 66½°, frigid to the poles. **Great circles** = Equator + **all meridians**; a parallel such as **60°N** is **not** a great circle.
13. **0° lat and 0° long** meet in the **Atlantic** (Gulf of Guinea), not on land. **Not** on Equator: **Mexico, Equatorial Guinea, Rwanda**. **Not** on Cancer: **Iran**. **Not** on Capricorn: **Bolivia**.
14. India’s area rank is **7th** (not 6th); land share $\approx$ **2.4%**. Tropic of Cancer runs through the **middle** of India — India is **not** wholly tropical. **Uttar Pradesh** does **not** lie on the Tropic of Cancer.
15. **IST** is based on **82°30’ E** near **Mirzapur (UP)** and is **GMT + 5 h 30 min** for the **whole** country. Time zones use $\sim 15^{\circ} = 1$ hour ($1^{\circ} = 4$ min). **IDL** $\approx$ **180°** (zigzag); the **west** side is a day ahead.
16. Earth’s spheres: **lithosphere** (crust + upper mantle), **atmosphere**, **hydrosphere**, **biosphere** (where the other three meet). Rocks cycle among igneous, sedimentary and metamorphic states.

17. Atmosphere upward order: **troposphere** → **stratosphere** → **mesosphere** → **thermosphere/ionosphere** → **exosphere**. Weather lives in the **troposphere**; **ozone UV shield** sits in the **stratosphere** (ideal for jet flight); mesosphere is coldest / meteors burn; ionosphere reflects radio waves and hosts auroras.
18. Dry-air abundance: **N₂ (~78%)** → **O₂ (~21%)** → **Ar (~0.93%)** → **CO₂ (~0.04%)**. Permanent gases = **N₂, O₂, Ar**; variable = vapour, CO₂, O₃, dust. **Argon is not** a greenhouse gas. **Ozone (O₃)** is variable.
19. Atmosphere is heated mainly by **long-wave terrestrial radiation**, not direct short-wave alone. Normal lapse ≈ **6.5°C/km**. **Albedo** = reflecting power (fresh snow highest). Max heating belt often ~**20°N**. Same-latitude **Agra vs Darjeeling** contrast is an **altitude** fact.
20. Heat budget (NCERT 100-unit picture): about **35/100** insolation is reflected (planetary albedo) and about **51/100** reaches the surface; surface returns **long-wave**. **Nuclear** and **geothermal** energy are **not** stored solar energy.
21. Greenhouse gases include water vapour, CO₂, CH₄, N₂O, O₃ and CFCs. **Montreal** = ozone / CFCs; **Kyoto / Paris** = climate / GHG path. Ozone-hole focus = **Antarctica** in spring.
22. **Weather** is short-term; **climate** is the long-term average. Light scattering / diffusion is classically linked to **dust**. Earth Day is **22 April**.
23. Cyclone regional names: **Hurricane**—USA, **Typhoon**—China Sea, **Willy-Willy**—Australia, **Baguio**—Philippines. Tornado scale = **Fujita**; hurricane = **Saffir-Simpson**. Cumulonimbus = thunder / intense rain.
24. **El Niño** = warm eastern Pacific / Peru phase with **less upwelling**, so **plankton and fish fall**; it can disturb the Indian monsoon. **La Niña** is the cold opposite phase.
25. **Coriolis** deflects to the **right in the NH** and **left in the SH**; it is **zero at the equator** (Ferrel's law).
26. Jet streams are mainly high-altitude **westerlies** (polar + subtropical). **Tropical Easterly Jet (TEJ)** is the summer easterly exception (~14°N). Speeds can reach **300–500 km/h**. "All jets are easterly" is false.
27. Pressure belts: **ITCZ** / doldrums = equatorial **low**; horse latitudes = subtropical **high** ~**30°**; subpolar low ~**60°**; polar high. Planetary winds = trades, westerlies, polar easterlies. Belts shift with the solstice seasons.
28. **Roaring Forties** = strong westerlies near **40°S** (Furious Fifties / Shrieking Sixties farther south). Mediterranean winter rain links to southward belt shift around NH winter solstice.
29. Interior shells: **Sial** (continental crust), **Sima** (oceanic crust), **Nife** (core). Discontinuities: crust–**Moho**–mantle–**Gutenberg** (~**2900 km**, S-wave stop)–**Lehmann**–inner core.
30. **P-wave** shadow ≈ **103–142°**; **S-wave** shadow beyond ~**103°** because the outer core is **liquid**. **P** waves are fastest (all states); **S** waves solids only; surface waves are slowest and most damaging.
31. **Wegener (1912)** proposed continental drift with **Pangaea** and **Panthalassa**. Plate margins: **divergent** (MOR / new crust), **convergent** (trenches / fold mountains), **transform** (side-slip). New ocean crust forms by **sea-floor spreading**. Himalaya = continent–continent collision.
32. **Fold mountains**: Himalaya, Alps, Andes, Rockies. **Fault / block mountains**: Vosges, Black Forest, Sierra Nevada, Rhine graben.
33. **Igneous** = cooled magma/lava (**no fossils**); **sedimentary** = layered + fossils; **metamorphic** = heat/pressure remake. Pairs: granite → **gneiss**; limestone → **marble**; sandstone → **quartzite**; shale → **slate**. Magma = underground; lava = surface melt.
34. Volcano types: shield (**Mauna Loa**), composite (**Fuji / Etna / Vesuvius**), cinder (**Paricutin**), fissure (**Deccan**), caldera (**Krakatoa**). Most abundant volcanic gas = **water vapour**. **Stromboli** = "lighthouse of the Mediterranean." **Barren Island** = India's only confirmed active volcano (Andaman).
35. Match pairs: **Rainier = USA**, **Etna = Italy**, **Paricutin = Mexico**, **Apo = Philippines**. **Kilimanjaro** = East African Rift (**dormant**), **not** Ring of Fire. **Aconcagua** is a peak, not an active-volcano match like Etna.
36. Earthquake–volcano belts: Pacific **Ring of Fire** (~**90%** of world quakes), Mid-Atlantic Ridge, Alpine–Himalayan belt, East African Rift. India's seismic map uses Zones **II–V**. NW India shakes mainly from **Indian–Eurasian** convergence.
37. Earthquake **focus / hypocentre** = rupture point inside Earth; **epicentre** = surface point above it. **Richter** = magnitude (~**10×** amplitude / ~**32×** energy per step); **Mercalli** = intensity. **Tsunami** = Japanese for harbour wave; classic cases = **2004** Indian Ocean and **2011 Fukushima**.

38. The Moon always shows roughly the same face because rotation $\approx$ revolution (**~ 27.3 days** sidereal). Weight on the Moon $\approx$ **$1/6$** of Earth weight. Apollo 11 (**1969**) landed in the **Sea of Tranquility**. Moon has **no atmosphere**, no twilight, no sound.
39. Humidity terms: absolute vs relative humidity; dew point is the temperature where air becomes saturated. Fog is a cloud on the ground; frost forms when dew freezes.
40. Local mountain winds often asked: **chinook** (warm downslope, Rockies), **mistral** (cold, Rhône), **bora** (cold Adriatic), **foehn** (Alps warm downslope), **loo** (hot dry North Indian summer wind).
41. Homosphere (mixed major-gas ratios) roughly reaches **$\sim 80\text{--}100$ km**; above that the **heterosphere** sorts gases more by molecular weight.
42. Temperature **inversion** = temperature rises with height in a short layer; common on calm valley nights and can trap pollution near the ground.
43. Prime Meridian land set in coaching lists includes UK, France, Spain, Algeria, Mali, Burkina Faso, Togo, Ghana — **not Niger / Nigeria**. Cairo $\approx$ **30° E $\approx$ $+2$ h** on GMT.
44. China keeps **one** official time zone despite its width; India also keeps **one** (IST). Russia has the most time zones among large states in the usual teaching map.
45. If the Pole Star is on your **left**, you are walking **east**. If stars rise **perpendicular to the horizon**, the observer is on the **Equator**.
46. Inner planets are rocky terrestrials; outer set is gas giants (**Jupiter, Saturn**) plus ice giants (**Uranus, Neptune**). Sun holds **$\sim 99.8\text{--}99.9\%$** of solar-system mass; H:He in the Sun is roughly **3:1**.
47. Kuiper Belt = icy leftovers beyond Neptune; **Oort Cloud** = far comet reservoir. A nebula is a star-forming cloud, **not** a solar-system “member” like planets/moons/asteroids/comets.
48. Earth’s axis makes about **$66\frac{1}{2}^\circ$** with the orbital plane (complement of $23\frac{1}{2}^\circ$). Sidereal day $\approx$ **23 h 56 min**; mean solar day = **24 h**. Century years are leap only if divisible by **400**.
49. India’s Tropic of Cancer states matter for map stems; do not park UP on Cancer. Nuclear fission energy and Earth’s internal heat (geothermal) are the classic “**not** from the Sun” pair in energy stems.
50. Plate-tectonic quick map: divergent = plates apart / MOR; convergent = trenches or collision mountains; transform = side-slip faults (e.g. San Andreas class). Himalaya $\neq$ fault-block Vosges type.

Must-score facts — motions, structure, latitudes

Earth motions / shape

Item	Lock
Rotation	Day–night; west→east
Revolution	Seasons; elliptical orbit
Shape	Oblate spheroid
Axial tilt	$\sim 23\frac{1}{2}^\circ$

Structure / circles

Item	Lock
Crust / mantle / core	Silicates · denser rocks · iron–nickel
Tropic of Cancer	$23\frac{1}{2}^\circ$ N (cuts India)
Equator	0°
International Date Line	$\sim 180^\circ$ (zigzag)
Atmosphere layers	Troposphere → Stratosphere → Mesosphere → Thermosphere

Topic 15 — Geomorphology & Landform Processes

Consolidated — 52 Must-Score Facts

1. **Endogenic** processes (folding, faulting, volcanism, uplift) build relief from inside the Earth. **Exogenic** processes wear and deposit at the surface. **Gradation** = degradation + aggradation.
2. Davis's cycle of erosion is framed as **structure–process–time** (youth–mature–old). Penck and King modify timing, but the landform vocabulary stays -useful.
3. **Igneous** rocks are primary: **granite** is intrusive (acidic); **basalt** is extrusive (basic). Fossils belong to **sedimentary** rocks.
4. Igneous bodies: batholith, laccolith, lopolith, phacolith; a **sill** is parallel to beds; a **dyke** cuts across beds.
5. Metamorphic pairs: **limestone** → **marble**, **sandstone** → **quartzite**, **granite** → **gneiss**, **shale** → **slate / schist**. Foliated rocks include slate–schist–gneiss; marble and quartzite are non-foliated.
6. **Weathering** breaks rock **in situ**. **Erosion** picks up and carries material. **Denudation** is weathering plus erosion. Chemical routes include carbonation, oxidation, hydrolysis and hydration.
7. River load moves by traction, saltation, suspension and solution. Agents of erosion are river, wind, glacier, sea, groundwater (karst) and gravity (mass wasting).
8. Fluvial stages: youth = **V-valley / waterfall / pothole**; mature = **meander / floodplain / levee**; old = **ox-bow / delta (डेल्टा) / peneplain**. Rejuvenation adds knickpoints, terraces and incised meanders.
9. On a meander, the **concave** bank erodes and the **convex** bank builds a **point bar**. A **yazoo** stream is deferred behind a levee. Mountain-foot dumps are **alluvial (जलोढ़) fans**; coalescing fans form a **bajada**.
10. Stream genetic types include consequent (अनुवर्ती), subsequent (परवर्ती), obsequent and resequent. **Antecedent** (पूर्ववर्ती) streams cut rising land; **superimposed** (अध्यारोपित) streams inherit a course from a cover; subsequent streams follow structure.
11. The **only** bird's-foot (पक्षी-पाद) example is the **Mississippi**. **Arcuate** (चापाकार) examples include the **Nile** and the **Ganga (गंगा)–Brahmaputra (ब्रह्मपुत्र)**. The usual cuspate example is the **Tiber**. East-flowing Indian mouths often build deltas; **Narmada (नर्मदा)** and **Tapi (तापी)** usually form estuaries.
12. Drainage patterns: **folded** beds → **trellis**; homogeneous rock → **dendritic** (वृक्षाकार); dome → **radial**; joints / faults → **rectangular**.
13. The classic fault-valley river tag is the **Damodar (दामोदर)**, not the Chambal (चंबल). **Narmada–Tapi** are rift / fault-line pairs as well.
14. The **Imphal** basin is **lacustrine** (lake-filled), not loess or glacial. Kashmir Vale also has lacustrine / structural pairs. **Loess** is wind-laid silt (China belt), not Imphal.
15. Structure landforms include cuesta, hogback, mesa and butte. A **mesa** is broader than a **butte**. Davis's humid old-age plain is a **peneplain** with leftover **monadnocks**; King's arid form is a **pediplain**.
16. Glacial erosional forms: cirque, arête, horn, **U-valley**, hanging valley, fjord, tarn. Depositional forms: moraine, drumlin, esker, kame, outwash, erratic.
17. A **fjord** is a drowned glacial trough. A **ria** is a drowned river valley. Do not treat them as the same estuary (ज्वारनदमुख) type.
18. Aeolian points: **barchan** horns point **downwind**; parabolic dunes often show horns upwind. Yardang, zeugen and mushroom rocks are wind-eroded forms.
19. Coastal deposition builds **spit, bar, tombolo and lagoon** with longshore drift. Coral reefs include fringing, barrier and atoll types. An **atoll's shape** is **horseshoe / ring** around a central lagoon (not triangular or rectangular).
20. Karst needs **limestone plus carbonation**. **Stalactite** hangs from the **ceiling**; **stalagmite** grows from the **floor**. Forms include doline, cave, uvala and polje.
21. River capture vocabulary includes pirate stream, beheaded stream, wind gap and elbow of capture.

22. East Indian rivers typically build **deltas**; west-flowing Narmada and Tapi typically form **estuaries** into the Arabian Sea.
 23. Conglomerate has rounded pebbles; breccia has angular fragments. Organic sedimentary rock pair includes coal.
 24. Intertrappean beds between Deccan (दक्कन) lava flows hold **land and freshwater** fossils, not marine plant–animal assemblages.
 25. Himadri (हिमाद्रि) is fossil-poor crystalline rock; Lesser Himalaya carries marine fossils; Shiwalik (शिवालिक) holds human remains — keep the three belts distinct when landform stems mix fossils.
 26. A gorge is a steep mountain cut; a canyon is often the arid stair-step equivalent. U-valley = glacier; V-valley = youthful river.
 27. Mekong source is Tibet; its delta is in **Vietnam**, not Cambodia.
 28. The rock cycle links magma $\rightleftharpoons$ igneous $\rightarrow$ sedimentary $\rightarrow$ metamorphic $\rightarrow$ melt, so the same material can reappear as a different rock type after burial, melting or recrystallisation.
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29. Mass wasting moves material downslope under gravity (creep, slide, flow, fall). It is an agent alongside river, wind, glacier, sea and groundwater.
 30. Glacial depositional set: **moraine, drumlin, esker, kame, outwash, erratic**. Erosional set: cirque, arête, horn, U-valley, hanging valley, fjord, tarn.
 31. Aeolian deposition builds **barchan** (horns downwind) and often **parabolic** dunes (horns upwind). Wind erosion forms yardang, zeugen and mushroom rocks.
 32. Coastal deposition builds spit, bar, tombolo and lagoon with longshore drift. Coral reefs: fringing, barrier and **atoll** (horseshoe / ring around a lagoon).
 33. If the stem calls a moribund delta a **subdivision**, the answer is the **Bengal Delta** (moribund, mature, active). If the stem asks which river's delta is the moribund **lobe**, the answer is the **Cauvery. Krishna (कृष्णा)–Godavari (गोदावरी)** and **Mahanadi (महानदी)** fit neither wording.
 34. The **Irrawaddy (इरावदी)** delta lies in **southern Myanmar**.
 35. The **Nile (नील)** delta lies in **northern Egypt**.
 36. The **Indus (सिंधु)** delta lies in **Sindh, Pakistan**.
 37. The **Danube (डेन्यूब)** delta meets the **Black Sea** in **Romania and Ukraine**.
 38. The **Rhine (राइन)** delta lies in the **Netherlands**.
 39. The **Volga (वोल्गा)** delta lies on the **Caspian Sea**.
 40. The **Niger (नाइजर)** has an inland delta in **Mali** and a sea delta in **Nigeria**.
 41. The **Godavari (गोदावरी)** delta lies in **Andhra Pradesh**.
 42. The **Krishna (कृष्णा)** delta lies in **Andhra Pradesh**.
 43. The **Mahanadi (महानदी)** delta lies in **Odisha**.
 44. The **Cauvery (कावेरी)** delta lies in **Tamil Nadu**.
 45. The **Red River** delta lies in **northern Vietnam**.
 46. The **Amazon** meets the Atlantic in **Brazil** as a broad estuary.
 47. Dendritic rivers include the **Ganga (गंगा)** and **Indus (सिंधु)** plains, the **Godavari (गोदावरी)**, the **Mahanadi (महानदी)**, the **Krishna (कृष्णा)**, and the **Kaveri (कावेरी)**.
 48. From **Amarkantak (अमरकंटक)**, the **Narmada (नर्मदा)** flows west and the **Son (सोन)** flows toward the Ganga. That spread is **radial**. The Mahanadi rises at **Sihawa**, so it is not in this radial set.
 49. The Indian **trellis** ground is the old folded belt of **Singhbhum (सिंहभूम)** on the Chotanagpur plateau.
 50. **Rectangular** streams follow joints on the **Vindhyan (विंध्य)** rocks.
 51. **Centripetal** streams drain into **Loktak (लोकतक)** and into **Sambhar (सांभर)**.
 52. The **Sharavati (शरावती)** follows a **parallel** pattern down the Western Ghats.
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Must-score facts — agents, landforms, rocks

Endogenic / exogenic

Item	Lock
Endogenic	Diastrophism, volcanism, earthquakes
Exogenic	Weathering, erosion, deposition
Weathering types	Physical, chemical, biological

Agent ↔ landform

Agent	Landforms
River	V-valley, meander, oxbow, floodplain, delta
Glacier	U-valley, cirque, moraine, fjord
Wind	Yardang, dune, loess
Wave	Cliff, beach, spit, stack
Karst	Sinkhole, cave, stalactite

Rock cycle

Igneous → **Sedimentary** → **Metamorphic** (with melting / uplift loops)

Topic 16 — Oceans

Consolidated — 40 Must-Score Facts

1. By area the oceans rank **Pacific > Atlantic > Indian > Southern > Arctic**. The Indian Ocean was the Greek **Erythraean Sea** and straddles **both sides of the Equator**.
2. Ocean-floor order from the shore is **shelf → slope → rise (or trench) → abyssal plain**. A **guyot** is a flat-topped seamount. The continental shelf is shallow (about **200 m**), rich in fish and oil, and covers roughly **7.5%** of the ocean floor.
3. India's **west shelf is wider than the east**. The Gujarat–Mumbai belt is widest and holds **Bombay High**. The east shelf is narrower but carries large delta (डेल्टा) fans.
4. The **Telegraphic Plateau** is part of the **North Atlantic Ridge**, not a separate land plateau.
5. Trench–ocean pairs: **Mariana–Pacific, Puerto Rico–Atlantic, Sunda/Java–Indian, and Molloy–Arctic**.
6. **Agulhas** and **Brazil** are **warm** currents. **Humboldt (Peru)** and **California** are **cold**. Do not mark Agulhas or Brazil as cold.
7. The **Benguela** is an **Atlantic** cold current off south-west Africa. It is **not** a Pacific current.
8. Among classic MCQ options, the current linked to the **Indian Ocean** is the **Agulhas**.
9. **El Niño** (अल नीनो) sends warm water off Peru, weakens upwelling, and cuts plankton and fish catch.
10. Open-ocean **salinity** is about **35‰**. The maximum sits near the **Tropics of Cancer and Capricorn**, not on the Equator. The **Red Sea** is high; the **Baltic** is low.
11. Seawater density rises when water is **colder** and **saltier**. Cold salty water sinks and drives the **thermohaline** circulation. **NaCl** is about **77%** of dissolved salts.
12. Surface temperature peaks near the equator. The **thermocline** lies roughly **300–1000 m**. Deep water is cold in every ocean.
13. A **spring tide** forms at **syzygy** (new/full moon) and is **large**. A **neap tide** forms at **quadrature** and is **small**. The tidal period is about **12 h 25 min**, not 12 h 30.
14. A **tidal bore** is famous on the **Hooghly (हुगली)** and the **Amazon**. The **Bay of Fundy** has the world's greatest tidal range. India's tidal-energy coast fact is the **Gulf of Khambhat**.
15. **Upwelling** is strongest on **west coasts**: Peru, California, Canary, Benguela, and Somalia.
16. **Constructive** waves build beaches; **destructive** waves erode them. A **tsunami** (सुनामी) is not a tide.
17. Ocean currents shape climate (mild or foggy coasts, west-coast deserts), concentrate fish where warm and cold meet or upwelling occurs, and affect navigation routes.
18. Deepest deposit is **red clay**. **Globigerina** and **pteropod** oozes are **calcareous**. **Diatom** and **radiolarian** oozes are **siliceous**. **Manganese nodules** sit on abyssal plains, especially in the Pacific and the Central Indian Ocean Basin.
19. The **Suez Canal** (Egypt (मिस्र), **1869**) joins the **Mediterranean** and the **Red Sea**. It is a **sea-level** cut — ships stay at one water height and do **not** climb stepped chambers. Lakes north→south: **Manzala → Timsah → Great Bitter → Little Bitter**. Ends: **Port (बंदरगाह) Said** (N) and **Suez** (S). India–Europe sea route is shorter by about **7000 km**. It does **not** itself touch the Atlantic or the open Indian Ocean.
20. The **Panama Canal** (**1914**) joins the **Atlantic/Caribbean** and the **Pacific**. Ships **climb stepped chambers** and use **Gatun Lake** to cross the isthmus. The **Kiel Canal** (Germany) joins the **North Sea** and the **Baltic**.
21. UNCLOS belts: **territorial sea 12 nm, contiguous zone 24 nm, EEZ 200 nm**. The continental shelf starts at **200 nm** and may extend to **350 nm**.
22. Classic fishery centres: **Grand Banks** (Labrador meets Gulf Stream), **Dogger Bank**, Peru upwelling, and the **Kuroshio–Oyashio** mix.
23. India's **operational** Antarctic stations are **Maitri** and **Bharati**. **Dakshin Gangotri (गंगोत्री)** is not the operational pair. **Himadri (हिमाद्रि)** is India's **Arctic** station, not Antarctic.

24. The **Antarctic Treaty** dates to **1959/61**. India joined in **1983**. India is an **observer** in the Arctic Council.
25. Key straits: **Hormuz** (Persian Gulf ↔ Gulf of Oman; **Iran north / Oman south**; ~25% seaborne oil), **Malacca**, **Gibraltar**, **Bering**, **Bab-el-Mandeb**, **Palk**, and the **10° Channel** (दस डिग्री). Persian Gulf coasts = Iran, Iraq, Kuwait, Saudi Arabia, Bahrain, Qatar, UAE — **not Oman**.
26. The **Carlsberg Ridge** lies in the north-west Indian Ocean. The **Ninetyeast Ridge** is not the Mid-Atlantic Ridge.
27. The International Seabed Authority manages the seabed **Area** beyond national zones (HQ Jamaica). India's inland waterway **NW-1** is the **Ganga** (गंगा) from Haldia to Prayagraj (प्रयागराज).
28. **Drake Passage** is not the Strait of Magellan. Panama opened in **1914** with stepped chambers; Suez opened in **1869** as a sea-level cut.
29. Hydrosphere ≈ **71%** of Earth. Usable fresh ≈ **<1%** of all water. Of fresh water, ice >> groundwater >> rivers/lakes.
30. **Datum line** = mean-sea-level height/depth zero. Abyssal plains ≈ most extensive ocean-floor province (~76%).
31. **Sargasso** = North Atlantic, **no coast**. **Red Sea** = axial trough. **NAD** = Blanket of Europe. **OMT** (26°C isotherm) helps monsoon rainfall forecast.
32. Lake salinity pairs: Assal / Van / Dead Sea / Great Salt (लवणाध्यक्ष) Lake. **Halocline** = salinity gradient with depth. **Diamantina** = Indian Ocean.
33. Türkiye clock: **N = Black**, **S = Mediterranean**, **W = Aegean**, **NW = Marmara**. West→east ladder: **Mediterranean** → **Black** → **Caspian** → **Aral** ("My Black Cat Ate").
34. **Gaza** faces the **Mediterranean**. Baltic coasts = Denmark, Germany, Poland, Lithuania, Latvia, Estonia, Russia, Finland, Sweden — **not Norway**. **Jordan** has no Mediterranean coast (only **Aqaba** / Red Sea).
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35. Warm currents raise coastal temperature and humidity; cold currents cool and dry coasts and help west-coast deserts. Fog is common where warm and cold currents meet.
36. Coral needs warm clear shallow water; bleaching pairs with about **2°C** sea-surface warming. India's reef facts: Andaman & Nicobar, Lakshadweep, Gulf of Mannar, Gulf of Kachchh.

Must-score facts — relief, currents, El Niño

Ocean relief

Feature	Lock
Continental shelf	Shallow margin
Slope / rise	Steep then gentle to floor
Abyssal plain	Deep flat floor
Trench	Deepest (e.g. Mariana)
Mid-ocean ridge	Divergent volcanism

Currents / climate

Item	Lock
Open-ocean salinity	~35‰; max near Tropics (not Equator)
Warm / cold currents	Affect coasts & fog / desert margins
Agulhas	Into Indian Ocean (S. Africa teaching)
El Niño	Warm water off Peru; weak upwelling; fish crash
Tides	Spring / neap; lunar–solar alignment

Topic 17 — World Rivers & Lakes

Consolidated — 34 Must-Score Facts

1. **New Orleans** stands on the **Mississippi**, not the Missouri. The Missouri joins the Mississippi at **St Louis**.
2. **Budapest** is on the **Danube**. **Cologne** is on the **Rhine**. India's **Hyderabad** (हैदराबाद) is on the **Musi**, not the Godavari (गोदावरी) or Paleru.
3. The **Mekong** rises in Tibet, flows **south / south-east**, and builds its delta (डेल्टा) in **southern Vietnam**. It does **not** flow south-west and the delta is **not** in Cambodia.
4. Direction pairs: **Amur** north-east; **Syr Darya** north-west into the Aral; **Angara** north out of Baikal; **Volga** from the Valdai Hills into the **Caspian**.
5. The main USA–Mexico border river is the **Rio Grande**. The **Colorado** is the usual trap option.
6. **Lake Onega** is in **Russia**, not Canada. **Lake Michigan** lies wholly in the **USA**. **Maracaibo** is in Venezuela. **Baikal** is in Russia.
7. Superlatives: the **Nile** is longest for Prelims; the **Amazon** has the largest discharge; the **Congo** is deepest and crosses the Equator **twice**; the **Yangtze** is Asia's longest; the **Volga** is Europe's longest.
8. Delta shapes: **Mississippi** = bird's-foot (पक्षी-पाद); **Nile**, **Hwang Ho**, and **Niger** = arcuate (चापाकार) (bow).
9. The **Danube** crosses the most countries. The **Rhine** is Europe's busiest navigation artery.
10. **Khartoum** is where the White and Blue Nile meet. **Aswan** created **Lake Nasser**. **GERD** sits on the **Blue Nile** in Ethiopia.
11. Dam pairs: **Three Gorges–Yangtze**, **Aswan–Nile**, **Itaipu–Paraná**, **Hoover–Colorado**, **Kariba–Zambezi**.
12. The **Caspian** is the largest lake. **Superior** is the largest freshwater lake by area. **Baikal** is deepest, holds the most fresh volume, and is among the oldest.
13. **Tanganyika** is second deepest and the longest freshwater lake. **Titicaca** is the highest navigable lake. The **Dead Sea** is the lowest land surface and is hypersaline (about **34%** salt (लवणाध्यक्ष)).
14. Great Lakes west to east: **Superior** → **Michigan** → **Huron** → **Erie** → **Ontario** (mnemonic: Super Man Helps Every One).
15. Falls: **Angel** (Venezuela) is highest; **Victoria** is on the Zambezi; **Niagara** is US–Canada; **Iguazu** is Brazil–Argentina.
16. The **St Lawrence** is the Great Lakes seaway. The **Rhine–Main–Danube** canal links the Black Sea system to the Rhine.
17. The **Aral Sea** shrinks mainly because the **Amu Darya** and **Syr Darya** were diverted. **Lake Chad** is also shrinking.
18. The **Niger** is the paradox river that first flows inland. The Congo is also known as the **Zaire**.
19. The **Darling Range** is a highland of south-west Australia. It is **not** the same as the **Murray–Darling** river system.
20. **Endorheic** lakes have no ocean outlet — Caspian, Aral, Dead Sea, Chad, and Eyre.
21. More city–river pairs: **Baghdad–Tigris**, **Paris–Seine**, **London–Thames**, **Cairo–Nile**.
22. Lake types include tectonic/rift, glacial, crater, lagoon, and oxbow. Do not treat every famous lake as simply a "fresh lake".
23. Continent spine: Nile (Africa), Amazon (South America), Yangtze (Asia), Mississippi–Missouri (North America), Volga (Europe), Murray–Darling (Australia).
24. Navigation classics include the Rhine, Danube, Volga, St Lawrence, Yangtze, and Mississippi.
25. **Limpopo** crosses the Tropic of Capricorn **twice**. **Congo/Zaire** crosses the Equator **twice**.
26. **Mahaweli** = longest Sri Lanka river; Sri Lanka drainage is **radial** from central highlands.
27. City–river desk: Phnom Penh–Mekong; Hanoi–Red; Yangon–Irrawaddy; Madrid–Manzanares; Bangkok–Chao Phraya; Basra–Shatt al-Arab.
28. **Grand Canyon** = Colorado. **Chisapani Gorge** = Nepal (Karnali). Largest delta = **Ganga (गंगा)–Brahmaputra (ब्रह्मपुत्र)**. Inselberg ≠ glacier.

29. Delta mouths: Mekong → **South China Sea**; Nile → **Mediterranean**; Mississippi → **Gulf of Mexico**; Danube → **Black Sea**; Indus → **Arabian Sea**.
30. Border rivers beyond Rio Grande: **Amur** (Russia–China), **Orange** (South Africa–Namibia), **Salween** (Myanmar–Thailand stretches), **Zambezi** (Zambia–Zimbabwe at Victoria Falls).
31. More dam–river pairs: **Guri–Caroní** (Venezuela), **Tucuruí–Tocantins** (Brazil), **Grand Coulee–Columbia** (USA), **Sayano-Shushenskaya–Yenisei** (Russia), **Tarbela–Indus** (Pakistan), **Nurek–Vakhsh** (Tajikistan).
32. Yangtze chain: Tibetan Plateau → China → **Three Gorges** → Shanghai belt → **East China Sea**.
33. Nile chain: Victoria / Tana → Sudan–Egypt → **Aswan High Dam** → Nile Delta → **Mediterranean**.
34. Only a marked stretch of a river is the international border — do not treat the whole Mekong or Colorado course as one continuous border.

29. International rivers to keep: Nile (Africa), Amazon (South America), Danube (most countries crossed), Rhine (busy European navigation), Mekong (SE Asia), Rio Grande (USA–Mexico border).
30. Lake genesis types: tectonic / rift, glacial, crater, lagoon, oxbow and artificial reservoir — do not treat every famous lake as simply “fresh”.

Must-score facts — city–river, deltas, lakes

City ↔ river

City	River
New Orleans	Mississippi (≠ Missouri)
Budapest	Danube
Cologne	Rhine
Hyderabad (India)	Musi (≠ Godavari)
Khartoum	White + Blue Nile meet

Direction / border

River	Lock
Mekong	Tibet → SE; delta S. Vietnam
Amur	North-east
Syr Darya	North-west into Aral
Rio Grande	Main USA–Mexico border (≠ Colorado trap)

Superlatives / deltas

Item	Lock
Longest (Prelims)	Nile
Largest discharge	Amazon
Bird's-foot delta	Mississippi
Arcuate	Nile, Hwang Ho, Niger
Onega	Russia (≠ Canada)
Michigan	Wholly USA

Topic 18 — World Landforms

Consolidated — 40 Must-Score Facts

1. Area order (largest → smallest): **Asia** → **Africa** → **North America** → **South America** → **Antarctica** → **Europe** → **Australia**. Australia is the **smallest** continent; Africa has the **most countries** (~54).
2. Mean elevation leader is **Antarctica** (~2300 m). Europe has the **highest share of plains** in its area. **Guyana** is in **South America**, not Africa.
3. The **Andes** are the world's **longest** fold chain. The **Himalaya** (हिमालय) are the **highest**. **Aconcagua** stands in the Andes of Argentina. Length ladder pair: Andes > Rockies > Great Dividing Range > Himalaya.
4. **Pyrenees** = Spain–France; **Alps** = Switzerland / central Europe (**not England**); **Apennine** = Italy; **Balkan** = Bulgaria; **Urals** mark Europe–Asia. **British Columbia** = "Sea of Mountains."
5. **Toubkal** (Atlas) is in **Morocco**. **Hoggar** is in **Algeria**. **Stanley / Rwenzori** is in Uganda. **Kilimanjaro** is in **Tanzania**. **Darling Range** = SW Australia.
6. **Chimborazo** is in **Ecuador**. The Atlas system is North African — do not place it in France or Spain. **Sierra Nevada** = block mountain (not young fold like Rockies/Alps/Himalaya).
7. **Kilimanjaro** sits on the **East African Rift** and is **not** in the Pacific Ring of Fire. **Etna** is in Italy / the Mediterranean. Tertiary young folds: Alps, Andes, Himalaya, Rockies, Atlas; Appalachians = **Caledonian / old**.
8. **Tibet** is the highest large plateau (~4500 m). The **Pamir** is the "**Roof of the world**". The **Altiplano** is Bolivia–Peru. The **Meseta / Madrid** is Spain. **Patagonia Plateau** = mineral storehouse pair.
9. The **Colorado Plateau** holds the Grand Canyon. The **Columbia Plateau** is a lava plateau. Do not swap the two. **Telegraphic Plateau** sits on the **North Atlantic Ridge** (not Indian Ocean).
10. Grassland names: **Pampas** = Argentina; **Campos** = Brazil; **Llanos** = Venezuela–Colombia; **Puszta** (पुस्ज़्टा) = Hungary; **Prairie** (प्रेयरी) = North America; **Steppe** (स्टेपी) = Eurasia; **Veld** = South Africa; **Downs** = Australia.
11. **Borneo** is shared by three countries and is **not** a volcanic dump. **Java** and **Sumatra** sit on a volcanic arc.
12. Island rank by size: **Greenland** > **New Guinea** > **Borneo** > **Madagascar** > **Baffin** > **Sumatra** > **Honshu** > **Victoria** > **Great Britain** > **Ellesmere**. Australia is a continent, not ranked as an island in that list. **Honshu** = Japan's largest; **Faroe** = Sheep Islands (Denmark).
13. The **Gobi** lies in **Mongolia and China** only. It is a cold desert, not Russia or Kazakhstan.
14. Hot-desert size order: **Great Sandy** < **Gobi** < **Arabian** < **Sahara**. The **Sahara** is the largest hot desert. **Thar** (थार) = densest populated desert; **Atacama** = driest; **Europe** has the least desertification (मरुस्थलीकरण) problem among continents.
15. **Gibson** = Australia (not Brazil). **Sonoran** = USA. **Taklamakan** = China. **Karakum** = Turkmenistan (also Kazakhstan fringe note).
16. The **Atacama** (N Chile / S Peru — not southern Chile alone) is the driest. The **Namib** is a fog coast. **Patagonia** is Argentina's rain-shadow / temperate desert.
17. Hot deserts favour **west coasts** because of the Horse Latitudes (हॉर्स अक्षांश) plus cold ocean currents (~15–30°).
18. "Land of Big Games" points to the tropical **savanna** (सवाना).
19. Sclerophyll scrub: **Maquis** = Mediterranean; **Fynbos** = South Africa; **Chaparral** = California; **Matorral** = Chile.
20. **Epiphytes** mark equatorial forest; **baobab** marks savanna; **cedars** mark the Mediterranean; **acacia** marks the Sahara fringe.
21. In Brazil, **Selva** (सेल्वा) is the rainforest and **Terra Roxa** is the famous coffee soil — both are Brazil facts.
22. The **Mediterranean** climate has **winter rain** on five **west coasts** near **30–45°**. Summer is dry under the subtropical high.
23. **Taiga** is boreal conifer forest. **Tundra** (टुंड्रा) is treeless. Climate letters: Savanna **Aw**, Steppe **BS**, Tundra **ET**.
24. Volcano pairs: **Rainier** USA; **Etna** Italy; **Paricutin** Mexico; **Apo** Philippines; also Fuji (Japan), Pinatubo (Philippines), St Helens (USA).

25. Peak continents: **Elbrus** = Europe (Caucasus / Russia); **Mont Blanc** = Alps; **Denali / McKinley** = North America; **Kosciuszko** = Australia mainland; **Cook / Aoraki** = New Zealand Southern Alps.
26. Temperate grasslands (Prairie, Steppe, Pampas, Veld, Downs) sit on **chernozem**-type wheat soils. Residual and dome mountains are worn or laccolith leftovers, not young fold belts.
27. **Death Valley** (California) = rift valley, extreme heat, Devil's Golf Course salt (लवणाध्यक्ष) pan. **Silicon Valley** = California chip belt. **Great Artesian Basin** = Australia. Blind valley / sinkhole = **karst**.
28. **Arakan Yoma** = Myanmar (India–Myanmar hills: Naga (नागा), Chin, Patkai (पटकाई) / Lushai — not Khasi (खासी) as the boundary trio). **Golan Heights** = Middle East (SW Syria / occupied note). **Black Forest** = Germany (east of Rhine; Vosges west).
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29. Plateau types in teaching: intermontane (**Tibet**), piedmont, and volcanic / lava (**Columbia**, Deccan). **Pamir** = "Roof of the world"; **Altiplano** = Bolivia–Peru.
30. Valley tags: rift valleys (East Africa, Rhine), glacial U-valleys, and structural vales. **Death Valley** = rift / extreme heat; do not confuse with Silicon Valley.
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Must-score facts — continents, mountains, grasslands

Continent area (large → small)

Asia → Africa → N. America → S. America → Antarctica → Europe → Australia

Mountain / plateau tags

Item	Lock
Longest fold chain	Andes
Highest chain	Himalaya
Pyrenees	Spain–France
Alps	Central Europe (≠ England)
Kilimanjaro	East African Rift (≠ Ring of Fire)
Roof of the world	Pamir
Highest large plateau	Tibet
Colorado vs Columbia Plateau	Grand Canyon vs lava plateau

Grasslands

Name	Region
Pampas	Argentina
Campos	Brazil
Llanos	Venezuela–Colombia
Puszta	Hungary
Prairie / Steppe / Veld	N. America / Eurasia / S. Africa

Topic 19 — World Regional Geography

Consolidated — 33 Must-Score Facts

1. Continent area order: **Asia > Africa > North America > South America > Antarctica > Europe > Australia/Oceania.**
2. The **Nobi** and **Kanto** plains are in **Japan**, not Korea.
3. Iraq's **Sunni Triangle** is **Baghdad, Tikrit, and Ramadi**. **Basra** (Shia south) is the trap.
4. West Asia mountains west to east: **Pontic → Zagros → Hindu Kush → Karakoram (काराकोरम).**
5. Central Asia capitals: Uzbekistan **Tashkent**, Tajikistan **Dushanbe**, Kyrgyzstan **Bishkek**, Turkmenistan **Ashgabat**, Kazakhstan **Astana**.
6. The **Kara Kum** desert is in **Turkmenistan**.
7. **Borneo** is shared by Indonesia, Malaysia, and Brunei and is **not** a volcanic island dump.
8. Philippines cane and coconut history (इतिहास) fact: **Spanish and Americans.**
9. Korea: **Seoul** south, **Pyongyang** north, roughly the **38th parallel**. Nobi/Kanto are not Korean plains.
10. Palestine map: **Gaza** on the Egypt (मिस्र) side, **West Bank** on the Jordan side, with the **Jordan River / Dead Sea** belt. Neighbours include Israel, Jordan, Egypt, Lebanon, and Syria.
11. Western Europe has **westerlies** and rain in **all months**.
12. The **Suez Canal** shortened the India–Europe sea route by about **7000 km**.
13. **Cape Verde**'s capital is **Praia**. **Bamako** is Mali's capital — do not swap them.
14. Maghreb capitals: Morocco **Rabat**, Algeria **Algiers**, Tunisia **Tunis**.
15. Brazil points: **Selva (सेल्वा)** rainforest and **Terra Roxa** coffee soil are both true.
16. Australia's **interior is desert**. The **north is tropical**, not temperate. The **Darling Range** lies in **south-west** Australia.
17. India's operational Antarctic stations are **Maitri** and **Bharati**. **Dakshin Gangotri (गंगोत्री)** is not the operational answer.
18. **Madeira** is Atlantic Portugal and is **not** Caribbean.
19. Indonesia west to east: **Sumatra → Java → Bali (बलि) → Lombok**.
20. The **Mekong** delta (डेल्टा) is in **southern Vietnam**, not Cambodia.
21. South America's landlocked pair is **Bolivia** and **Paraguay**. Uruguay, Peru, and Suriname have coasts.
22. **Igarka** is in **Russia**, not China.
23. New Zealand's capital is **Wellington**. NZ has North and South Islands with **Cook Strait** and is **not** an Australian state.
24. **Ethiopia** is landlocked after Eritrea's secession; **Eritrea** holds the Red Sea coast. The Horn is Ethiopia–Somalia–Eritrea–Djibouti.
25. Only **double-landlocked** states in the usual set are **Uzbekistan** and **Liechtenstein**.
26. Capitals that are not the country's most-famous tourist city: Turkey **Ankara**, Australia **Canberra**, Brazil **Brasília**, UAE **Abu Dhabi**, NZ **Wellington**.
27. Boundary parallels: **38th** ≈ Korea; **49th** ≈ Canada–USA.
28. South Asia map: Nepal and Bhutan are landlocked; Maldives and Sri Lanka are islands; **Kabul** is Afghanistan's capital. The Caucasus trio is Georgia–Armenia–Azerbaijan; Istanbul is not Turkey's capital.
29. **Esperanto** is an artificial world auxiliary language. **Tamil** is an official language of **Singapore**. Spanish is official in Chile/Colombia/Cuba — **not** Congo. **Bahasa** = Indonesia (not Thailand). Mandarin leads classic L-1 speaker counts; English often leads total L-1+L-2.
30. Great Lakes west→east: **Superior → Michigan → Huron → Erie → Ontario**. The St Lawrence is the seaway outlet.

31. Maghreb = Morocco–Algeria–Tunisia (sometimes Libya in wider use). Sahel is the semi-arid belt south of the Sahara — not the same as Maghreb.
32. USA–Canada land border roughly follows the **49th parallel** in the west. Alaska is USA; Greenland is Denmark politically / North America geographically.
33. East Asia flash cards: Japan's **Honshu** is the main island; Korea split near the **38th parallel**; China's capital is **Beijing** (not Shanghai).

Must-score facts — capitals, deserts, regional traps

Central Asia capitals

Country	Capital
Uzbekistan	Tashkent
Tajikistan	Dushanbe
Kyrgyzstan	Bishkek
Turkmenistan	Ashgabat

Regional locks

Item	Lock
Nobi / Kanto plains	Japan (≠ Korea)
Sunni Triangle	Baghdad–Tikrit– Ramadi (≠ Basra)
W Asia mountains W→E	Pontic → Zagros → Hindu Kush → Karakoram
Kara Kum	Turkmenistan
Borneo	Indonesia + Malaysia + Brunei
Korea	Seoul south / Pyongyang north (~38th parallel)
Gaza / West Bank	Egypt side / Jordan side

Topic 20 — World Agriculture

Consolidated — 22 Must-Score Facts

1. In plantation MCQs the classic plantation crop is **tea**, not wheat, rice, or maize. Plantation means estate, capital, monoculture, hired labour, and export. **Fazenda** = Brazil plantation; it is **not** jhum.
2. Coffee producer order often taught for **2016** is **Brazil > Vietnam > Colombia > Indonesia**. Brazil is mainly **arabica** on Terra Roxa; Vietnam is bulk **robusta**. **Mocha** = Yemen; **Kona** = Hawaii.
3. Major cocoa producers are **Côte d'Ivoire, Ghana (घन), and Cameroon**. **Latvia** is not a cocoa producer. Cocoa originated in the Amazon; chocolate manufacturing centres (Switzerland, Belgium) are not the same as growers.
4. India's rubber fact is **Kerala**. Rubber (*Hevea*) originated in the Amazon; **Henry Wickham** moved seed stock to Ceylon / Malaya. Malaysia's **Kinta Valley** is **tin**, not rubber. Tree tapping starts after about **6–7 years**.
5. Shifting-cultivation names: **Jhum** (NE India), **Podu** (Andhra/Odisha), **Ladang** (Malaysia), **Milpa** (Mexico), **Roca** (Brazil), **Chena** (Sri Lanka), **Caingin** (Philippines), **Taungya** (Myanmar). It needs low density, short crop years and long fallow — not estate tea.
6. The world **citrus** / vine / olive belt is **Mediterranean** on five **west coasts** near **30–45°**, not equatorial. **Olive** majors = Spain, Italy, Greece.
7. Philippines cane and coconut history fact: **Spanish and Americans**. Wrong Indian crop–state dumps include Gujarat–tea, UP–jute and Assam–wheat. Right pairs: **Kerala–rubber; Assam–tea** volume.
8. **Oil palm** majors are **Indonesia and Malaysia**. World **sugarcane** leader is **Brazil**; **sugar beet** belongs to temperate **Europe**.
9. For rice and wheat, **China and India** lead in **volume**, but classic **exporters** are often other countries (Thailand / Vietnam for rice; Russia / USA / Canada and others for wheat). **Golden rice** carries **Vitamin A**.
10. Rice types: **Indica** (long-grain South / SE Asia), **Japonica** (stickier East Asia), **Javanica** (Indonesia). **IRRI** sits at **Los Baños, Philippines**; classic HYV rice = **IR-8**.
11. Wheat needs cool growth and bright ripening (~**50–75 cm** rain). **Winter wheat** = autumn sow in mild-winter belts; **spring wheat** = spring sow in harsh-winter Prairie / Siberia; **durum** = pasta wheat of the Mediterranean.
12. **Intensive subsistence** = monsoon wet rice. **Extensive commercial grain** = Prairie / Pampas wheat. **Mixed farming** = crops plus livestock (West Europe / US Midwest). Ranching belts = Pampas, Prairie, Veld, Downs.
13. **Black tea** is fermented; **green tea** is not. The pluck fact is **two leaves and a bud**. Tea likes **20–30°C, 150–300 cm** rain, slope and acid soil.
14. **Von Thünen** = market rings around a city (distance rent). **Whittlesey** = about **thirteen** world agricultural region types. Do not merge the two models.
15. Cotton: long staple = **Egypt / Sudan** (and Sea Island); medium = USA Upland; short staple = India–Pakistan–China belt. Maize volume order often **USA > China > Brazil**; **US Corn Belt** is the classic tag.
16. Banana: India produces heavily; **Ecuador** is a classic shipper. Silk and wool volume often centre on **China**; Australia is famous for **Merino** wool. **Shahtoosh** = Chiru wool.
17. Soy export triangle = **USA–Brazil–Argentina**. India leads world **milk volume**; New Zealand and the Netherlands dominate the dairy-**export** story. World Food Day = **16 October**.
18. **IRRI** (Philippines) drove HYV rice; **CIMMYT (Mexico)** and **Norman Borlaug** drove HYV wheat — Nobel for **Peace (1970)**, not "agriculture." **FAO HQ** = **Rome**.
19. Term cards: **apiculture** = bees; **viticulture** = grapes; **sericulture** = silk (**China** leads volume); **floriculture** hub = Netherlands; **olericulture** = vegetables; **pisciculture** = fish rearing.
20. India often leads milk + banana + castor; China leads tea / tobacco / silk volume stories; Brazil leads cane + coffee #1; Thailand is a rubber story. **Golden Crescent** = Afghanistan–Iran–Pakistan opium.
21. Jute world is almost entirely **India and Bangladesh** on Ganga–Brahmaputra alluvium — do not treat UP as a jute state.

22. Plantation vs shifting: plantation = capital estate monoculture for export; shifting = humid-tropics slash-and-burn subsistence. They are not the same system.

 Must-score facts — plantation, shifting, leaders

Crop / system tags

Item	Lock
Classic plantation crop (MCQ)	Tea (estate/capital/labour)
Coffee order (2016 teaching)	Brazil > Vietnam > Colombia > Indonesia
Cocoa majors	Côte d'Ivoire, Ghana, Cameroon
India rubber	Kerala (<i>Hevea</i> Amazon origin)
Oil palm majors	Indonesia, Malaysia
Olive majors	Spain, Italy, Greece
Sugarcane leader	Brazil ; sugar beet = temperate Europe
Citrus belt	Mediterranean

Shifting cultivation names

Name	Region
Jhum	NE India
Podu	Andhra / Odisha
Ladang	Malaysia
Milpa	Mexico

Topic 21 — World Minerals & Energy

Consolidated — 26 Must-Score Facts

1. Coalfield–country pairs: **Appalachian–USA**, **Lancashire–England**, **Ruhr–Germany**, **Kuzbass–Russia**. **Donetsk / Donbass** = Ukraine. **Karaganda** = Kazakhstan coal city.
 2. **Mount Newman / Pilbara / Hamersley** = **iron** in Australia. **Krivoy Rog** = Ukraine iron. **Lorraine** = France. **Kiruna** = Sweden. **Mesabi** = USA. Germany–Normandy iron is **wrong** — **Normandy** is in **France**.
 3. Iron exporters = Australia (**Pilbara**) and Brazil (**Carajás**). China mines iron and still imports heavily. **Hematite** = red bulk ore; **magnetite** = black highest grade.
 4. **Chile** leads copper with northern Andes **porphyry** deposits (**Chuquibambilla, El Teniente**). Also fact Katanga (DRC), Copperbelt (Zambia), and Bingham (USA).
 5. Malaysia's **Kinta Valley** = **tin** (cassiterite). Also tin centres: Bangka–Belitung (Indonesia) and the Andes of Bolivia. Myanmar's **Pegu Yoma** = **mineral oil**, not tin.
 6. **Kashagan** oil is in **Kazakhstan**, not Kuwait. **Burgan** = Kuwait. Also fact Ghawar / Dhahran (Saudi), Kirkuk / Zubair (Iraq), Haft Kel (Iran), Baku (Azerbaijan). **Brent** = North Sea light crude.
 7. Natural gas's main constituent is **methane**. **LPG** (propane–butane) is not the same as **CNG**. Qatar's **North Field** and Iran's **South Pars** are one continuous Gulf gas giant.
 8. **Postmasburg** (South Africa) = **manganese** (not uranium / mica / bauxite). Also Mn: Moanda (Gabon), Groote Eylandt (Australia), Nikopol (Ukraine).
 9. Bauxite → aluminium: **Weipa / Gove / Darling Range** (Australia); **Guinea** for reserves renown; Les Baux (France) is the name origin. Mount Newman is **iron**, not bauxite.
 10. Gold: **Witwatersrand** (South Africa historic); Kalgoorlie (Australia); modern volume often **China**. Diamond: **Kimberley** / kimberlite (South Africa); Golconda is historical **India**.
 11. Nickel: **Sudbury** (Canada) and **Norilsk** (Russia). Chromite: **Bushveld** (South Africa) and Great Dyke (Zimbabwe). Lead–zinc: **Broken Hill** / Mount Isa (Australia).
 12. Phosphate rock king = **Morocco**. Potash names include Canada (Saskatchewan) and the Dead Sea belt. **Iodine** (2018 spelling "Iodine") fact = **Chile** Atacama caliche.
 13. Uranium volume often facts on **Kazakhstan**; Canada **Athabasca**; Australia **Olympic Dam** (U + Cu + Au). **Uranium City** = **Canada**. Thorium / monazite sands = India (Kerala–TN), Brazil, Australia.
 14. **Lithium Triangle** = Chile–Argentina–Bolivia (Brazil is **not** in it). Cobalt volume = **DRC**. REE processing fact = **China**. Tungsten volume often **China**.
 15. Nuclear and geothermal energy are **not** "stored solar". Wind, biomass and hydro are solar-linked renewables. Coal, oil and gas are conventional non-renewables.
 16. **OPEC** HQ = **Vienna**. Original OPEC five (**1960**): Iran, Iraq, Kuwait, Saudi Arabia, Venezuela. North Sea oil/gas = UK–Norway; West Siberia = Russia's giant hydrocarbon province.
 17. **Itaipu** = Brazil–Paraguay hydel. **Three Gorges** = China on the Yangtze. Steel tonnage top = **China**.
 18. Ancient **shields** host many metals. Sedimentary **basins** host coal, oil and gas. Andean porphyry belts host copper, with lithium brines nearby in the same broad story.
 19. **German silver** contains **no silver**. **Peace Pipeline** = Iran–Pakistan. Sheet **mica** classic = **India**. Platinum = Bushveld / Norilsk.
 20. Energy classification check: coal / oil / gas = conventional non-renewable; wind / biomass / hydro = renewable solar-linked; nuclear and geothermal stand apart from "stored solar" wording.
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Must-score facts — coalfields, ores, energy types

Coalfield ↔ country

Field	Country
Appalachian	USA
Lancashire	England
Ruhr	Germany
Kuzbass	Russia

Ore / oil tags

Item	Lock
Pilbara / Newman / Hamersley	Australia iron
Krivoy Rog	Ukraine iron
Chile copper	Andes porphyry (Chuquicamata)
Kinta Valley	Malaysia tin
Pegu Yoma	Myanmar oil (≠ tin)
Kashagan	Kazakhstan oil (≠ Kuwait)
Burgan	Kuwait
Postmasburg	South Africa manganese
Hematite / magnetite	Red bulk / black high-grade

Energy

Item	Lock
Natural gas main	Methane
LPG ≠ CNG	Different fuel forms
Nuclear / geothermal	Not "stored solar"

Topic 22 — World Industries

Consolidated — 25 Must-Score Facts

1. City–industry pairs: **Osaka–cotton**, **Detroit–auto**, **Cuba–cigar**, **St Petersburg–shipbuilding**.
2. **Akron** = tyres; **Toulouse** = Airbus; **Silicon Valley** (California) = electronics / IT — not Detroit.
3. **Igarka** is in **Russia** (Yenisei timber), not China. **Rotterdam** is in the **Netherlands**. **Montevideo** is Uruguay. **Jakarta** is Indonesia.
4. **Duisburg** is an inland Rhine port, not a Dutch sea mouth. **Entrepôt** classics = Singapore, Rotterdam, Hong Kong. Break-of-bulk points concentrate at ports.
5. The **Suez Canal** joins the **Mediterranean** and **Red Sea** and cut India–Europe distance by about **7000 km**. It is a **sea-level** cut (no stepped chambers). Lakes north to south: **Manzala** → **Timsah** → **Great Bitter** → **Little Bitter**.
6. Port Said is at the **north** end of Suez; Suez town is at the **south**.
7. **Panama** joins Atlantic / Caribbean and Pacific **with stepped chambers** and **Gatun Lake**. **Kiel** joins the North Sea and Baltic.
8. Location logics: **weight-losing** industries (steel) sit near ore / coking coal; **market-oriented** (bottling) sit near consumers; **aluminium** seeks cheap **hydel**; **footloose** example = electronics / software; oil refining often seeks the port or pipeline end.
9. Soft location teaching still turns on raw material, power, labour, market and transport — not a single “any city works” rule.
10. **Ruhr** = Germany heavy industry. **Lancashire** = cotton. **Yorkshire** = wool. **Pittsburgh–Great Lakes** = steel. **Detroit** = autos.
11. Italy’s industrial triangle is the **Po Basin / Milan–Turin–Genoa** belt.
12. Japan belts: **Keihin** = Tokyo–Yokohama; **Hanshin** = Osaka–Kobe; **Chukyo** = Nagoya autos. Nicknames: Osaka = Manchester of Japan; Nagoya = Detroit of Japan; Kawasaki = Pittsburgh of Japan; Ivanovo = Russian Manchester.
13. China’s classic coastal manufacturing belt in this chapter = **Pearl River Delta**. Shanghai leads container volume among common port MCQs; Rotterdam sits at the Rhine mouth.
14. Shipbuilding volume leaders are **China–South Korea–Japan**. St Petersburg remains the classic European city match.
15. Textiles often seek labour and market; steel is the classic weight-losing case.
16. **Chinook** is a warm dry wind of the Rockies; **Foehn** is the Alps equivalent. Both can be true together.
17. **Mistral** = southern France (**not** Australia). **Shamal** = Arabia (**not** Austria). **Brickfielder** = Australia.
18. A **willy-willy** is a cyclone name, not a local wind.
19. More wind pairs: **Santa Ana** = California; **Haboob** = Sudan; **Yamo** = Japan; **Leveche** = Spain; **Bora** = Adriatic cold; **Harmattan** = West Africa; **Sirocco** = Sahara→Med; **Khamsin** = Egypt.

Must-score facts — city–industry, canals, winds

City ↔ industry

City	Industry
Osaka	Cotton
Detroit	Auto
Cuba	Cigar
St Petersburg	Shipbuilding
Ruhr	Heavy industry (Germany)

City	Industry
Lancashire / Yorkshire	Cotton / wool
Pittsburgh–Great Lakes	Steel

Japan belts

Belt	Lock
Keihin	Tokyo–Yokohama
Hanshin	Osaka–Kobe
Chukyo	Nagoya autos

Canals / winds / industry type

Item	Lock
Suez	Med ↔ Red Sea; ~7000 km shorter India–Europe
Panama	Atlantic/Caribbean ↔ Pacific
Kiel	North Sea ↔ Baltic
Chinook / Foehn	Rockies / Alps warm dry winds
Mistral	S. France (≠ Australia)
Footloose	Electronics; aluminium seeks hydel; steel seeks ore+coal
Entrepôt	Singapore, Rotterdam, Hong Kong

Topic 23 — Political & Map-Based Geography

Consolidated — 40 Must-Score Facts

1. UNCLOS: **territorial sea 12 nm, contiguous zone 24 nm, EEZ 200 nm**. The continental shelf may extend to **350 nm**, but that does **not** push EEZ water beyond 200.
2. **Innocent passage** applies in the territorial sea. **Transit passage** applies in international straits.
3. **McMahon Line** = India–China (1914). **Durand Line** = Pakistan–Afghanistan (1893). **Radcliffe Line** = 1947 India–Pakistan/Bangladesh.
4. India has **seven** land neighbours (Pakistan, Afghanistan via Wakhan/PoK, China, Nepal, Bhutan, Myanmar, Bangladesh). Maritime neighbours are **Sri Lanka** and the **Maldives**.
5. Longest Indian **state** coastline = **Gujarat**. Longest land border = **Bangladesh**. Shortest land border = **Afghanistan**. Mainland plus islands coastline $\approx$ **7516 km**.
6. The **Suez Canal** joins Med and Red Sea, shortens India–Europe by about **7000 km**, and lakes run Manzala → Timsah → Great Bitter → Little Bitter.
7. **Panama** = Atlantic–Pacific canal. **Kiel** = North Sea–Baltic.
8. Straits: **Hormuz** = Gulf oil; **Malacca** = Indian Ocean–South China Sea; **Gibraltar** = Med–Atlantic; **Bosporus** = Black Sea–Marmara; **Bering** = Russia–USA.
9. Area ladder: Russia > Canada > USA > China > Brazil > Australia > **India (7th)**. Longest world coastline = **Canada** (then Indonesia, Russia, Philippines, Japan, Australia in the usual set).
10. Central Asia capital set: Uzbekistan **Tashkent**, Tajikistan **Dushanbe**, Kyrgyzstan **Bishkek**, Turkmenistan **Ashgabat**.
11. **Bolivia** is landlocked among common South America traps. **Laos** = only SE Asia landlocked. **Nobi/Kanto** = Japan. **Igarka** = Russia.
12. **Thorntwaite** (थॉर्न्थ्वेट) is the true **vegetation** climate index. **Köppen** gives letter-code climate classes. Mediterranean = **winter rain**. Western Europe = rain **all months** plus westerlies (**Cfb**).
13. India is the **seventh**-largest country, about **2.4%** of world land, with the Tropic of Cancer (कर्क रेखा) through the middle — so India is **not** wholly tropical. The Tropic does **not** cross **Uttar Pradesh** (उत्तर प्रदेश).
14. **Cape Verde** capital = **Praia**. **Bamako** = Mali. Only common double-landlocked states: **Uzbekistan** and **Liechtenstein**.
15. Uttar Pradesh's only foreign neighbour is **Nepal**.
16. Largest landlocked country by area = **Kazakhstan**. Most populous landlocked = **Ethiopia**. **Lesotho** is an enclave inside South Africa.
17. **49th Parallel** $\approx$ USA–Canada. **38th Parallel** $\approx$ Koreas. Maginot $\approx$ France–Germany; Rio Grande $\approx$ USA–Mexico.
18. Equator traps: Egypt (मिस्र) and Mexico are **not** equatorial countries in the usual MCQ sense.
19. India's nine coastal states include Gujarat as the longest; Telangana is not coastal.
20. Oder–Neisse is the historic (ऐतिहासिक) Germany–Poland line; do not confuse it with McMahon or Durand.
21. Shelf rights beyond 200 nm (up to 350) are seabed rights — they do not create a wider EEZ water column.
22. Horn of Africa = Djibouti, Eritrea, Ethiopia, Somalia (**not Sudan**). Balkans = Albania, Bosnia and Herzegovina, Bulgaria, Croatia, Greece, Kosovo, Montenegro, North Macedonia, Romania, Serbia, Slovenia, European Türkiye — **Austria not** Balkan. Oceania excludes **Indonesia**. Caspian five exclude Armenia/Iraq.
23. Capitals $\neq$ famous cities: Australia **Canberra**; Japan **Tokyo**; Nigeria **Abuja**; Myanmar **Naypyidaw**; Slovenia **Ljubljana** (Bratislava = Slovakia).
24. Old names: Siam→Thailand; Formosa→Taiwan; Gold Coast→Ghana (घना); Dutch Guiana→**Suriname**; Southern Rhodesia→Zimbabwe.
25. Greenland = Denmark politically / N America geographically. Gaza borders **Egypt + Israel**. Afghanistan does **not** border Russia.
26. Most megacities are coastal as **ocean gateways**. Alaska = USA; Malta = Mediterranean; Baikonur = **Kazakhstan**.

27. **Norway** = Land of the Midnight Sun (Arctic Circle). **Japan** = Land of the Rising Sun. **Finland** = Thousand Lakes. **(South) Korea** = Morning Calm. **Thailand** = White Elephants.
28. **South America** = Bird Continent. **Sri Lanka** = Mistress of the Eastern Sea / Pearl of the Indian Ocean. **Singapore** = Gateway to Asia. **Istanbul** = Gateway to the West.
29. City tags: **Venice** = canals; **Osaka** = Manchester of the East; **San Francisco** = Golden Gate; **Chicago** = City of Smoke; **Buenos Aires** = Paris of South America; **St. Petersburg** = Venice of the North.
30. **Pamir** = Roof of the World. **Baikal** = Pearl of Siberia. **Bahrain** = Island of Pearls. **Aberdeen** = Oil Capital of Europe. **Ninety East Ridge** = Indian Ocean.
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31. India's land neighbours are seven; maritime neighbours are Sri Lanka and Maldives. Longest land border = **Bangladesh**; shortest = **Afghanistan** (via Wakhan / PoK teaching).
32. Sobriquet diet extras: **Venice** = canals; **Osaka** = Manchester of the East; **Chicago** = City of Smoke; **Pamir** = Roof of the World; **Baikal** = Pearl of Siberia.
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Must-score facts — UNCLOS, borders, straits

UNCLOS distances

Zone	Distance
Territorial sea	12 nm
Contiguous zone	24 nm
EEZ	200 nm
Continental shelf max	Up to 350 nm (conditions)
Innocent passage	Territorial sea
Transit passage	International straits

Borders / lines

Line	Lock
McMahon	India–China (1914)
Durand	Pakistan–Afghanistan (1893)
Radcliffe	1947 India–Pakistan
Longest Indian state coastline	Gujarat
Longest land border	Bangladesh
Shortest land border	Afghanistan (teaching)
Area ladder	Russia > Canada > USA > China > Brazil > Australia > India (7th)

Straits

Strait	Lock
Hormuz	Gulf oil
Malacca	Indian Ocean–South China Sea
Gibraltar	Med–Atlantic
Bosporus	Black Sea–Marmara

Topic 24 — Geography of Uttar Pradesh

Consolidated — 28 Must-Score Facts

1. UP is India's most populous state and has the **largest rural** population. Area rank is **4th** (~2.4 lakh km²). There are **75** districts and **18** divisions. Capital is **Lucknow**; High Court is at **Prayagraj**. The State name date is **24 January 1950**; **Uttarakhand** was carved on **9 November 2000**.
2. Latitudes run roughly **23°52'–31°28' N**; longitudes **77°–84° E**. The **Tropic of Cancer does not** enter UP. Present UP has **no Himalayan high peaks** after 2000 — **Nanda Devi** is in **Uttarakhand**.
3. Neighbours are Uttarakhand, Haryana, Delhi, Rajasthan, Madhya Pradesh, Chhattisgarh, Jharkhand, Bihar and **Nepal**. The only foreign land neighbour is **Nepal**. Seven Nepal-border districts are Pilibhit, Lakhimpur Kheri, Bahraich, Shravasti, Balrampur, Siddharthnagar and Maharajganj.
4. District extremes: north **Saharanpur**, south **Sonbhadra** (सोनभद्र), east **Ballia**, west **Shamli**. **Lalitpur** has only **one** UP-district neighbour (**Jhansi**). **Devipatan** division HQ is **Gonda** (Gonda–Bahraich–Shravasti–Balrampur) — **not Basti**.
5. Physiographic map: **Bhabar** = pebble soak where streams sink; **Terai** = marshy re-emergence belt (Dudhwa / Katarniaghat / Suhelwa). **Doab** = Ganga–Yamuna Green Revolution heart; **Awadh** = Gomti–Ghaghara; **Rohilkhand** = Ramganga / mentha; **Purvanchal** = eastern rice plain; **Bundelkhand** = drought plateau; **Vindhyan** = Sonbhadra–Mirzapur energy fringe.
6. Köppen tag for the Gangetic plain is **Cwg** (monsoon with dry winter). Rain generally rises **west to east** (west doab ~**60–80 cm**; east / Terai ~**100–140 cm**). Classic wettest belt tag = **Gorakhpur**; driest = **Mathura**.
7. May–June **Loo** hits the western plain. Hottest summer belt tags include **Jhansi, Banda and Agra**. **Western Disturbances** give winter rain that supports **rabi wheat**. After Uttarakhand's creation, UP has **no alpine / snow climate**.
8. About **90%** of UP is **alluvial**. **Khadar** = new fertile floodplain; **Bangar** = older alluvium with **kankar**; **Bhur** = sandy western-doab ridges. **Usar / reh / kallar** on west–central canal tracts needs **gypsum plus drainage**, not lime alone.
9. Bundelkhand soils: **Mar / Kabar** = **black**; **Parua / Rakar (pathari)** = **red**. **Bhonta / Monta** = Vindhyan red–yellow (Mirzapur–Sonbhadra). Ravines mark the **Agra–Etawah–Jalaun** Chambal–Yamuna belt.
10. **Ramganga** meets the Ganga at **Kannauj**. **Betwa** meets the Yamuna at **Hamirpur** (not Prayagraj). **Ken** joins near **Banda**. Yamuna right-bank order west to east: **Chambal–Sind–Betwa–Ken**.
11. **Gomti** rises at **Madho Tala / Fulhar, Pilibhit** — Lucknow stands on it (2018 pollution / “biological disaster” tag). **Ghaghara / Saryu** passes Ayodhya; **Rapti** passes Gorakhpur. **Hindon** is a Yamuna **left**-bank NCR river.
12. National **Chambal** Sanctuary spans **UP–MP–Rajasthan** (Etawah–Agra) for gharial — **not** Haryana. **Badua** dam is **not** in UP. **Matatila** is on the **Betwa** in **Lalitpur**. **Ken–Betwa** is the live Bundelkhand interlinking project. Eastern UP hydel is blocked because the Himalaya to the north lies in **Nepal**.
13. Irrigation is dominated by **tubewells** (~**75%**; canals ~**15%**). Western UP faces groundwater **dark-zone** pressure. UP leads absolute tubewell area and groundwater resources for irrigation.
14. Canal classics: Upper Ganga offtake **Bhimgoda / Haridwar (1854, P.T. Cautley)**; Lower Ganga **Narora (1878)**; **Sharda** at **Banbasa** (longest UP system length); East Yamuna at Tajewala / Hathinikund; Agra Canal at Okhla; **Saryu Canal** National Project for eastern UP; plus Sarda Sahayak.
15. Crop map: **wheat** = west–centre rabi GR belt; **sugarcane** = west–centre (UP often **#1 quantity**, Maharashtra often higher **productivity**); **rice** = east and Terai; **potato** = UP leads quantity; **mentha** = Rohilkhand; oilseeds / pulses = Bundelkhand. UP–jute / UP–tea / UP–rubber are **wrong** pairs.
16. CIP potato centre fact = **Agra (Singna)**, not Aligarh. **IRRI-SARC** rice centre = **Varanasi**. Malihabad pairs with **Dasheri** mango.
17. **Sonbhadra** is the energy / mineral capital (non-coking Singrauli-side coal, limestone, thermal). **HINDALCO Renukoot** = aluminium. **Shankargarh** silica feeds **Firozabad** glass. **Chunar–Churk–Dalla** = cement limestone.

- Sonrai (Lalitpur)** = copper occurrence. **Mathura** has the oil refinery. Obra, Anpara and Rihand are thermal hubs.
18. City–industry pairs: **Kanpur** textile / leather; **Moradabad** brass; **Aligarh** padlocks / hardware; **Firozabad** glass; **Meerut** sports goods; **Bhadohi** carpets; **Khurja** pottery; **Saharanpur** woodcraft; **Varanasi** silk; Noida–Ghaziabad = NCR engineering. ODOP Moonj pair = **Amethi + Sultanpur**.
19. Forest cover is about **6%** of TGA (not 12.8%). ISFR >20% districts: **Chandauli** and **Shravasti**. Jaunpur–Ghazipur–Ballia have near-nil forest land. Terai–bhabar vegetation is moist deciduous (**sal**); Vindhyan south is dry deciduous.
20. UP's **only National Park** is **Dudhwa** (Lakhimpur Kheri). Tiger Reserves are Dudhwa (NP + Kishanpur + **Katarniaghat**), **Pilibhit**, **Amangarh (Bijnor)** and **Ranipur (Chitrakoot)** — do not invent extra NPs. **Katarniaghat** WLS = Bahraich, **1975**. **Chandraprabha** WLS = **Chandauli**.
21. Other sanctuary tags: **Hastinapur** (Meerut–Bijnor–Amroha; swamp deer); **Kaimoor** (Sonbhadra–Chandauli); **Suhelwa** (Balrampur–Shravasti–Gonda); **Sohagibarwa** (Maharajganj); **Mahavir Swami** (Lalitpur).
22. UP has **13** Ramsar sites; India's **100th** was **Surha Tal / JP Narayan Bird Sanctuary, Ballia**. Sites **not** in UP: Rudrasagar (Tripura), Sultanpur (Haryana), Surinsar–Mansar (J&K). **Sarsai Nawar** and **Samaspur** are in UP.
23. State animal = **swamp deer**; state bird = **Sarus crane**; state flower = **Butea / Dhak / Palash** (*Butea monosperma*).
24. Census 2011 freeze: UP ~**16.5%** of India; density **829**; sex ratio **912**; literacy ~**67.7%**; urban ~**22.3%**. Lowest female literacy among common facts = **Shrawasti**. Highest literacy belt = **Gautam Buddha Nagar**. Smallest-area trap among listed four = **Hapur**. Largest-area standard = **Lakhimpur Kheri**.
25. Density extremes: highest **Ghaziabad**; lowest belt **Lalitpur / Sonbhadra**. UA 2011 order **Kanpur > Lucknow > Ghaziabad > Agra**. Kanpur crossed a million in **1971**; Lucknow in **1981**.
26. Central / Mission Smart Cities = **10** (Lucknow, Kanpur, Prayagraj, Agra, Varanasi, Aligarh, Bareilly, Jhansi, Moradabad, Saharanpur). **Ghaziabad is not** in that Central-10. **HRIDAY** in UP = **Varanasi and Mathura** only.
27. NER HQ = **Gorakhpur**; NCR HQ = **Prayagraj**. No major seaport; waterway fact = **NW-1** on the Ganga. **Kushinagar** international airport opened **24 June 2020**. Jewar = Noida International. Amrit Bharat UP stations fact = **149**.
28. Expressways: **Yamuna** (Noida–Agra) was the first; also Agra–Lucknow, **Purvanchal** (Lucknow–Ballia via **Mau**), Bundelkhand, Ganga Expressway and Gorakhpur Link. Built Purvanchal does **not** run through Basti.

Must-score facts — parks, rivers, soils, Census

Parks / wetlands

Item	Lock
Only National Park	Dudhwa (Lakhimpur Kheri)
Chandraprabha	Sanctuary — Chandauli
Forest cover	~ 6% (not 12.8%)
Ramsar count (teaching)	13 in UP
Not in UP	Rudrasagar, Sultanpur, Surinsar–Mansar

Rivers / soils / agri

Item	Lock
Ramganga–Ganga meet	Kannauj
Betwa–Yamuna meet	Hamirpur (≠ Prayagraj)
Bundelkhand soils	Mar/Kabar = black ; Parua/Rakar = red
Alluvial share	~ 90% of UP

Item	Lock
Matatila	Betwa — Lalitpur
Potato	UP leads quantity; CIP Agra (Singna)
Mentha	Rohilkhand / Tarai belts

Admin / Census traps

Item	Lock
Devipatan HQ	Gonda (≠ Basti)
Lowest female literacy (common)	Shrawasti
NER HQ	Gorakhpur
Smart Cities	10; Ghaziabad out of central list teaching

Uttarakhand — Topic 1 — Location, Relief & Structure

Consolidated — 22 Must-Score Facts

1. Uttarakhand occupies the **central Himalayan** sector of India, roughly **28°43'–31°27' N** and **77°34'–81°02' E**.
2. **Himachal Pradesh** borders the west; **Nepal** the east; the **Tibetan** frontier the north; **Uttar Pradesh** plains the south.
3. Historic-administrative divisions: **Garhwal** is west; **Kumaon** is east.
4. Garhwal districts include Dehradun, Tehri, Uttarkashi, Chamoli, Rudrapur, and Pauri with related hill tehsils.
5. Kumaon districts include Nainital, Almora, Bageshwar, Champawat, and Pithoragarh with related hill tehsils.
6. **Haridwar** and **Udham Singh Nagar** sit on the southern Tarai fringe of this dual map.
7. South→north relief stack: **Tarai → Bhabar → Shiwalik → Dun → Lesser Himalaya → Great Himalaya**.
8. Compact four-name lists often read **Tarai → Shiwalik → Lesser Himalaya → Great Himalaya**; placing Tarai last is the usual inversion trap.
9. National plains teaching reads the same foothill belts southward as **Shiwalik → Bhabar → Terai** — do not reverse either list without checking where the stem starts.
10. **Bhabar** is porous gravel-boulder at the Himalayan foot where streams often **sink** underground.
11. **Tarai** is the marshy belt south of Bhabar where groundwater **reappears** and soils support dense cultivation and forest.
12. **Shiwaliks** are young, soft foothills of unconsolidated sediments and are landslide-prone.
13. **Duns** are longitudinal flat-floored valleys in the Shiwalik belt; **Dehradun** is the classic dun.
14. **Main Boundary Thrust (MBT)** separates the Shiwaliks from the Lesser Himalaya.
15. **Main Central Thrust (MCT)** separates the Lesser Himalaya from the Higher / central crystalline belt.
16. **South of the MBT** = Shiwalik foothills and duns; **between MBT and MCT** = **Lesser Himalaya**.
17. **North of the MCT** = Higher / central crystalline **Great Himalaya** (snow line, major peaks, glacier heads).
18. Tarai and Bhabar sit **south of the Shiwalik**, not between the thrusts.
19. Farther north, **Trans-Himalaya / Tethys Himalaya** is a separate structural family beyond the Great Himalayan crest.
20. **Nanda Devi** is the highest major UK peak at about **7816–7817 m** in **Chamoli**; **Kamet** is next among major peaks at about **7756 m**.
21. Named Garhwal crest peaks also include **Trisul**, **Dunagiri**, and **Chaukhamba**.
22. Great Himalayan glaciers (Gangotri, Pindari, Milam and related ice fields) feed Bhagirathi–Alaknanda headstreams and shape pilgrimage–tourism geography.

Uttarakhand — Topic 2 — Climate & Drainage

Consolidated — 28 Must-Score Facts

1. **Kali (Sharda)** is the longest river in Uttarakhand at **252 km**; source Lipulekh / Kalapani; it forms the international boundary with Nepal.
2. **Bhagirathi (205 km)** and **Alaknanda (195 km)** unite at **Devprayag** to form the **Ganga**.
3. West→east river alignment: **Yamuna** → **Bhilangana** → **Alaknanda** → **Gori Ganga**.
4. Bhagirathi rises at **Gaumukh** from the **Gangotri Glacier** (Uttarkashi); Alaknanda rises from **Satopanth** and **Bhagirath Kharak** (Chamoli).
5. **Yamuna** rises from the Yamunotri Glacier (Bandarpunch, Uttarkashi); **Tons** carries about **twice the water volume** of the Yamuna.
6. **Western Ramganga** rises in the Dudhatoli range and traverses Corbett National Park into UP; **Kosi (Kaushiki)** drains Almora–Ramnagar and joins Ramganga in UP.
7. **Saryu** rises at Sarmool (Bageshwar) and meets Kali at **Pancheshwar**; **Mandakini** rises from Chaurabari Glacier and meets Alaknanda at Rudraprayag.
8. **Bhilangana** rises from Khatling Glacier (Tehri) and is submerged in the Tehri Dam reservoir.
9. Panch Prayag north→south: **Vishnuprayag** → **Nandaprayag** → **Karnaprayag** → **Rudraprayag** → **Devprayag**.
10. Vishnuprayag = Alaknanda + Western Dhauliganga; Nandaprayag = Alaknanda + Nandakini; Karnaprayag = Alaknanda + Pindar.
11. Rudraprayag = Alaknanda + Mandakini; Devprayag = Alaknanda + Bhagirathi → Ganga.
12. Longest river = **Kali (252 km)**; river with greatest water volume among common traps = **Alaknanda**.
13. **Gangotri Glacier** (Uttarkashi) is the largest in Uttarakhand (~**30 km** long, 2 km wide); snout **Gaumukh** (~3,900 m).
14. **Milam Glacier** (Pithoragarh) is the largest in Kumaon (~**16 km**); source of **Gori Ganga**.
15. **Pindari Glacier** (Bageshwar) feeds the **Pindar**; discovered / crossed by **G. W. Traill** via Traill's Pass (1830).
16. **Khatling Glacier** (Tehri) feeds Bhilangana; **Chaurabari Glacier** (Rudraprayag) feeds Mandakini and Chaurabari Tal (Gandhi Sarovar) — linked to the June 2013 Kedarnath disaster.
17. **Roopkund** (Chamoli, ~5,029 m) is the "Mystery / Skeleton Lake" with 9th-century skeletal remains on the Nanda Devi Raj Jat route.
18. **Naukuchiatal** (Nainital) is the deepest lake in Kumaon (~40 m, nine corners); **Bhimtal** is the largest lake in Kumaon.
19. **Nainital Lake** is pear-shaped, discovered by **P. Barron** in 1841; **Dodital** (Uttarkashi) is the six-cornered Ganesha lake feeding Asiganga.
20. **Hemkund (Lokpal)** (Chamoli) is sacred among seven snow peaks; **Deoria Tal** (Rudraprayag) reflects the Chaukhamba peaks.
21. Panch Kedar body parts: Kedarnath = hump; Madhyamaheshwar = navel; **Tungnath = arms** (highest Shiva temple, **3,680 m**); Rudranath = face; Kalpeshwar = hair (open all year).
22. All **Panch Badri** seats are in **Chamoli**: Vishal Badri (Badrinath), Yogdhyan Badri, Bhavishya Badri, Vridha Badri, Adi Badri.
23. Kali major junctions include Gori Ganga at **Jauljibi** and Saryu at **Pancheshwar**; Gori Ganga is the eastern Pithoragarh trunk in west→east lists.
24. Glacier-fed Himalayan rivers and steep gradients shape both drainage and hydropower potential of the state.
25. High-altitude lakes and glacial tarns (Sahasratal on the Khatling ridge; Roopkund; Hemkund) sit on pilgrimage and adventure circuits.
26. Do not swap Gangotri (largest in UK) with Milam (largest in Kumaon).
27. Do not swap Vishnuprayag (Alaknanda + Western Dhauliganga) with Devprayag (Alaknanda + Bhagirathi).

28. Do not swap Tungnath (highest Shiva temple / arms) with Kedarnath (supreme Jyotirlinga / hump).

Uttarakhand — Topic 3 — Vegetation & Wildlife

Consolidated — 24 Must-Score Facts

1. Forest richness of Uttarakhand is explained primarily by **topography and climate** packing many belts into a short horizontal distance.
2. Altitude stack: foothill **sal** → **chir** → oak–rhododendron → **deodar / fir** → alpine **birch–juniper** → **bugyal**.
3. Foothill **sal** and mixed deciduous forests occupy Tarai–Bhabar and the lower Shiwalik face.
4. Sub-tropical **chir** pine commonly sits about **900–2000 m** on outer slopes.
5. **Deodar** is abundant in the western Himalayan temperate belt at about **1500–2500 m**.
6. Among deodar / pine / sal / teak options, **deodar** grows at the **highest** elevation from sea level.
7. **Birch** belongs to Himalayan alpine / moist alpine scrub and has paper-like bark once used for writing.
8. Above the tree-line, alpine meadows called **bugyals** support seasonal pastures and high-altitude herbs.
9. Southern Himalayan slopes get more rain and sunlight than the rain-shadow north face, so vegetation is denser on the south.
10. Tree-line is generally **lower** in the Western Himalaya than in the Eastern Himalaya.
11. **Lisa** is pine resin; distillation yields turpentine and residual resin used in paint, varnish, and related industries.
12. **Khair** is a **sub-tropical** forest tree of Shiwalik / western Himalaya scrub and also yields **katha**.
13. **Burans** is the rhododendron flowering hill shrub — do not confuse it with Lisa or deodar.
14. **Nanda Devi Biosphere Reserve** is a UNESCO World Heritage property with Valley of Flowers; fauna includes **snow leopard, musk deer, and blue sheep (bharal)**.
15. **Askot** sanctuary = **Pithoragarh**; **Binsar** sanctuary = **Almora** — classic district swap trap.
16. **Govind Pashu Vihar / NP** = **Uttarkashi**; **Nandhaur** sanctuary = **Nainital**.
17. National Park year spine: **Corbett 1936, Rajaji 1983, Gangotri 1989, Govind 1990** (Govind is not 1989).
18. Corbett carries India's first national-park tradition; keep year pairs stable across sanctuary–NP match stems.
19. Forests remain the major natural resource of the western Himalayan region and support rural livelihoods through resin, timber, and NTFP.
20. Oak–rhododendron and mixed broad-leaved forests mark the temperate mid-hill transition below the deodar–fir belt.
21. Higher belts carry **fir and spruce** before alpine birch–juniper scrub toward the tree-line.
22. Keekar and acacia appear with khair in sub-tropical scrub above about **1000 m**.
23. Swapping Askot and Binsar districts loses sanctuary–district match codes.
24. Protected-area size crowns and Ramsar detail live mainly in the Environment UK parks chapter; this card keeps vegetation belts, products, and district–sanctuary pairs.

Uttarakhand — Topic 4 — Minerals, Power & Industry

Consolidated — 18 Must-Score Facts

1. Uttarakhand has scattered Himalayan–foothill minerals; it does **not** have one giant coal–iron belt.
2. **Dolomite** pairs with **Dehradun** and **Pithoragarh**.
3. **Soapstone** pairs with **Chamoli** and **Bageshwar**.
4. **Graphite** pairs with **Almora** and **Nainital**.
5. **Silica sand** pairs with **Dehradun, Nainital, and Tehri Garhwal** — **not Pauri Garhwal**.
6. Pauri Garhwal is the planted wrong match for silica sand in the standard pairing.
7. Limestone and magnesite appear as minor coaching occurrences; the four named pairs above carry the heaviest recall weight.
8. Hydropower dominates the power story because of steep gradients and glacier-fed rivers.
9. Major project belts follow the **Bhagirathi, Alaknanda, and Yamuna** systems.
10. The **Tehri** hydropower complex on the Bhagirathi near the Bhilangana confluence is the flagship multipurpose project (power, flood moderation, storage).
11. Named coaching heads also include run-of-river and storage schemes such as **Vishnuprayag** and **Srinagar** on the Alaknanda belt, plus Yamuna-side projects in western Garhwal.
12. High alpine sites face glacial and seismic constraints; **thermal coal is not** the core identity of the UK power map.
13. Industry concentrates in **Tarai / plains** districts and the **Dehradun** dun more than in high alpine tehsils.
14. **SIIDCUL** anchors planned industrial estates in the plains–dun belt.
15. Coaching industrial nodes include **Haridwar, Pantnagar / Rudrapur** (Udham Singh Nagar), and **Sitarganj**-side Tarai estates.
16. **Udham Singh Nagar** hosts agro-based and light manufacturing linked to the Tarai farm belt.
17. **Dehradun** carries administration, education, pharma, IT, and service-industry pockets; forest-based and tourism services matter in mid-hill towns.
18. Heavy smokestack industry is limited by terrain, ecology, and transport cost — light manufacturing and services dominate instead.

Uttarakhand — Topic 5 — Agri, AH & Irrigation

Consolidated — 18 Must-Score Facts

1. Cropped area concentrates in the southern plains and Tarai where continuous fields are possible on gentler slopes.
2. **Haridwar** and **Udham Singh Nagar** form the granary / high cultivated-area belt.
3. Series B for 2025 marks **Haridwar** as the district with the **highest agricultural land**; many notes also name **Udham Singh Nagar** as the Tarai rival — keep both names.
4. Hill agriculture is fragmented and **terrace-based** on steep slopes.
5. Mid-hill farms grow cereals, millets, pulses, and oilseeds on small holdings.
6. Horticulture matters in suitable altitudes — apples, stone fruits, citrus, and off-season vegetables.
7. Plains and Tarai emphasise **rice, wheat, sugarcane**, and oilseeds under denser irrigation.
8. Hills emphasise **mandua (finger millet)** and **jhangora**, with barley and other hardy cereals on terraces.
9. Pulses and potatoes appear widely as catch or kitchen crops in mid-hills; orchard and vegetable belts add cash income where roads allow.
10. Crop choice follows altitude, slope, and water availability — the state does not run on a single statewide monoculture.
11. Animal husbandry is a **supplementary** occupation to agriculture through milk, wool, manure, and household nutrition — not the sole livelihood for most households.
12. Cattle and buffalo support milk and draught in plains and mid-hills; sheep and goats matter in higher pastures.
13. In the 2025 multi-statement AH stem, Series B accepts **Only statement 1** (supplementary occupation); the sheep-rearing environmental-degradation claim is not accepted in that key.
14. **Tarai / plains** irrigation relies on **canals and tubewells** (Haridwar and Udham Singh Nagar type districts).
15. Mid-hill systems rely more on **springs** and small gravity channels called **guls**; tanks appear in some mid-hill systems.
16. Steep terrain limits large canal networks in the higher Himalaya, so irrigation intensity falls from Tarai granary districts toward high alpine valleys.
17. Plains lead cultivated-area stories; hills lead terrace / horticulture stories — do not invert the contrast.
18. Fresh land-use releases can shuffle absolute Haridwar vs Udham Singh Nagar ranks; prefer the Series B letter when that paper is cited.

Uttarakhand — Topic 6 — Population, SC/ST & Settlements

Consolidated — 26 Must-Score Facts

1. Census 2011 total population of Uttarakhand is **1,00,86,292** (~1.01 crore).
2. Highest population district = **Haridwar** (~18.90 lakh); lowest = **Rudraprayag** (~2.42 lakh).
3. State decadal growth (2001–11) = **18.81%** (national 17.70%); highest growth = **Udham Singh Nagar (33.45%)**.
4. Two districts registered **negative** decadal growth: **Pauri Garhwal (-1.41%)** and **Almora (-1.28%)** — the ghost-village / *palayan* story.
5. State density = **189** persons/sq km; highest = **Haridwar (801)**; lowest = **Uttarkashi (41)**.
6. State sex ratio = **963** (national 943); highest = **Almora (1142)**; lowest = **Haridwar (880)**.
7. Sex-ratio descending trap set: **Almora (1142) → Rudraprayag (1114) → Tehri Garhwal (1077) → Haridwar (880)**.
8. Seven hill districts have sex ratio above **1000** due to heavy male out-migration; plains districts Haridwar (880) and Dehradun (889) are among the lowest.
9. Child sex ratio (0–6) = **890** (national 919); Almora highest (922); Pithoragarh lowest (816).
10. Overall literacy = **78.82%**; highest = **Dehradun (84.25%)**; lowest = **Udham Singh Nagar (73.10%)**.
11. Highest male literacy = **Rudraprayag (93.90%)**; lowest female literacy extreme = **Uttarkashi (59.74%)** (only district under 60% female literacy).
12. SC share = **18.76%** of total population; highest SC % = **Bageshwar (27.73%)**; lowest SC % = **Dehradun (14.5%)**.
13. ST share = **2.89%** of total population; highest ST % = **Udham Singh Nagar (7.46%)**; lowest ST % = **Rudraprayag (0.15%)**.
14. **Udham Singh Nagar** holds about **42.1%** of the state's tribal population (mainly **Tharu** and **Buksa**).
15. **Dehradun** holds about **38.2%** of the tribal population (mainly **Jaunsari** in Jaunsar-Bawar).
16. **Pithoragarh** holds **Bhotia** (Johar/Darma) and the primitive **Raji (Vanrawat)**; **Chamoli** holds Marchha and Tolchha Bhotias; **Uttarkashi** holds Jad Bhotias.
17. **Vanrawats / Raji** are forest-dwelling groups concentrated in **Pithoragarh** and historically practised **shifting cultivation** until restricted by forest laws.
18. Urban population = **30.23%** (rural 69.77%); most urbanized = **Dehradun (55.52%)**; least urbanized = **Bageshwar (3.49%)**.
19. Mid-Himalayan settlements are often nucleated on southern spurs for sunlight with terraced *khet* cultivation.
20. Doon and Tarai valleys show dispersed and linear settlement ribbons along highways and canal banks.
21. Alpine zones retain transhumant seasonal dwellings (*Mait* in summer pastures; *Gunda/Munsa* in winter valleys).
22. Highest SC population count is in **Haridwar**; highest ST population count is in **Udham Singh Nagar**.
23. Do not swap highest overall literacy (**Dehradun**) with highest male literacy (**Rudraprayag**).
24. Do not swap highest SC % (**Bageshwar**) with highest ST % (**Udham Singh Nagar**).
25. Do not swap highest density (**Haridwar**) with lowest density (**Uttarkashi**).
26. Negative growth belongs to **Pauri** and **Almora** — not to Chamoli/Rudraprayag alone.

Uttarakhand — Topic 7 — Transport, Tourism & Hazards

Consolidated — 28 Must-Score Facts

1. **Silkyara–Barkot** bi-directional tunnel is **4.531 km** long on NH-134 under the Char Dham project and shortens the Gangotri–Yamunotri highway distance by more than **20 km**.
2. Airports / airstrips: **Jolly Grant** (Dehradun), **Pantnagar** (Udham Singh Nagar), **Naini Saini** (Pithoragarh), **Chinyalisaur** (Uttarkashi), **Gauchar** (**Chamoli**).
3. **Gauchar** is in **Chamoli** — not Rudraprayag (classic airport mismatch).
4. **Lipulekh Pass** (~5,334 m) links Pithoragarh with Tibet on the Kailash–Mansarovar route (India–Nepal–China tri-junction).
5. **Mana (Dungri-La)** (~5,608 m) and **Niti** (~5,068 m) link Chamoli with Tibet; Niti closed after the 1962 war.
6. **Sinla Pass** connects **Darma** and **Byans** valleys inside Pithoragarh (internal inter-valley pass).
7. **Traill's Pass** (~5,312 m) links Bageshwar (Pindari) with Pithoragarh (Milam); crossed by G. W. Traill in **1830**.
8. **Kalindi Pass** (~5,950 m) is a severe glaciated mountaineering pass linking **Gangotri (Uttarkashi)** with **Badrinath (Chamoli)**.
9. **Shrinkinganth Pass** connects Uttarkashi with Kinnaur / Himachal Pradesh.
10. Other Tibet-facing Garhwal–Kumaon passes include Thang-La, Unta Dhura, Chorhoti, and related high *la* / *dhura* corridors.
11. **Bugyals** are alpine meadows above the tree-line (about **3,000–4,500 m**); called Bugyal in Garhwal and Payar / Bugyal in Kumaon.
12. **Bedni Bugyal** (Chamoli) is the **largest** bugyal in the state and lies on the Nanda Devi Raj Jat route; Brahma Kamal flourishes here.
13. **Auli Bugyal** (Chamoli) is the premier winter skiing destination with a ropeway from Joshimath.
14. **Dayara Bugyal** (Uttarkashi) hosts the **Anduri Utsav / Butter Festival** on Bhado Sankranti.
15. **Chopta** (Rudraprayag) is called the “Mini Switzerland of Uttarakhand” and is the base for Tungnath–Chandrashila treks.
16. State Council for Climate Change (**SCCC**) was established in **2011**; UAPCC drafting belongs to the **Forest & Environment** side — not the Revenue Department.
17. Uttarakhand lies in **Seismic Zones IV and V** with high vulnerability along the Central Seismic Gap and MBT–MCT belt.
18. **Uttarkashi earthquake** of **20 October 1991** (~6.6 M) killed over **768** people along the MCT belt.
19. **Malpa landslide** of **17–18 August 1998** (Pithoragarh, Kali valley) killed about **221**, including Kailash–Mansarovar pilgrims and dancer **Protima Bedi**.
20. **Chamoli earthquake** of **29 March 1999** (~6.8 M) caused over **100** fatalities around Chamoli / Gopeshwar / Rudraprayag.
21. **Kedarnath** disaster of **16–17 June 2013** combined cloudburst with **Chaurabari Lake (Gandhi Sarovar)** breach and Mandakini deluge; over **5,700** fatalities and later SDRF strengthening.
22. **Tapovan / Rishiganga** disaster of **7 February 2021** (Chamoli) was a rock-ice avalanche that destroyed Rishi Ganga hydel and damaged Tapovan–Vishnugad; over **204** dead / missing.
23. **Joshimath land subsidence** (January 2023) affected wards built on ancient landslide moraine debris and forced large evacuations.
24. Do not swap **Sinla** (internal Darma–Byans) with **Lipulekh** (international to Tibet).
25. Do not swap **Dayara** (Butter Festival, Uttarkashi) with **Bedni** (largest bugyal, Chamoli).
26. Do not swap **Malpa 1998** (landslide / rockfall on Kali) with **Kedarnath 2013** (GLOF / cloudburst on Mandakini).
27. Silkyara exact length trap distractors are **3.531 km** and **6.531 km** — correct is **4.531 km**.
28. Char Dham corridor, border roads, and alpine tourism all sit on the same fragile slope–seismic hazard map.